Mirrleesian Carbon Taxation*

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Alleviating the economic damages from climate change is, to first order, a problem of efficiently limiting firms' emissions. We analyze a neoclassical general-equilibrium model in which fossil-energy use generates climate damages. The model crucially incorporates substantial cross-firm heterogeneity in emission intensity that we document using a novel firm-level dataset spanning 151 countries. Firms choose energy and other inputs and self-report emissions that are otherwise privately observed. Our central result is a simple formula for the optimal carbon tax that modifies marginal externality damages to account for firms' incentives to distort reported emissions. Unlike standard carbon-tax proposals, the optimal tax varies markedly across firms as a function of emission intensities and exceeds common uniform-tax benchmarks, on average. Quantifying the model shows large welfare gains from internalizing reporting incentives: the constrained optimal tax recovers around 80% of the potential welfare benefits; a uniform tax that does not internalize reporting incentives, by contrast, recovers only around 20% of them.

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1 Introduction

A defining challenge of our time is climate change. Given the central role that firms' carbon emissions play in its development, it is important to have a theory of carbon taxation that is effective. There is reason to believe that such a theory needs to be richer than the benchmark model of Pigouvian taxation. Building on the work of Pigou (1912, 1920), the dominant model of carbon taxation over the past three decades has been that of a tax on firms' unpriced carbon emissions, to ensure that firms internalize the true social cost of their actions (e.g., Nordhaus, 1977, 2007). This approach has, in turn, led to several real-world carbon tax proposals (e.g., Statement on Carbon Dividends, 2019).[1] Nevertheless, the practical implementation of carbon taxation faces several challenges.

An important obstacle—long recognized by policymakers (e.g., European Commission, 2023a) and practitioners (e.g., ECA, 2015) alike—is that firms' carbon disclosures are based on *firm-specific information* and primarily *self-reported*.[2] Combined with the presence of a carbon tax, this creates an incentive for firms to distort their carbon disclosures, hindering the effectiveness of carbon taxation.[3] In part as a consequence of these concerns and several high-profile court cases that have substantiated their relevance, trust in the effectiveness of carbon taxation is low among the general public (e.g., Dechezleprêtre et al., 2025).[4] Our aim with this paper is to fill the gap between the macroeconomic theory that addresses firms' climate externality through carbon taxation and its practical implementation, plagued by imperfect knowledge about firms' emissions.

To attain this objective, we propose a Mirrleesian modification of the standard Pigouvian approach that recognizes the need to internalize firms' reporting incentives. Within the context of a workhorse dynamic integrated-assessment model, we show that a simple, firm-specific adjustment to a uniform carbon tax optimally internalizes firms' disclosure incentives. Our constrained optimal tax varies markedly across firms as a function of their emission intensity and the social cost of carbon, and exceeds common uniform-tax benchmarks, on average. Quantifying our tax formula, we find substantial welfare and

[1]See, for example, Stiglitz et al. (2017), Parry et al. (2021), and Shultz and Baker (2017).

[2]See also GHG Protocols (2004), International Sustainability Standards Board (2023), ECA (2015), and SEC Carbon Disclosures (2024), among others.

[3]We discuss firms' carbon disclosures in the EU, US, and the UK in Section 2 and Appendix A.3.

[4]In Europe, recent enforcement actions have highlighted failures in emissions reporting and verification, including the Brikel/Bobov Dol EU ETS investigation (2021), the Scottish Environment Protection Agency v. Diageo plc (2022), and the Scottish Environment Protection Agency v. ExxonMobil Chemical Ltd (Mossmorran) (2010). We discuss these three representative legal cases in detail in Section 2, with additional institutional background and documentation provided in Appendix A. Even observed non-compliance often does not result in sizable fines (Calel et al., 2025). As acknowledged by policymakers, due to difficulties verifying carbon disclosures, these presumably represent the tip of the iceberg.

environmental gains from internalizing reporting incentives: the constrained optimal tax recovers around 80% of the potential social benefits; a uniform tax that does not internalize reporting incentives, by contrast, recovers only around 20%.

To motivate our analysis, we provide new evidence on firms' reported emissions. Using a comprehensive dataset on public and private firms' carbon-equivalent emissions (CO_2e) from 151 countries over 15 years—*the ICE ESG Company Database*—we show that firms' *direct, production-based* (scope 1) emissions account for at least half of overall emissions into the atmosphere—a lower bound, given that our estimate covers only ICE firms. Addressing climate change is, to a large extent, a problem of curtailing firms' emissions.[5]

Crucially, we document substantial *heterogeneity in firms' emission intensity*, defined as emissions per unit of output or revenue. Consistent with Lyubich et al. (2018), emission intensities are widely dispersed, even after controlling for country-sector-time fixed effects, and the spread is large relative to other familiar sources of firm heterogeneity such as TFP (Section 2 and Appendix A.3 discuss in detail the technological reasons behind this heterogeneity).[6] Finally, we exploit a novel feature of the ICE ESG database: we compare reported emissions for firms whose disclosures are *third-party verified* (that is, for which the internal emissions calculations are checked by external consultants) with those that are not. We find that non-verified firms report, on average, 20% lower emissions than otherwise similar verified firms, with the gap being larger in sectors with greater heterogeneity in emissions intensity. Our findings are therefore consistent with policymaker concerns about systematic over-optimistic reporting of emissions.

Our core contribution is to develop a theory of carbon taxation consistent with firms' incentives. We consider a workhorse dynamic integrated-assessment model in the spirit of Golosov et al. (2014) and Barrage and Nordhaus (2024). The distinguishing feature of our model is that firms are *heterogeneous*, both in their *productivity* and their *emission intensity*. The latter measures the extent to which a firm's energy use converts into carbon-equivalent emissions. Importantly, a firm's emission intensity, and hence its emissions, are partially *private information* to the firm, unknown to the social planner or other firms in the economy. This information friction creates a conflict between the social planner's desired disclosures for a firm and the firm's own optimal disclosure. The rest of the model is standard. In the model, firms produce output using capital, labor, and energy—with energy produced indirectly from output in a roundabout fashion—and invest in abatement to curb emissions. Emissions reduce economy-wide productivity through a

[5]The ICE database covers substantially more firms than the more commonly used S&P Trucost dataset, which is why our estimates are larger than those reported in e.g., Bolton and Kacperczyk (2023).

[6]Other papers documenting pervasive heterogeneity in emissions intensity include Capelle et al. (2023), De Lyon and Dechezleprêtre (2025), and Fontaine et al. (2026).

standard damage function (i.e., an *environmental externality*) that firms do not internalize. Within this environment, we derive two main results.

Our first main result shows that the combination of firm-specific information and heterogeneity in emission intensity challenges conventional carbon tax proposals. We show that a standard, homogeneous carbon tax—as implied in our economy by the Pigouvian optimal tax when the planner has full information about firms' emission intensities—introduces profit differentials between equally productive firms with different emission intensities. The tax scheme leads to larger tax bills for firms with higher emission intensity. In an economy in which emission intensities are partially private information, homogeneous carbon taxes are therefore *incentive-incompatible*; they generate strong incentives for firms to misreport their emissions. We show that the social cost of neglecting these incentives can be substantial, depending on the extent of heterogeneity.

The second main result builds on this challenge. We derive the optimal carbon tax when the social planner is constrained to choose allocations that are ex-post incentive-compatible. We refer to our constrained optimal tax as the *Mirrleesian carbon tax*. On the one hand, the Mirrleesian planner desires to curb emissions, to raise economy-wide productivity; but, on the other hand, efforts to reduce emissions also distort input use, creating misallocation that lowers productivity. The optimal full-information tax balances this *efficiency–misallocation trade*-off with a constant, homogeneous tax on emissions equal to the social cost of carbon (e.g., Golosov et al., 2014). By contrast, in the Mirrleesian case, the optimal constrained tax instead proportionally adjusts the full-information tax as a function of a firm's location in the carbon-intensity distribution, to ensure incentive compatibility. The Mirrleesian tax is, thus, *linear within firms but heterogeneous across firms*.

Indeed, a central feature of the Mirrleesian carbon tax is that it *declines* with a firm's emission intensity. A firm's incentive to misreport is unidirectional—high-emission-intensity firms wish to pretend to be less carbon intensive. Combined with the Mirrleesian tax schedule resulting in allocations—conditional on productivity—in which all firms pay the same total carbon tax as a share of revenue, this removes any incentive to misreport. We show that our constrained optimal tax varies markedly across firms as a function of their emission intensity and exceeds common uniform benchmarks, on average.

To explore the quantitative benefits of Mirrleesian carbon taxation, we calibrate the model to the within-sector, within-time heterogeneity in emission intensities for the verified U.S. ICE ESG sample. Absent information frictions, we estimate the optimal carbon tax to be \$60 per ton of CO_2 equivalent emissions (tCO_2e), in line with recent estimates of the social cost of carbon (e.g., IWG, 2021). By contrast, the Mirrleesian carbon tax schedule implies substantially higher marginal rates. The optimal tax is around \$200 per tCO_2e for the median emission-intensity firm, rising to roughly \$1,000 per tCO_2e for firms 1 standard deviation below the mean, and declining to about \$40 per tCO_2e for firms 1 standard deviation above. Notwithstanding this variation in marginal rates, total tax payments are *identical across firms with the same productivity* and on average amount to c. 1.5% of revenue. This contrasts with the sharply rising tax bills faced by high emission-intensity firms under the full-information tax. The Mirrleesian schedule thus pairs heterogeneous rates with equalized, modest total payments, removing any gains from misreporting.

Crucially, we document that the Mirrleesian adjustment extracts *most* of the potential benefits from carbon taxation. Compared to the (first-best) full-information tax, the Mirrleesian tax recovers around 80% of the potential welfare gains and 90% of the potential decline in emissions. A tax equal to its optimal value under full information—to which firms optimally respond by misreporting—by contrast, recovers only 20% of the social and environmental benefits attainable in the first best case.

We show that the reason behind this comparative performance is that most of the benefits from carbon taxation arise due to higher aggregate productivity driven by reduced emissions. Changes in capital and energy use, as well as the misallocation costs of carbon taxation, are in contrast small. We document through an exact decomposition of firms' emissions that the Mirrleesian tax attains a similar decline in overall emissions to that in the first-best, full-information case whenever the first-best correlation between firms' energy use and carbon intensity is close to zero—i.e., whenever productivity (as in our model) is the dominant driver of first-best energy choices. Our quantitative results thus underscore the importance of jointly accounting for productivity differences and carbon-intensity heterogeneity within one unified framework.

We conclude our analysis by showing that the relative performance of Mirrleesian taxation extends across a range of robustness exercises and extensions. First, we allow for a more complex climate specification in which the climate system is characterized by both emissions and temperature, following Miftakhova et al. (2020) and Folini et al. (2024)—an approach commonly used in the natural sciences (e.g., Allen et al., 2009; Matthews et al., 2009; Joos et al., 2013) and increasingly popular in macroeconomics (e.g., Dietz et al., 2021; Fernández-Villaverde et al., 2024). Second, we undertake a sensitivity analysis of the parameters that govern the climate block of the economy, focusing in particular on the abatement-cost elasticity and the climate-damage elasticity. Third, we introduce a renewable energy sector into the model, to account for one important driver of the heterogeneity in emission intensity. This follows the approach in Hassler et al. (2016), among others. We also allow for the possibility that abatement not only reduces emissions but also plays a productive role for firms, as in Lanteri and Rampini (2025). And lastly, we vary the accuracy of the planner's information about firms' emission intensities. The latter further allows us to compute ballpark estimates for "break-even monitoring costs". Crucially, conditional on the estimated benefits from Mirrleesian and full-information carbon taxation, emission verification does not easily become cost effective. Verification buys little when the modified tax schedule already does most of the work.

Finally, a premise of our analysis is that policymakers often have only crude estimates of firms' emission intensity. Greenstone et al. (2025) document a multi-decade attempt to gain more accurate information about firms' emissions in India and the costs associated with it.[7] Recognizing this informational gap and its impact on the efficacy of carbon taxation, policymakers across developed and developing economies have called for a combination of (i) *tougher compliance enforcement* (EC; ECA; WB) and (ii) *better disclosure standards plus independent assurance* (IFRS; IAASB; UNFCCC)—in effect attempting to decrease the information gap between policymakers and firms.[8] However, as also recognized by policymakers, the cost of doing so is often substantial—both in terms of the added regulatory burden placed on firms, resources spent on measurement devices, and the time spent constructing carbon estimates. In this paper, we instead propose a different approach, one that accepts an informational gap between policymakers and firms, and instead adjusts the carbon tax schedule to account for any incentive concerns. Our analysis suggests that information frictions should be treated as a first-order design concern for carbon taxation, much like uncertainty about social damages or discounting is currently done.

Related Literature. In addition to the works cited above, this paper relates to three strands of research. We review these in order of proximity below.

First, our paper builds on the work that followed Pigou's seminal contributions to the *optimal taxation of externalities* (Pigou, 1912, 1920). The general theory behind Pigouvian taxation is extended in Kang (2020) and Pai and Strack (2022). Important applications to climate change include Weitzman (1974), Nordhaus (1977, 1993), Acemoglu et al. (2012, 2016), Golosov et al. (2014), Barrage (2020), Van den Bremer and Van der Ploeg (2021), Barrage and Nordhaus (2024), Bilal and Känzig (2024), Bertolotti et al. (2026), among others. We complement this work by showing that, when emissions are highly heterogeneous and partially private information to firms, the standard Pigouvian approach results in

[7] A system referred to as continuous emissions measurement systems (CEMS) can in principle be fitted on every firm burner. But, as documented by European Commission (2023b) almost all installations instead use a calculation-based methodology. Indeed, only 145 installations (1.7%) in 22 countries reported using continuous emissions measurement systems (CEMS) in 2025, nine installations fewer than in 2021.

[8] These are: the European Commission (EC), the European Court of Auditors (ECA), the World Bank (WB), the International Financial Reporting Standards (IFRS), the International Auditing and Assurance Standards Board (IAASB), and the United Nations Framework Convention on Climate Change (UNFCCC).

incentive-incompatible allocations with distorted disclosures. The optimal tax formula that we derive shows how to optimally adjust the standard carbon tax for such concerns. In complementary and closely-related work, Cicala et al. (2022) study the design of allocation mechanisms when firms hold private information about their emissions. While Cicala et al. (2022) elicit firms' private information through an allocation mechanism relying on costly third-party emissions verification, we show that when such verification is prohibitively costly or not credible, a simple modification of the standard Pigouvian tax can recover much of the efficiency gains from climate taxation.

Second, our work rests on important insights from the literature on *optimal taxation under private information*. The literature on household income taxation in the presence of private information was pioneered by Mirrlees (1971) and Diamond (1998), among others, and later extended to the presence of household externalities by Rothschild and Scheuer (2016) and to dynamic contexts in Golosov et al. (2006), Kocherlakota (2010), and Stantcheva (2020). The literature on the optimal regulation of firms with private information is reviewed in Tirole (1988) and Laffont (1994), and with a recent dynamic application to firm R&D investments in Akcigit et al. (2022). Our contribution within this context is to show that utilizing the insights from this literature can restore the effectiveness of the standard Pigouvian approach to carbon taxation: a simple adjustment, based on firms' reported carbon intensities under the tax schedule, recovers most of the benefits of the textbook Pigouvian approach while ensuring compliance.

Lastly, our work is related to that which followed Allingham and Sandmo (1972) contribution on *tax evasion* and *misreporting incentives*. Recent applications to business taxation include Di Nola et al. (2021), Fernandez-Bastidas (2023), and Bhandari et al. (2024). We contribute to this literature by showing that its insights about the complex interaction between reporting incentives and taxation extend to Pigouvian taxes when agent-types are partially private information. But unlike this work, we do not address this challenge through improvements in auditing technology; instead, we propose a modification of the otherwise standard Pigouvian approach that recognizes its impact on firms' disclosure incentives. In this sense, our approach presents a natural evolution of the Pigouvian approach to carbon taxation—one that internalizes firms' carbon-disclosure incentives.

2 Motivating evidence

In this section, we document the *scope* of firms' carbon-equivalent emissions and the *heterogeneity* in their emission intensities. We then examine the drivers of this heterogeneity and how disclosure practices can translate into systematic misreporting at the firm level.

2.1 Firm emissions and heterogeneity

We present new evidence on firms' carbon-equivalent emissions. To start, we use micro data from *Intercontinental Exchange's (ICE's)* ESG company database. The ICE-ESG database provides one of the most comprehensive datasets on private and public firms' emissions in existence and spans the period 2009-2024 for 46,928 firms across 151 countries. The database reports, for an individual firm-year, estimates of CO_2-equivalent (CO_2e) direct emissions from production (i.e., *scope 1* emissions). We further validate our results using publicly-listed firms' CO_2e emissions data from *S&P Trucost*. Appendix A.1 provides more information on sample construction and the ICE-ESG emission measures.[9]

We begin by illustrating the importance of firms' carbon-equivalent emissions for overall emissions into the atmosphere. Figure 1 shows the share of global emissions accounted for by firms' emissions in the ICE-ESG sample. We plot estimates separately for the top 20 emitting countries. Data on world-wide emissions are taken from the *Global Carbon Budget report* (2024 estimates). Overall, firms' emissions account for at least half of aggregate emissions into the atmosphere—with U.S. and Chinese firms contributing by far most to the total. Importantly, notice that these estimates, although substantial, present only a lower bound on the importance of firms' emissions, as they only cover firms in the ICE-ESG sample. We conclude that firms' direct emissions are a key driver of overall emissions into the atmosphere.[10] Addressing climate change is, to a large extent, a problem of addressing firms' direct, production-based emissions.

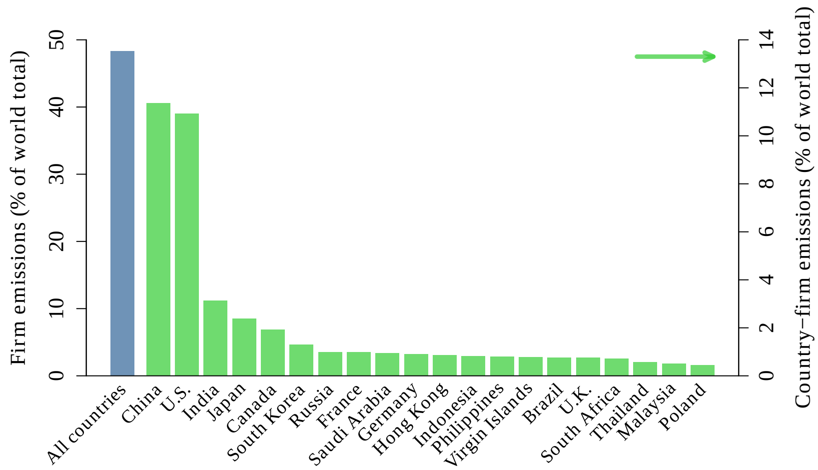
Such emissions are, in turn, the by-product of firms' supply of goods and services to the economy. The size of overall emissions attributable to firms, visible in Figure 1, thus raises the question of whether—and to what extent—firms differ in the efficiency of their carbon use. Figure 2 addresses this question by plotting the distribution of *emission intensities* in the ICE-ESG and Trucost samples. We measure emission intensity at the firm level by tons of CO_2e (tCO_2e) emissions per million $ of revenue. Panel (a) shows substantial heterogeneity in emission intensities in the raw data—similar to that documented in Lyubich et al. (2018), among others.[11] The larger heterogeneity visible in the ICE-ESG sample when compared to Trucost reflects in part the presence of non-publicly-listed firms.

[9]When referring to *carbon emissions*, we in all cases mean *carbon-equivalent emissions*. The ICE-ESG database provides emission estimates broken down into three components: scope 1 (direct emissions from production), scope 2 (indirect emissions from energy use), and scope 3 (all other indirect emissions). We focus on scope 1 emissions throughout our analysis to avoid double-counting emission flows. Figure A.1 in the Appendix reports the average share of scope 1, 2, and 3 emissions for a firm in the ICE-ESG database.

[10]This is in line with several technical reports on the topic (e.g., Report, 2017; IPCC, 2022). Figure A.2 in the Appendix shows the equivalent figure to that in Figure 1 for the S&P Trucost sample.

[11]Capelle et al. (2023), De Lyon and Dechezleprêtre (2025), and Fontaine et al. (2026) make a related point.

Figure 1: Estimated share of overall emissions



Note: Data from the ICE-ESG sample. The figure shows overall firm emissions in 2024 (measured by tons of CO_2e emissions) divided by total world emissions. Estimates for world emissions are taken from the *Global Carbon Budget* website (2024 estimates). Top-20 country contributions are shown on the right-side axis.

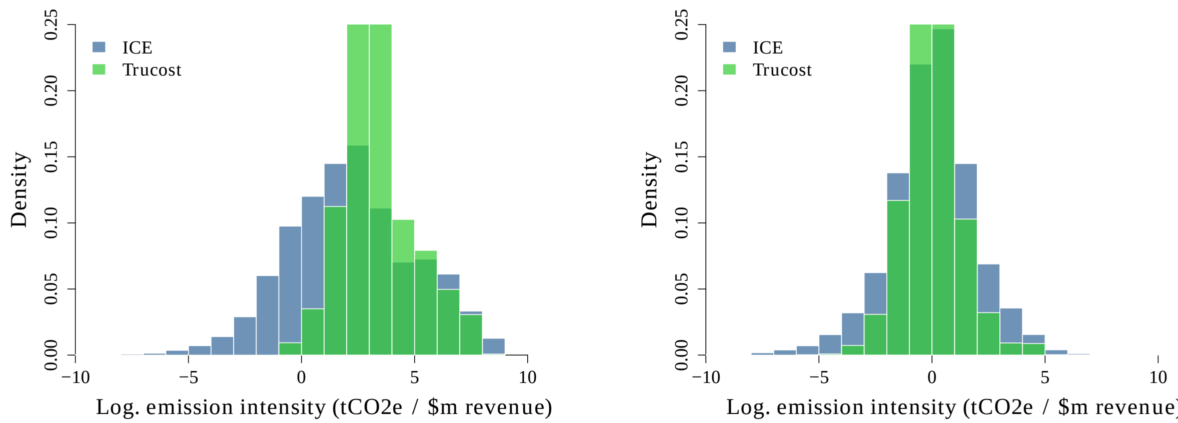
Clearly, part of this heterogeneity is due to differences in output and the production processes used in different sectors and countries at different points in time. To address this issue, Panel (b) in Figure 2 residualizes emission intensities for sector, country, and time fixed effects. Although a substantial share of the variation can be attributed to these factors—the standard deviation of the distribution decreases by 30%—the bulk of the heterogeneity remains.[12] Most of the differences in emission intensity are attributable to factors associated with a firm *within* a given sector, country, and time period. Crucially, we show that large-scale heterogeneity is also present in narrowly-defined sectors with relatively homogeneous output, further alleviating concerns that output differences drive the observed heterogeneity.[13] We perform the same analysis for the U.S. energy, airline, and steel industries in Figure A.3 in the Appendix. We measure output by megajoules produced, thousand round-trip flights flown, and tons of steel created. All distributions show substantial variation within sectors, as well as clear differences between them.

A natural question that arises is whether the documented heterogeneity is large compared to other sources of firm heterogeneity. Figure 3 gauges the overall size of this heterogeneity for the case of the U.S. by comparing it to the within-sector heterogeneity that exists in firm-level TFP. All else equal, the figure shows a substantially wider distribution for emission intensities than TFP: the level and the dispersion of the unit-free measure—

[12]Figures A.4 and A.5 document heterogeneity in emission intensities across sectors and countries. Consistent with earlier work, the most emission intensive sectors are energy production, agriculture, and resource extraction. Emerging economies are also, all else equal, more carbon intensive.

[13]Petersen (2025) finds large heterogeneity in emission intensities within narrowly defined industries in the German manufacturing sector.

Figure 2: Heterogeneity in emission intensities



(a) Overall variation　　**(b)** Residualized variation

Note: Data from the ICE-ESG and TruCost samples. The figure shows the histogram of log emission intensities. Emission intensities are measured by tons of CO_2e emissions divided by revenue (millions of $). Panel (a) shows the raw data, while Panel (b) residualizes the data wrt. sector, time and country fixed effects. We use ICE or NAICS-2 sectoral definitions, respectively. We consider only firms with high-quality reporting and trim emission intensities at the top and bottom 1% level (Appendix A.1).

the log within-industry 90/10 difference (Lyubich et al., 2018)—is almost 2-3 times larger than that for TFP.[14] Obviously, this comparison is not informative of the size of the aggregate consequences of this heterogeneity—our model will instead be informative of this—but what it does highlight is that there are indeed large-scale differences in the emission intensities when compared to other, standard measures of firm heterogeneity.

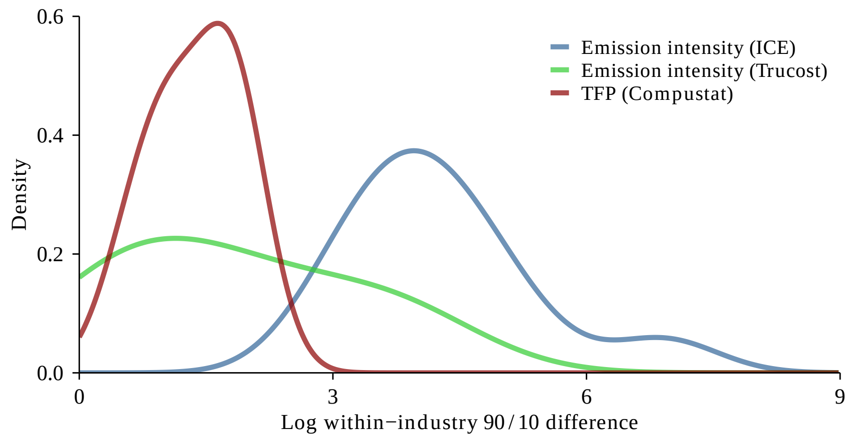
Finally, a related question is whether differences in emission intensity are associated with other measures of firm-level efficiency, such as TFP. Capelle et al. (2023) provide some evidence in this direction. Table A.2 in Appendix shows that, all else equal, more productive firms are also more carbon efficient; however, the overall contribution of this relationship to emission intensities is small ($R^2 \approx 5\%$): TFP is a poor predictor. This will be important later for the quantification of our baseline framework.

2.2 Sources of heterogeneity and disclosures

The above evidence documents substantial heterogeneity in emission intensities. As argued in the engineering and climate literatures, and embedded in the disclosure standards of major ETS markets, the ultimate source of this heterogeneity can be broken down into 3 components: (i) differences in the *amount of fuel* used in production (e.g., tons of coal

[14]Notice that Compustat and Trucost rely on the same sample of firms: U.S. publicly-traded firms.

Figure 3: Emission intensities and TFP



Note: Data from the ICE-ESG, Trucost, and Compustat samples for the U.S.. The figure shows the estimated kernel densities of the log within-industry 90/10 difference of emission intensities and total factor productivity. We use NAICS 2 industry codes for the Trucost and Compustat samples and ICE industry codes for the ICE-ESG sample. The top and bottom 1 percent of emission intensities and total factor productivity estimates have been removed. Total factor productivity is estimated as in Ottonello and Winberry (2020). We consider only firms with high-quality reporting in the ICE sample (Appendix A.1).

or m^3 of natural gas); (ii) differences in the *fuel mix* used (e.g., different types of coal or natural gas); and (iii) differences in the *technological adjustment* (e.g., differences in the oxidation rates of burners, filtration technology employed, or carbon-equivalent capture).[15] Combined, these factors can cause two firms, who otherwise produce a similar type and quantity of output, to feature remarkably dissimilar emission intensities. We discuss how each of these factors lead to heterogeneity in emission intensities in Appendix A.3.

An important feature of the emission data used in Figures 1-3 and those available from other sources is that they are *self-reported* (e.g., ICE, 2024; SEC Carbon Disclosures, 2024). Because the factors highlighted above turn on the quantity and specific blend of fuels used, the condition of individual burners, and the filtration technology installed at each installation, among others, a firm's true emission intensity depends on a vector of operational details that outside parties cannot easily assess. Consultancies and outside organizations are sometimes hired to validate the computations by which emission estimates have been arrived at, but the overall inputs into the process, and thus the final outcome, are in all cases up to a firm's discretion (GHG Protocols, 2004; International Sustainability Standards Board, 2023; SEC Carbon Disclosures, 2024). This private-information nature of the data used to compute firms' emissions has, as mentioned in the introduction, raised

[15]For more, see, for example, the reports from the European Commission Directorate-General for Climate Action (2025) and UK Department for Energy Security and Net Zero (2025).

concerns about systematic misreporting among policymakers and practitioners alike (e.g., ECA, 2015; European Commission, 2023a). Such concerns have been compounded by recent enforcement cases that have exposed two distinct modes of disclosure failure—duplicitous disclosures and non-disclosure altogether. Three cases have been particularly influential and are all in the context of the ETSs that operate in Europe:

- *The case of Brikel (EU ETS, 2021-2023):* Two coal-fired power plants in Bulgaria, linked by a common owner, reported coal emission factors of between 76.2 and 90.5 over 2018–2020, used to compute overall emissions, although they were allegedly burning coal with European Environment Agency reference value of 104.2, in addition to undisclosed industrial waste with a higher emission content. The Organized Crime and Corruption Reporting Project (2021) estimates that the under-declaration saved the two plants at least €26.6 million in EU ETS allowance costs over the period. Both plants' disclosures had been certified as accurate by a third-party verifier. As of early 2026, the investigation is still ongoing; Brikel was briefly closed in 2022 but continued to operate during court proceedings, incurring administrative fines of only a few thousand euros, a small fraction of the estimated ETS savings.

- *The case of Diageo plc (UK ETS, 2022):* Between 2013 and 2019, three Diageo whisky-production sites in Scotland carried out regulated combustion activity without the necessary permits, with the consequence that emissions from these sites went unreported for over 6 years. Closely-situated Diageo sites held such permits. Under the U.K. ETS only emission sources over a certain size require permits, and thus payments for carbon allowances. The breach was identified through an externally-triggered audit in 2019, and in March 2022 the Scottish Environment Protection Agency imposed a fine of £1.2 million—against group profits of £3.7 billion in 2021.[16]

- *The case of ExxonMobil (EU ETS, 2010):* In its 2008 disclosure under the EU ETS, ExxonMobil omitted three emission sources at its Mossmorran ethylene plant in Fife, altogether, understating reportable CO2e by approximately 33,000 tonnes per year. The Scottish Environment Protection Agency, warned of the under-reporting by a whistleblower, imposed a fine of £2.77 million in September 2010.[17]

[16] Furthermore, notice that, according to Calel et al. (2025), the observed overall non-compliance in the EU ETS should have resulted in €13 billion in total fines, but only €2.1 billion appear to have been collected.

[17] Notice that the UK implementation of the EU ETS already embeds a per-tonne under-reporting penalty well above the allowance market price. The mandatory civil penalty for undeclared emissions is €100 per tCO2e, irrespective of intent—several multiples of the prevailing allowance price over most of the sample period, and the rate at which SEPA's 2010 £2.77m fine against ExxonMobil Chemical was mechanically computed. That the Brikel and Diageo cases nevertheless document multi-year disclosure failure indicates that the effective deterrent depends jointly on the penalty rate and the probability of detection.

Table 1: Emission disclosures and emission intensity

	Log. emission intensity			
	(1)	(2)	(3)	(4)
No third-party validation	-0.193^{***}	-0.214^{***}	-0.244^{***}	-0.205^{***}
	(0.031)	(0.023)	(0.047)	(0.023)
No third-party validation × log(90/10)	.	.	.	-0.051^{***}
				(0.017)
Fixed effects				
Time		✓	✓	✓
Sector		✓	✓	✓
Country		✓		✓
U.S. only			✓	
Observations	32,268	32,268	6,520	30,298

Notes: Least-squares estimates from the ICE-ESG sample. ICE sectoral definitions and verification status are used. log(90/10) denotes the log 90th-to-10th percentile ratio of emission intensity, as defined in Figure 3. The top and bottom 1 percent of emission intensities (tons of CO_2e emissions divided by revenue in millions of dollars) are removed. Robust standard errors clustered by sector are in parentheses.

Appendix A.3 discusses each case in detail and explains how such failures could arise within existing disclosure practices and regulations governing ETS protocols.

Although the overall scope of misreporting is obviously unknown, a clear advantage of the ICE-ESG database is that it contains information about which firms have their disclosures validated (i.e., have their emission calculations checked). Indeed, we restricted Figures 2 and 3 to observations that ICE classifies as "higher quality"—that is, have their disclosures third-party validated. Table 1 shows that this restriction matters: firms without third-party validation tend to report systematically lower and more optimistic emission intensities (by around 20%), all else equal, suggesting scope for systematic under-reporting.[18] Moreover, focusing on the U.S., a major economy without carbon pricing, Table 1 shows that these results are unlikely to be driven by the interaction between third-party verification and country-level carbon pricing. Finally, the table shows that firms that are in sectors with more dispersed emission intensities appear to under-report more, consistent with these sectors having larger "scope for misreporting" in the first place.

In summary, in this section, we have documented that firms' emissions are a key driver of overall emissions into the atmosphere and that there is substantial heterogeneity in

[18]Notice that self-selection into validation seems less of a concern: if clean firms verify, verified firms should report lower intensities, not higher, as we find in Table 1.

firms' emission intensities, even within narrowly-defined sectors. Motivated by these findings and the self-reported nature of firms' emissions, in the next section, we extend a workhorse macro-climate model to allow for heterogeneity in emission intensity and asymmetric information. We then proceed to characterize and quantify the joint implications of these features for the optimal taxation of firms in general equilibrium.

3 Baseline model framework

We begin by developing our baseline economy with two key features; that firms are heterogeneous in their emission intensity and that firms' emission intensities are firm-specific information, unknown to the social planner and other firms in the economy.

3.1 Environment

Preferences and endowments. We consider an infinite-horizon economy with discrete time. The economy is populated by a representative competitive household with preferences over consumption streams, described by the utility function:

$$\mathcal{U} = \sum_{t=0}^{\infty} \beta^t \cdot u(C_t), \tag{1}$$

where $\beta \in (0, 1)$ is the discount factor, $u(\cdot)$ is the standard concave period utility function, and C_t is consumption at time t. The household is endowed with N units of labor, which it supplies inelastically to the labor market every period, and an initial stock of capital, $K_0 > 0$. The capital stock accumulates according to the recursion:

$$K_{t+1} = (1 - \delta_K) \cdot K_t + I_t, \tag{2}$$

where $\delta_K \in (0, 1)$ is the capital depreciation rate and I_t is the household's investment.

Technology, climate, and markets. Output is produced by a final-goods sector comprised of a continuum of heterogeneous competitive firms, who live for L periods, owned by households. Each firm is endowed at entry with *productivity* a_i and *carbon intensity* z_j, where pairs $\{a_i, z_j\} \in \mathcal{A} \times \mathcal{Z} \subseteq \mathbb{R}^2$ are jointly distributed with mass ω_{ij}, satisfying $\sum_{ij} \omega_{ij} = 1$. We assume $\mathcal{A} \times \mathcal{Z}$ is compact and index a firm of type (i, j) with age $\ell \leq L$ by (i, j, ℓ). Upon exit at age L, firms are replaced by an entering cohort of age $\ell = 1$ of equal size, implying a stationary age distribution with mass $1/L$ at each ℓ. We let $w_{ij\ell} \equiv \omega_{ij}/L$

denote the joint p.d.f. over types and ages, so that $\sum_{ij\ell} w_{ij\ell} = 1$. Firm (i, j, ℓ) produces quantity $y_{ij\ell,t}$ of output at time t according to the production technology:

$$y_{ij\ell,t} = A(G_t) \cdot \exp^{a_i} \cdot F\left(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}\right), \tag{3}$$

where $n_{ij\ell,t}$ and $k_{ij\ell,t}$ are the units of labor and capital employed by the firm, respectively, $e_{ij\ell,t}$ is the firm's energy use, and $F(\cdot)$ is a standard neoclassical production function with decreasing returns-to-scale. Economy-wide productivity $A(G_t)$ depends negatively on the stock of carbon in the atmosphere G_t ($A_G < 0$), so that increases in carbon concentration damage the productive capabilities of the economy. Given an initial concentration $G_0 > 0$, the stock of carbon in the atmosphere evolves in accordance with:

$$G_{t+1} = (1 - \delta_G) \cdot G_t + \sum_{ij\ell} w_{ij\ell} \cdot g_{ij\ell,t}, \tag{4}$$

where $g_{ij\ell,t}$ are firm (i, j, ℓ)'s emissions and $\delta_G \in [0, 1)$ is the rate of depreciation of carbon in the atmosphere. Firm (i, j, ℓ), in turn, produces emissions due to its energy use:

$$g_{ij\ell,t} = \exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t}, \tag{5}$$

where $\exp^{z_j - x_{ij\ell,t}}$ is the firm's *effective carbon intensity* and $x_{ij\ell,t}$ is the firm's expenditure on the *abatement* of emissions at cost $Q\left(x_{ij\ell,t}\right)$ with $Q' > 0$, $Q'' > 0$ for $x > 0$, and $Q(0) = 0$.[19] Energy is available to firms in infinite supply at price $P > 0$.[20] Finally, firms rent capital and labor in competitive markets with factor prices equal to R_t and W_t, respectively.

Information structure. A central friction in our economy is that a firm's carbon intensity z_j is *private information* to the firm. When choosing its energy use $e_{ij\ell,t}$, a firm of type (i, j, ℓ) knows z_j and thus the emissions $g_{ij\ell,t}$ that arise as a result of its energy use and abatement. However, this information is not shared with other firms in the economy or the social planner.[21] Clearly, this assumption presents a natural limit case of an information friction that exists between firms and the social planner about firms' carbon emissions, and we later study other cases in which the planner instead has partial (but

[19] Section 5.4 considers the case in which abatement costs also have a productive role for firms, as in Lanteri and Rampini (2025) and Bertolotti et al. (2026), in which they directly increase firm-specific TFP.

[20] This is equivalent to energy being produced by a competitive sector that produces energy from final goods at rate P—i.e., a so-called "round-about production function" for energy.

[21] Consistent with the literature on information frictions, we nevertheless assume that the distribution of z_j is known to all in the economy. Given our focus on emission disclosures, we also assume that abatement costs are known by all, unlike in the earlier work by Kwerel (1977) and Roberts and Spence (1976).

still incomplete) information about firms' carbon-intensity type (Section 5.4).

3.2 Discussion of the environment

The above environment includes four features central to the carbon taxation of firms. First, in the above environment, firms differ along two dimensions, in addition to their age: *carbon intensity* z_j and *productivity* a_i. We show below that jointly accounting for both sources of heterogeneity is crucial for the effective design of corrective policy. In particular, the effectiveness of corrective policy interventions will depend centrally on the relative importance of productivity and carbon intensity in shaping the first-best allocation of energy and abatement across firms.

Second, our baseline framework links firms' emissions to their energy use through an effective carbon intensity $\exp^{z_j - x_{ij\ell,t}}$ that varies across firms (Equation 5). As we shall see below, this heterogeneity maps one-for-one into differences in the emission intensity measure—emissions per unit of revenue—shown in Figure 2. Cross-firm variation in emissions, in turn, reflects both differences in the amount of energy use and in the composition of fuels and technologies used (Section 2.2 and Appendix A.3), captured here by the term $\exp^{z_j - x_{ij\ell,t}}$. While abatement is the primary margin through which firms reduce their emissions intensity in our baseline model, we extend the framework in Section 5.4 to also allow firms to choose between *renewable energy*, which is carbon-free, and *fossil-based energy*, which emits carbon. This follows the approach in, e.g., Golosov et al. (2014).

Third, our environment allows firms to live for multiple periods L, enabling us to study the static and dynamic incentives faced by a social planner in designing optimal corrective policy. Specifically, it allows us to analyze how current and future disclosures interact, and potentially modify the design of optimal carbon policy.

Finally, our baseline model views the climate as well-approximated by one variable: the current stock of carbon in the atmosphere G_t. Following Golosov et al. (2014) and others, we argue that this is a reasonable approximation given available medium-complexity climate models used in the natural sciences. Section 5.4 analyzes the case in which the damage function instead depends negatively on mean temperature, which in turn depends in a near non-stationary manner on atmospheric carbon, in addition to its own lagged value. This follows the approach taken in Miftakhova et al. (2020) and Folini et al. (2024), among others, which has recently found some popularity in the natural sciences (e.g., Allen et al., 2009; Matthews et al., 2009; Joos et al., 2013), as well as in macroeconomics (e.g., Dietz et al., 2021; Fernández-Villaverde et al., 2024). We nevertheless stress that our main results do not hinge on whether climate-related damages arise directly from carbon concentration or are mediated by temperature, nor on the speed at which these damages materialize over time. Our assumption that it is only the production-side of the economy that is affected is made to keep our analysis comparable to those in Nordhaus (2007), Golosov et al. (2014), Barrage and Nordhaus (2024), and others.

4 Equilibrium and planner allocations

We next turn to the equilibrium and social planner allocations for our economy, the central difference between them being the extent to which firms' energy and abatement use internalize the emission externality, $A(G_t)$. We start with the decentralized equilibrium.

4.1 Decentralized equilibrium

Household problem. The representative household chooses consumption C_t and capital K_{t+1} to maximize its utility in (1) subject to the per-period budget constraint:

$$C_t + K_{t+1} = W_t \cdot N + (R_t + 1 - \delta_K) \cdot K_t + \sum_{ij\ell} w_{ij\ell} \cdot \pi_{ij\ell t}, \tag{6}$$

where $\pi_{ij\ell,t}$ denotes the period-t profits of firm (i, j, ℓ). The household takes the wage W_t and the rental rate of capital R_t as given when solving its problem. The solution yields:

$$u'(C_t) = \beta \cdot (R_{t+1} + 1 - \delta_K) \cdot u'(C_{t+1}). \tag{7}$$

Firm problem. The problem of firm (i, j, ℓ) is similarly standard, as firms in the decentralized equilibrium do not internalize the damages their emissions have on the productive capabilities of the economy. Instead, firm (i, j, ℓ) chooses labor $n_{ij\ell,t}$, capital $k_{ij\ell,t}$, energy $e_{ij\ell,t}$, and abatement $x_{ij\ell,t}$ to maximize its present discounted value of profits:

$$v_{ij\ell,t} = \pi_{ij\ell,t} + \mathbb{1}_{\{\ell<L\}} \cdot \beta \cdot \frac{u'(C_{t+1})}{u'(C_t)} \cdot v_{ij,\ell+1,t+1}, \tag{8}$$

where per-period profits are given by:

$$\pi_{ij\ell,t} = y_{ij\ell,t} - W_t \cdot n_{ij\ell,t} - R_t \cdot k_{ij\ell,t} - P \cdot e_{ij\ell,t} - Q(x_{ij\ell,t}), \tag{9}$$

subject to (3). The firm thus optimally equates the *marginal revenue product* of labor and capital with the wage W_t and the rental rate of capital R_t, respectively:

$$mrpn_{ij\ell,t} \equiv A(G_t) \cdot \exp^{a_i} \cdot F_n(\cdot) = W_t \quad \text{and} \quad mrpk_{ij\ell,t} \equiv A(G_t) \cdot \exp^{a_i} \cdot F_k(\cdot) = R_t, \quad (10)$$

while the relevant conditions for energy $e_{ij\ell t}$ and abatement $x_{ij\ell t}$ are:

$$mrpe_{ij\ell,t} \equiv A(G_t) \cdot \exp^{a_i} \cdot F_e\left(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}\right) = P \quad \text{and} \quad Q'\left(x_{ij\ell,t}\right) = 0. \quad (11)$$

Equation (10) presents a main departure point for our analysis: a value-maximizing firm (i, j, ℓ) optimally equates the marginal revenue product of energy $mrpe_{ij\ell,t}$ to its price P and chooses zero abatement, $x_{ij\ell t} = 0$. Importantly, notice that Equations (10) and (11) show that a firm's value-maximizing input choices are independent of its carbon intensity z_j and age ℓ. The reason is that neither carbon intensity nor age alter the firm's productive capabilities. We are now ready to define the equilibrium of our economy:

Equilibrium notion and characterization. A decentralized (competitive) equilibrium consists of household allocations $\{C_t, K_{t+1}\}_{t=0}^{\infty}$, firm allocations $\left\{\{n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t}\}_{i,j,\ell}\right\}_{t=0}^{\infty}$, factor prices $\{W_t, R_t\}_{t=0}^{\infty}$, and firm-driven emissions $\{\{g_{ij\ell,t}\}_{i,j,\ell}, G_t\}_{t=0}^{\infty}$ such that:

- Given factor prices, W_t and R_t, the household and firm allocations solve the household and firm problem, respectively (i.e, Equations (6)-(11) hold).
- Given allocations, the labor, capital, and goods market clears; i.e., for all t:

$$\sum_{ij\ell} w_{ij\ell} \cdot n_{ij\ell,t} = N \quad \text{and} \quad \sum_{ij\ell} w_{ij\ell} \cdot k_{ij\ell,t} = K_t, \quad (12)$$

  and:

$$C_t + K_{t+1} = (1 - \delta_K) \cdot K_t + \sum_{ij\ell} w_{ij\ell} \cdot \left(y_{ij\ell,t} - P \cdot e_{ij\ell,t} - Q(x_{ij\ell,t})\right). \quad (13)$$

- Emissions $\{\{g_{ij\ell,t}\}, G_t\}$ and capital K_{t+1} evolve in accordance with (2) and (4)-(5).

The decentralized equilibrium features a tight link between a firm's carbon intensity and the emission-intensity measure shown in Figure 2. Indeed, it follows directly from Equation (11) that in the case in which $F(\cdot)$ admits the Cobb-Douglas form: $\log(g_{ij\ell,t}/y_{ij\ell,t}) = \log(\nu/P) + z_j$, where $\nu = \partial \log y_{ij\ell,t} / \partial \log e_{ij\ell,t}$.[22] Variations in carbon intensity z_j translate one-for-one into variations in emission intensity $\log(g_{ij\ell,t}/y_{ij\ell,t})$ and are the ultimate

[22]In particular, we have that: $\log\left(\frac{g_{ij\ell,t}}{y_{ij\ell,t}}\right) = \log\left(P^{-1} \cdot \frac{\partial \log y_{ij\ell,t}}{\partial \log e_{ij\ell,t}}\right) + z_j = \log\left(\frac{\nu}{P}\right) + z_j$.

source behind this heterogeneity in the model. This tight relationship between z_j and the emission-intensity measure used in the data extends more generally to any neoclassical production function. We summarize this result and our discussions above in Lemma 1:

Lemma 1. *In the decentralized equilibrium (DE):*

i. Firm (i, j, ℓ)'s energy use $e^{DE}_{i,t}$ and abatement $x^{DE}_{i,t}$ satisfy:

$$mrpe^{DE}_{i,t} = P \quad \text{and} \quad Q'\left(x^{DE}_{i,t}\right) = 0, \quad \forall j, \ell, \tag{14}$$

and are independent of the firm's carbon-intensity type j and age ℓ.

ii. Variations in carbon intensity z_j translate one-for-one into variations in emission intensity:

$$\log\left(g^{DE}_{ij,t}/y^{DE}_{i,t}\right) = \log(\nu/P) + z_j + o(1), \tag{15}$$

where $\partial \log y_{i,t}/\partial \log e_{i,t} \equiv \nu + o(1)$.

Finally, notice that a key feature of the decentralized equilibrium is that firms do not internalize the damages from emissions, $A(G_t)$, when making their energy and abatement choices. A firm directly equates its marginal benefit of energy, the marginal revenue product of energy, with its marginal private cost, equal to the market price, and chooses zero abatement. This, in turn, follows from the profit function in (9) being independent of emissions and carbon intensity, $\partial\pi_{i,t}/\partial g_{ij,t} = \partial\pi_{i,t}/\partial z_j = 0$, leading a firm's input choice to be independent of its carbon-intensity type. The decentralized equilibrium, as such, features a conflict between profit maximization and the overall productivity of the economy, leading, as we will see next, to a suboptimal allocation of all factors of production.

4.2 The full-information planning problem

We proceed with the planning problem in which the social planner has full information about firms' carbon intensity $\{z_j\}$, and thus their emissions $\{g_{ij\ell,t}\}$. Although our ultimate interest lies with the solution to the planning problem in which firms have private information, it will be useful to start with the full-information case, as the comparison between the two will allow us to sharply characterize the costs of information frictions.

The full-information planner solves the problem:

$$\max_{\{C_t, K_{t+1}, \{k_{ij\ell,t}, n_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t}\}_{i,j,\ell}\}_{t=0}^{\infty}} \mathcal{U} = \sum_{t=0}^{\infty} \beta^t \cdot u\left(C_t\right) \tag{16}$$

subject to Equations (3), (4), (5), (12), and (13), as well as the usual non-negativity constraints (Appendix B.1). To describe the solution to the problem, it will be helpful to first define two objects, which combined describe how the solution departs from the decentralized outcome. The first measures the relevant notion of the *marginal social cost of energy* for an individual firm; the second, the related *marginal social benefit of abatement*.

Definition 1. *Let* $\Lambda_{G,t} \equiv \sum_{h=0}^{\infty} \beta^{h+1} \cdot \frac{u'(C_{t+h+1})}{u'(C_t)} \cdot (1-\delta_G)^h \cdot \left(-\frac{A'(G_{t+h+1})}{A(G_{t+h+1})}\right) \cdot \mathcal{Y}_{t+h+1} > 0$ *denote the* ***social cost of carbon****, where* $\mathcal{Y}_t \equiv \sum_{ij\ell} w_{ij\ell} \cdot y_{ij\ell,t}$ *is* ***gross output*** *in the economy. Firm* (i,j,ℓ)*'s* ***marginal social cost of energy*** *is* $\lambda_{G,ij\ell,t} \equiv \Lambda_{G,t} \cdot \exp^{z_j - x_{ij\ell,t}} > 0$*, while its* ***marginal social return on abatement*** *is denoted by* $\Delta_{G,ij\ell,t} \equiv \lambda_{G,ij\ell,t} \cdot e_{ij\ell,t} > 0$.

Let $\Lambda_{K,t}$ denote the constraint multiplier on the market-clearing condition for capital at time t. The first-order conditions for K_{t+1} and C_t then once more yield the standard consumption Euler equation in (7) with R_{t+1} replaced by $\Lambda_{K,t+1}$.

Let $\Lambda_{N,t}$ analogously denote the constraint multiplier on the market-clearing condition for labor at time t. The full-information planner then further optimally equates the marginal revenue product for labor and capital to the multipliers $\Lambda_{N,t}$ and $\Lambda_{K,t}$, respectively, akin to (10). The relevant conditions for energy $e_{ij\ell,t}$ and abatement $x_{ij\ell,t}$ are:

$$mrpe_{ij\ell,t} = P + \underbrace{\Lambda_{G,t} \cdot \exp^{z_j - x_{ij\ell,t}}}_{=\lambda_{G,ij\ell,t}} \quad \text{and} \quad Q'\left(x_{ij\ell,t}\right) = \underbrace{\Lambda_{G,t} \cdot \exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t}}_{=\Delta_{G,ij\ell,t}}. \tag{17}$$

Equation (17) characterizes the planner allocations of energy and abatement and can be directly compared to Equation (14), which describes the decentralized equilibrium. Notice that these equations—combined with the conditions for labor and capital—once more provides solutions for firms' input allocations that are independent of firm age ℓ, as age does not alter the productive capabilities of firms.[23]

The central difference between the two sets of conditions arises because $\Lambda_{G,t} > 0$. This term captures the relevant notion of the *social cost of carbon* for our economy: the present discounted value of marginal productivity damages created by an additional unit of emissions (Definition 1). Multiplying it by firm (i,j,ℓ)'s effective carbon intensity $\exp^{z_j - x_{ij,t}}$ yields firm (i,j,ℓ)'s *marginal social cost of energy* $\lambda_{G,ij,t}$ (Definition 1). Equation (17) therefore shows that the full-information planner equates a firm's marginal revenue product of energy to its *overall social cost*—the market price P *plus* the marginal social cost $\lambda_{G,ij,t}$—whereas firms in the decentralized equilibrium account only for the market price. Similarly, the planner equates the marginal cost of abatement to its *marginal social return* $\Delta_{G,ij,t}$, whereas firms in the decentralized equilibrium always choose zero abatement.

[23] We thus suppress the use of the subscript ℓ for the rest of this subsection when it does not cause confusion.

Thus, because of the internalization of the emission externality—driven by the positive social cost of carbon ($\Lambda_{G,t} > 0$)—the full-information planner allocation differs from that of the decentralized equilibrium. Indeed, it follows from Equations (17) that:

Lemma 2. *In the full-information planner (first-best, FB) allocation:*

i. Firm (i, j, ℓ)'s energy use $e^{FB}_{ij,t}$ and abatement $x^{FB}_{ij,t}$ satisfy:

$$mrpe^{FB}_{ij,t} = P + \lambda_{G,ij,t} \quad \text{and} \quad Q'\left(x^{FB}_{ij,t}\right) = \Delta_{G,ij,t}, \quad \forall \ell, \tag{18}$$

and are independent of the firm's age ℓ.

ii. For given factor prices (multipliers), the planner depresses energy and increases abatement efforts heterogeneously across firms. In particular, for any i and any (j', j) with $z_{j'} > z_j$:

$$e^{FB}_{ij',t} < e^{FB}_{ij,t} < e^{DE}_{i,t} \quad \text{and} \quad x^{FB}_{ij',t} > x^{FB}_{ij,t} > x^{DE}_{i,t} = 0. \tag{19}$$

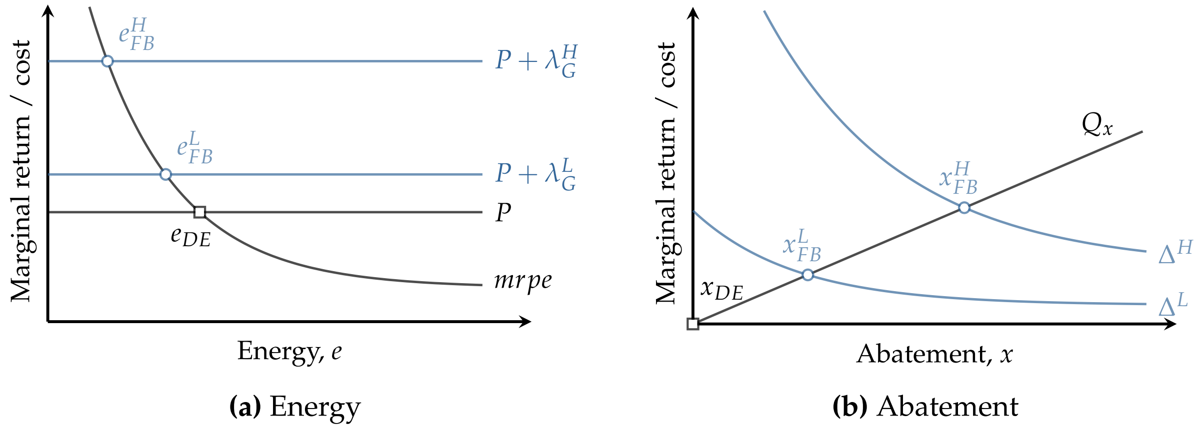
Lemma 2 highlights two consequences of the first-best internalization of the emission externality that will be important later. First, because the marginal social cost of energy $\lambda_{G,ij,t}$ is positive, the full-information planner depresses all firms' energy uses relative to what firms would choose to maximize their value, $e^{FB}_{ij,t} < e^{DE}_{ij,t}$ (Equation 14 and 18). The planner likewise commands firms to abate their emissions, $x^{FB}_{ij,t} > x^{DE}_{i,t} = 0$. Both adjustments reduce emissions and thereby raise economy-wide productivity.

Second, these adjustments are *heterogeneous across the carbon-intensity distribution*. In particular, because a firm's marginal social cost of energy increases with its carbon intensity, the planner allocates less energy and more abatement to more carbon-intensive firms. Figure 4 shows graphically how the energy and abatement allocations are modified by the full-information planner relative to what a firm would choose in equilibrium.

Moreover, notice that these heterogeneous distortions imply *factor misallocation*: the marginal revenue products of energy are no longer equalized across firms. The planner accepts this misallocation as the productivity gains from lower emissions exceed its cost.

Finally, because firms' marginal revenue products and the marginal cost of abatement are equated to type-specific objects, it follows that the planner also creates *profit and value differentials*, even among equally productive firms. In particular, as the full-information planner allocates more energy to less carbon-intensive firms, it pushes their allocations closer to those in the decentralized equilibrium and thereby allows them to attain higher profits and values. Indeed, it follows that:

Figure 4: The full-information social planner allocation



Note: The figure illustrates the determination of energy (Panel a) and abatement (Panel b) in the decentralized equilibrium (black) and the full-information allocation (blue). The figure considers the case in which firms have either a high or low marginal social cost of energy, $\lambda^H_G > \lambda^L_G$.

Corollary 1 (Incentive-incompatibility). *In the full-information allocation, the implied value of firm (i, j, ℓ) is higher than that of firm (i, j', ℓ), $v^{FB}_{ij\ell,t} > v^{FB}_{ij'\ell,t}$ for all (i, ℓ) when $z_j < z_{j'}$.*

As we show further below, the full-information allocation can be implemented by a uniform carbon tax. In the absence of full information, however, any value differential across otherwise similar firms generates incentives for high carbon-intensive firms—those with larger values of z_j—to misreport their emissions intensity to increase firm value. Such behavior in fact becomes part of firms' optimal disclosure strategy and thereby undermines the effectiveness of the tax, as we demonstrate next.

A naïve planner. Suppose a *naïve planner (NV)* was to act as the full-information planner presented above but without having direct information on firms' carbon intensities $\{z_j\}$. In such an economy, all firms would find it optimal to report having the lowest carbon intensity, i.e., $\underline{z} \equiv \min_{z\in\mathcal{Z}} z_j = \arg\max v_{ij\ell,t}$.[24] A firm would always have a *dominant reporting strategy* given by type $(i, \underline{j}, \ell)$, where $\underline{j}$ is the index that corresponds to $\underline{z}$. The first-order conditions for energy and abatement would then become:

$$mrpe^{NV}_{ij,t} = P + \lambda_{G,i\underline{j},t}, \quad Q'(x^{NV}_{ij,t}) = \Delta_{G,i\underline{j},t}, \tag{20}$$

where $\lambda_{G,i\underline{j},t} = \Lambda_{G,t} \cdot \exp^{\underline{z}-x_{ij,t}}$ is the *misreported marginal social cost of energy* and $\Delta_{G,i\underline{j},t} = \lambda_{G,i\underline{j},t} \cdot e_{ij,t}$ is the *misreported marginal return of abatement*. Clearly, both conditions imply

[24] Recall that Section 2 showed that firms who do not have their emission calculations checked report systematically lower emissions, and that firms who operate in sectors with more dispersed emission intensities "mistate" their emissions by more.

that firms would use more energy and do less abatement than in the full-information case ($\lambda_{G,ij,t} > \lambda_{G,i\underline{j},t}$), implying a smaller internalization of the emission externality.

4.3 The Mirrleesian planning problem

We now turn to our main problem of interest: the problem faced by a social planner that is uncertain about firms' carbon intensity $\{z_j\}$. This Mirrleesian planner faces the same problem as the full-information planner but, in addition, faces two constraints: (i) an *incentive-compatibility constraint* and (ii) an *individual-rationality constraint*.

Since a firm of type (i, j, ℓ) can always pretend to be of type (i, j', ℓ), where $j \neq j'$, in the planner's allocation, each firm-type's value must exceed those from another carbon-intensity type (*incentive compatibility*):[25]

$$v(\iota_{ij\ell,t}, v_{ij,\ell+1,t+1} \mid a_i, z_j, \ell, W_t, R_t) \geq v(\iota_{ij'\ell,t}, v_{ij',\ell+1,t+1} \mid a_i, z_j, \ell, W_t, R_t) \quad \forall j' \neq j, \tag{21}$$

where $\iota_{ij\ell,t} \equiv \left[n_{ij\ell,t}\ k_{ij\ell,t}\ e_{ij\ell,t}\ x_{ij\ell,t}\right]$ denotes the vector with the firm's input choices, and the right-hand side of (21) is the value of firm (i, j, ℓ) when it receives the allocation intended for firm (i, j', ℓ). Moreover, to be willing to participate in the mechanism, a firm of type (i, j, ℓ) must obtain non-negative value from it (*individual rationality*):

$$v(\iota_{ij\ell,t}, v_{ij,\ell+1,t+1} \mid a_i, z_j, \ell, W_t, R_t) \geq 0. \tag{22}$$

The problem faced by the planner subject to these additional constraints resembles the classical problem faced by all Mirrleesian planners in the household taxation literature (e.g., Mirrlees, 1976). The central difference is that a firm's unobserved type is not directly payoff-relevant: conditional on its inputs and abatement, a firm's profits in (9) do not depend on its carbon-intensity type, $\partial\pi_{ij\ell,t}/\partial z_j = 0$.[26] This contrasts with the household case in which the household's unobserved type (e.g., often the effort-type) matters *directly* for the household's payoff through its earned income.

We impose one further restriction on the planner's problem. An important feature of our framework is that, although the use of energy has an environmental externality, the decentralized use of labor and capital is efficient. The full-information planner equates the marginal revenue product of these inputs to their respective prices. This, in turn, im-

[25] Because we are assuming that productivity a_i is known, or alternatively that all inputs are known and thus productivity can be recovered from the production function accordingly, no firm of type (i, j, ℓ) can pretend to be of type (i', j, ℓ) and thus no incentive compatibility constraints are needed on this dimension.

[26] Thus, the *single-crossing condition*—often used as a sufficient condition for the existence of a truth-telling mechanism, because it allows one to focus on local (nearest) deviations—is not satisfied in our setting.

plies that the Mirrleesian planner has an incentive to distort labor and capital markets to relax the above incentive and rationality constraints. A "Theory of the Second Best" logic applies to our framework (Lipsey and Lancaster, 1956)—a logic that may be challenging to implement in practice. Rather than allowing such allocations, we focus in the following on the more constrained (and arguably more realistic) allocations in which:

$$W_t = \Lambda_{N,t} \quad \text{and} \quad R_t = \Lambda_{K,t}, \tag{23}$$

where $\Lambda_{N,t}$ and $\Lambda_{K,t}$ once more denote the multipliers on the market-clearing conditions for labor and capital. That is, we assume that the (implied) wages and interest rates used by firms in Equations (21) and (22) to compute their values equal the corresponding shadow costs, thus ensuring that the planner leaves labor and capital markets undistorted.[27] As we show in Section 5.2, imposing such "no-implied transfer" constraints does, in either case, not matter much for the relative benefits of the Mirrleesian approach. Consistent with a no-transfer approach, we also abstract from any lump-sum transfers in the incentive and participation constraints, respectively (Equations 21 and 22).

The Mirrleesian planner thus solves the problem:

$$\max_{\{C_t, K_{t+1}, W_t, R_t, \{k_{ij\ell,t}, n_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t}\}_{i,j,\ell}\}_{t=0}^{\infty}} \mathcal{U} = \sum_{t=0}^{\infty} \beta^t \cdot u\left(C_t\right) \tag{24}$$

subject to the aforementioned Equations (3), (4), (5), (12), and (13), in addition to Equations (21), (22), and (23) above, as well as the usual non-negativity constraints.

At first pass, the Mirrleesian planner's problem may seem daunting. The non-convexity of the constraint set, induced by the incentive constraints, precludes the use of standard methods. However, the direct pay-off irrelevance of a firm's carbon-intensity type, $\partial \pi_{ij\ell,t} / \partial z_j = 0$, combined with the monotone ranking of firm profits in z_j in the full-information allocation (Corollary 1), provides a way forward. A consequence of the two is that each firm has a dominant strategy: to (mis)report its carbon-intensity type and claim to be of type $\underline{z} \equiv \min_{z \in \mathcal{Z}} z_j$, as is the case under the naïve planner. It follows that a firm's profits in the Mirrleesian case must be independent of its carbon-intensity type, $\pi_{ij\ell,t} = \pi_{i\ell,t}$ for all j, and so too its value, $v_{ij\ell,t} = v_{i\ell,t}$ for all j. We show in Appendix B.1 that such value equalization—the fact that the incentive-compatibility constraint binds for all j—together with optimality, in turn, implies that the planner chooses a *symmetric allocation* such that $\{k_{ij\ell,t}, n_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t}\} = \{k_{i,t}, n_{i,t}, e_{i,t}, x_{i,t}\}$ for all (j, ℓ).[28]

[27] See also, for example, Armstrong and Sappington (2007) for discussions of this and related approaches.
[28] The Inada condition on $F(\cdot)$ ensures the individual-rationality constraint in Equation (22) never binds.

Crucially, notice that the allocation is also independent of firm age ℓ. The planner never *experiments* with allocations across firm age, to learn a firm's carbon-intensity type, and then change the allocation later on. Instead, as we show through a backwards induction argument in Appendix B.1, the Mirrleesian planner always sets allocations independent of firm age ℓ. This follows from the technologies available to a firm being independent of its age, and the fact that we consider a planning problem under *full commitment*. Although the case *without commitment* is interesting in its own right, we view the case *with commitment* to ultimately be the case of greater applied interest.

A simple example can help with the intuition behind the symmetry of the optimal allocation. Suppose energy is the only input into production, that firms share the same productivity, live for only one period, and that there is no abatement. Suppose further that for a firm of type j, there are two energy allocations, $e_{1,jt}$ and $e_{2,jt}$, which satisfy incentive-compatibility, $\pi(e_{1,jt}|z_j) = \pi(e_{2,jt}|z_j)$, with $e_{1,jt} < e_{2,jt}$. This is generically the case. With energy as the sole input, consumption equals value-added in the economy, which in turn equals total profits. Thus, any allocation that uses only $e_{1,jt}$ and $e_{2,jt}$ is incentive compatible and delivers the same welfare absent the externality. Yet, because emissions enter through Equation (3), and thus create a negative externality, $\partial A(G_t)/\partial G_t < 0$, the Mirrleesian planner prefers the lower energy level $e_{1,jt}$ for all type-j firms.

Using this symmetry of allocations, it is straightforward to derive the solution to the Mirrleesian problem. To describe its solution, it will, however, be useful to first define:

Definition 2. *Let $\overline{\lambda}_{G,i,t} \equiv \mathbb{E}[\lambda_{G,ij,t} \mid a_i] > 0$ be type i's* ***average marginal social cost of energy*** *and $\overline{\Delta}_{G,i,t} \equiv \overline{\lambda}_{G,i,t} \cdot e_{i,t} > 0$ be type i's* ***average marginal social return on abatement****.*

Recall that $\Lambda_{K,t}$ and $\Lambda_{N,t}$ denote the constraint multipliers on the market-clearing condition for capital and labor. The Mirrleesian planner, as the full-information planner, equates the marginal revenue product for capital $k_{i,t}$ and labor $n_{i,t}$ to the multipliers $\Lambda_{K,t}$ and $\Lambda_{N,t}$, respectively. The planner's conditions for K_{t+1} and C_t also again yield the standard Euler equation for consumption in (7), while the social cost of carbon $\Lambda_{G,t}$ equals that in Definition 1. The difference between the full-information and the Mirrleesian allocation only manifests itself in the firm-specific conditions for energy and abatement:

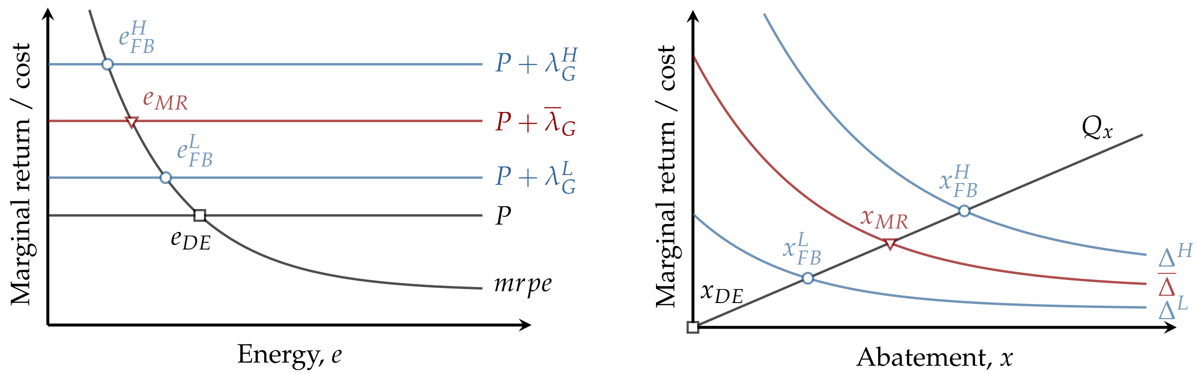
Lemma 3. *In the Mirrleesian planner (MR) allocation:*

i. Firm (i, j, ℓ)'s energy use $e_{i,t}^{MR}$ and abatement $x_{i,t}^{MR}$ satisfy:

$$mrpe_{i,t}^{MR} = P + \overline{\lambda}_{G,i,t}, \qquad Q'\left(x_{i,t}^{MR}\right) = \overline{\Delta}_{G,i,t}, \quad \forall j, \ell, \tag{25}$$

and are independent of the firm's carbon-intensity type j and age ℓ.

Figure 5: The Mirrleesian social planner allocation



(a) Energy

(b) Abatement

Note: The figure shows the determination of energy (Panel a) and abatement (Panel b) in the decentralized equilibrium (black), the full-information optimum (blue), and the Mirrleesian optimum (orange). The figure considers the case in which firms have either a high or low marginal social cost of capital, $\lambda_G^H > \lambda_G^L$, implying respectively a high or low marginal benefit of abatement. The average value is denoted by $\overline{\lambda}_G$.

ii. *For given factor prices (multipliers), the Mirrleesian planner depresses energy and increases abatement, while bunching firms across carbon intensities and age. In particular:*

$$e_{i,t}^{MR} < e_{ij\ell,t}^{DE} \quad \textit{and} \quad x_{i,t}^{MR} > x_{ij\ell,t}^{DE} \quad \textit{for any } i,j,\ell.$$

Figure 5 illustrates how the Mirrleesian planner allocations differ from those considered above. Like the full-information planner, the Mirrleesian planner depresses energy uses for all firms relative to the decentralized equilibrium and abates emissions. However, because the Mirrleesian planner does not know a firm's carbon intensity z_j, she cannot equate a firm's marginal revenue product of energy to its marginal social cost of energy. Instead, the Mirrleesian planner optimally averages and equates a firm of type (i,j)'s marginal revenue product of energy to group-i's *average* marginal social cost of energy. A similar averaging occurs for abatement. In this sense, the differences between the type-specific marginal costs and returns, $\lambda_{G,ijt}$ and $\Delta_{G,ijt}$, and their average values, $\overline{\lambda}_{G,it}$ and $\overline{\Delta}_{G,it}$, capture the bite that the information friction has on the economy.[29]

Finally, notice that because allocations are bunched within productivity types, the Mirrleesian planner's allocation features less factor misallocation than its full-information counterpart. This comes at a cost, however: by preventing the planner from tailoring energy and abatement to firms' carbon intensities, the associated information friction limits the planner's ability to reduce the economic damages from emissions.

[29]Notice that the average marginal social cost of energy $\overline{\lambda}_{G,i,t}$ is, in effect, the social planner's "best guess" (Bayesian optimal) for the individual marginal costs $\lambda_{G,ij,t}$.

4.4 Implementation and carbon taxation

Both the Mirrleesian allocation and its full-information counterpart can straightforwardly be implemented by a simple, linear carbon tax in the decentralized equilibrium. To see this, let $\tau_{ij\ell,t}(g)$ denote the carbon-tax payments made by firm (i, j, ℓ) at time t when its emissions equal g. Firm (i, j, ℓ)'s after-tax profits then become:

$$\pi_{ij\ell,t} = y_{ij\ell,t} - W_t \cdot n_{ij\ell,t} - R_t \cdot k_{ij\ell,t} - P \cdot e_{ij\ell,t} - Q(x_{ij\ell,t}) - \tau_{ij\ell,t}\left(g_{ij\ell,t}\right). \tag{26}$$

The maximization of firm (i, j, ℓ)'s value subject to the profit expression in (26) now results in the following *tax-adjusted conditions* for energy $e_{ij\ell,t}$ and abatement $x_{ij\ell,t}$:

$$mrpe_{ij\ell,t} = P + \tau'_{ij\ell,t}\left(g_{ij\ell,t}\right) \cdot \exp^{z_j - x_{ij\ell,t}}, \quad Q'\left(x_{ij\ell,t}\right) = \tau'_{ij\ell,t}\left(g_{ij\ell,t}\right) \cdot \exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t}; \tag{27}$$

while the conditions for labor $n_{ij\ell,t}$ and capital $k_{ij\ell,t}$ are identical to those in Section 4.1. Comparing Equation (27) with the corresponding equations in Lemma 2 and 3, combined with the similarity of the aggregate conditions, then immediately shows that:

Proposition 1. *Let the carbon tax schedule for firm (i, j, ℓ), $\tau_{ij\ell,t}(g)$, equal:*

$$\tau_{ij\ell,t}(g) = \Lambda_{G,t} \cdot g \equiv \tau_t^{FB}(g) \quad or \quad \tau_{ij\ell,t}(g) = \Lambda_{G,t} \cdot \frac{\mathbb{E}[\exp^{z_j} | a_i]}{\exp^{z_j}} \cdot g \equiv \tau_{ij,t}^{MR}(g), \tag{28}$$

and let tax proceeds be rebated lump-sum to the representative consumer. Then, the decentralized equilibrium allocation coincides with: (i) the full-information planner allocation when $\tau_{ij\ell,t}(g) = \tau_t^{FB}(g)$ and firms disclose truthfully without explicit verification; and (ii) the Mirrleesian planner allocation when $\tau_{ij\ell,t}(g) = \tau_{ij,t}^{MR}(g)$. The carbon tax bill as a share of firm revenue is:

$$\frac{\tau_t^{FB}\left(g_{ij,t}^{FB}\right)}{y_{ij,t}^{FB}} = \nu \cdot \frac{\lambda_{G,ij,t}}{P + \lambda_{G,ij,t}} \qquad or \qquad \frac{\tau_{ij,t}^{MR}\left(g_{i,t}^{MR}\right)}{y_{i,t}^{MR}} = \nu \cdot \frac{\overline{\lambda}_{G,i,t}}{P + \overline{\lambda}_{G,i,t}}. \tag{29}$$

Proposition 1 shows that a simple, linear carbon tax, equal to the social cost of carbon $\Lambda_{G,t}$ implements the full-information allocation when firms truthfully disclose. The proposition, as a result, recovers the main insight of Golosov et al. (2014) but extends it to the case with heterogeneity in productivity and carbon intensity. The result arises because the marginal damage to production from any additional unit of carbon is identical across firms: $A(G_t)$ is a function of aggregate emissions G_t alone; firm-level emissions are perfect substitutes. This, in turn, implies that the planner optimally sets a *homogeneous* tax across firms, equal to the social cost of carbon $\Lambda_{G,t}$, to adjust for the externality.

When heterogeneity is coupled with information frictions, however, the optimal carbon tax becomes *type-specific*.[30] Indeed, Proposition 1 shows that the Mirrleesian planner's allocation can still be implemented by a linear carbon tax, but crucially one that is *heterogeneous* across firms: the optimal marginal tax $\Lambda_{G,t} \cdot (\mathbb{E}[\exp^{z_j} | a_i] / \exp^{z_j})$ is still constant within a firm but depends on the carbon intensity z_j of the firm, and through the conditional expectation term may also vary with the productivity type.

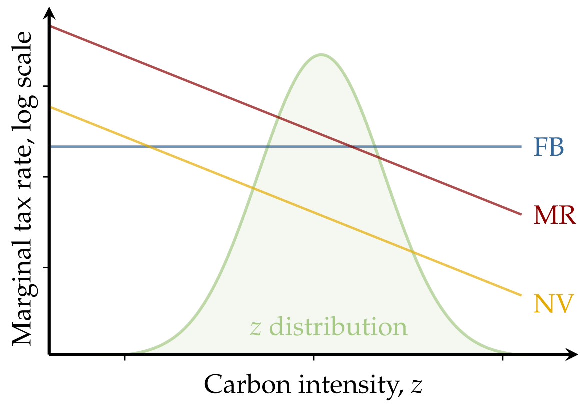
This dependence arises because of the incentive-compatibility constraint in (21). Under the full-information tax, the size of the overall tax bill rises with a firm's carbon intensity z_j through the social cost of energy $\lambda_{G,ij,t}$ (Equation 29). This generates an incentive for more carbon-intensive firms to report a lower z_j. To eliminate this incentive, the Mirrleesian planner equalizes tax payments across carbon-intensity types within productivity groups. However, because less carbon-intensive firms emit less, equalizing total payments requires a *higher marginal tax rate for less carbon-intensive firms*. The optimal marginal tax, as a result, declines with a firm's carbon intensity (Equation 28).

Figure 6 illustrates the two marginal tax rates in Proposition 1 as a function of carbon intensity. Proposition 1 further implies that the Mirrleesian rate *exceeds* the full-information rate on average, $\frac{\mathbb{E}[\exp^{z_j} | a_i]}{\exp^{z_j}}$ is on average above 1, as well as for firms whose carbon intensity is below their group average. By contrast, the Mirrleesian rate falls *below* the full-information rate for firms whose carbon intensity is sufficiently above the group average. Importantly, the two tax schedules are conceptually similar: the Mirrleesian schedule merely proportionally adjusts the full-information tax by $\frac{\mathbb{E}[\exp^{z_j} | a_i]}{\exp^{z_j}}$ to ensure incentive compatibility. This ratio-based (or "predictive-relative-to-actual") adjustment determines the distinctive shape in Figure 6. In this sense, the Mirrleesian tax is no harder to implement than its full-information counterpart. Given a firm's carbon-intensity type—which firms have no incentive to misreport under the tax scheme—the tax is still linear and a simple modification of the uniform Pigouvian tax. However, as we will see in the next section, this simple adjustment goes a long way in recovering the potential benefits of carbon taxation in environments in which firms can misreport emissions. In part, this will be because the size of the tax bill in (29) will remain small.

Finally, Figure 6 also shows the effective tax rate that would prevail with the naïve planner encountered earlier, who acts as the full-information planner but without having accurate information on carbon-intensity types. It follows from comparing (20) with (27) that the effective tax implied by the naïve planner allocation is also linear within firms

[30] Recall that we constrain the planner not to distort labor and capital markets through the additional constraint in Equation (23). Absent this restriction, the planner would generally choose a corrective policy that could not be implemented through a single carbon tax alone.

Figure 6: Optimal marginal tax rates



Note: The figure shows the marginal tax rates that implement the full-information social planner allocation (in blue), the Mirrleesian planner allocation (in red), and the naïve tax schedule (in orange).

and equal to:

$$\tau_{j,t}^{NV}(g) \equiv \Lambda_{G,t} \cdot \exp^{z-z_j} \cdot g. \tag{30}$$

Notice that the naïve tax rate *declines* in z_j, as the benefits of misreporting, which are reflected in the effective tax, are increasing in carbon intensity. It is also always lower than the marginal rates charged by the full-information and the Mirrleesian planner.

Before proceeding to quantify the benefits of a Mirrleesian approach to taxation, we should mention that other, more elaborate policies exist that also implement the planners' allocations. A tax on energy combined with a non-linear subsidy on abatement can, for example, also attain this aim in the full-information and Mirrleesian cases (Appendix B.2).[31] Changing the basis of taxation does, however, not overcome the information friction—that carbon intensities are, ultimately, firm-specific information.

4.5 A welfare decomposition

The above implementation is informative about the manner in which firms' reporting incentives change the optimal carbon tax on firms. In this subsection, building on our earlier analysis, we turn to studying the trade-offs inherent to the planner allocations. To do so, we will make use of the following decomposition of consumption, which helps showcase the bite firms' reporting incentives have on the economy:

[31] We focus specifically on carbon taxes for two reasons: (i) it is a single instrument, whereas the alternative requires coordinating two or more instruments; and (ii) a carbon tax is robust to the exact functional form of the emissions function, whereas the energy tax-abatement subsidy combination works only under the (common but restrictive) assumption that emissions depend solely on energy.

Proposition 2. *Let $F(n,k,e) = k^{\alpha} \cdot n^{\gamma} \cdot e^{v}$ with $\alpha, \gamma, v > 0$ and $\alpha + \gamma + v < 1$, and $Q(x) = (1+\theta)^{-1} \cdot x^{1+\theta}$ with $\theta > 0$. Then, in each equilibrium $\mathcal{P} \in \{DE, FB, MR, NV\}$:*

$$C_t^{\mathcal{P}} = \Omega_t^{\mathcal{P}} \cdot \Theta_t^{\mathcal{P}} \cdot A(G_t^{\mathcal{P}}) \cdot \exp^{a} \cdot (K_t^{\mathcal{P}})^{\alpha} \cdot (N^{\mathcal{P}})^{\gamma} \cdot (E_t^{\mathcal{P}})^{v}, \tag{31}$$

where $\Omega_t^{\mathcal{P}} \in (0,1]$ measures factor misallocation (Equation A46), $\Theta_t^{\mathcal{P}}$ is the intermediate input share (Equation A47), and $\exp^{a}$ denotes the relevant firm-level productivity average (Equation A46). All objects in Equation (31) are a function of the distribution of taxes $\{\tau_{ij\ell,t}^{\mathcal{P}}(g)\}$

Proposition 2 uses Proposition 1 to characterize consumption, the determinant of household welfare, as the outcome of the decentralized economy with carbon taxes. The decentralized equilibrium without carbon taxation corresponds to the case where $\tau_{ij\ell,t}(g) = 0$; the full-information planner, the Mirrleesian planner, and the naïve planner allocations correspond to $\tau_{ij\ell,t}(g) \in \{\tau_t^{FB}(g), \tau_{j,t}^{MR}(g), \tau_{ij,t}^{NV}(g)\}$, respectively.

Because carbon taxes are zero in the original decentralized equilibrium, $\Omega_t^{DE} = 1$: there is no factor misallocation in the original decentralized equilibrium. However, because emissions are universally higher in this case, aggregate productivity is also lower. The naïve planner delivers an outcome close to that of the decentralized equilibrium. By imposing only minimal distortions, it avoids the bulk of misallocation but at the cost of only modest emission reductions and thus limited productivity gains. In contrast, the full-information planner optimally balances the *efficiency-misallocation trade-off*. By setting $\tau_{ij\ell,t}(g) = \tau_t^{FB}(g)$, the planner optimally internalizes the emission externality, and thus attains the highest level of productivity and consumption possible in welfare terms; yet the planner does so at the cost of creating misallocation along more dimensions.

Finally, compared to the full-information planner, the Mirrleesian planner, in addition, faces the key incentive-compatibility constraint in (21). As shown above, this constraint leads allocations to be bunched across carbon-intensity types. Consequently, the Mirrleesian allocation features less misallocation, but at the cost of larger emissions and thus lower economy-wide productivity. In effect, the four allocations considered—the decentralized equilibrium, the full-information planner allocation, the naïve planner allocation, and the Mirrleesian planner allocation—can be seen as tracing out different points on an *efficiency-misallocation curve*, one that naturally adjusts for the intermediate and accumulated input use. We will later return to the decomposition in Proposition 2 when evaluating the benefits from Mirrleesian taxation.

5 Quantification and extensions

We have demonstrated that accounting for firms' reporting incentives changes the optimal carbon tax on firms. Our results show that internalizing firms' reporting incentives, all else equal, leads to higher marginal taxes for the median carbon-intensity firm and a decreasing carbon-tax schedule in carbon intensity. In this section, we use our analysis to shed light on the quantitative implications of our results. Does accounting for firms' reporting incentives *quantitatively* alter the environmental and welfare benefits of carbon taxation? And if so, to what extent and through which channels? To answer these questions, we proceed to parametrize and quantify our baseline economy using our data and empirical findings. As climate change generates damages over multiple decades, we throughout account for the full-transition dynamics of the model starting from $t = 0$ (year 2024) rather than compare steady states. Appendix C.2 describes the solution algorithm.

5.1 Functional forms and parametrization

The aim of our functional-form assumptions and parameterization is to ensure that our model quantitatively replicates key features of firm-level outcomes and the macroeconomy, as well as captures the rich heterogeneity in emissions documented earlier.

Functional forms. To this end, we retain the specification of the production technology and the abatement cost curve used in Section 4.5: we let $F(n,k,e) = k^{\alpha} \cdot n^{\gamma} \cdot e^{v}$, where $\alpha, \gamma, v > 0$ and $\alpha + \gamma + \nu < 1$, and set $Q(x) = (1+\theta)^{-1} \cdot x^{1+\theta}$ with $\theta > 0$. The per-period utility function is of the logarithmic form, $u(C_t) = \log(C_t)$. Similar to Golosov et al. (2014), we assume that the aggregate damage function for emissions is $A(G) = \exp^{-\phi \cdot G}$, where ϕ governs the semi-elasticity and thus the implied aggregate productivity loss from carbon-equivalent emissions. Finally, we assume that firm productivity and carbon intensity are normally distributed, $a_i \sim \mathcal{N}(\mu_a, \sigma_a^2)$ and $z_j \sim \mathcal{N}(\mu_z, \sigma_z^2)$, and allow for a non-zero correlation ρ_{az} between the two types. We later truncate both distributions to ensure that their supports are compact, consistent with our above assumptions.

Parametrization. We assume a model period corresponds to a decade. Following Golosov et al. (2014), we set the discount factor β to 0.985^{10}, a value that lies between standard macro calibrations and the lower discount rates advocated by Stern (2008).[32] We normalize the energy price P to 1 and set the energy share of gross output to 4%. Assuming

[32] We set the firm's lifespan L to 1, corresponding to one decade, in line with the median estimate reported in World Bank (2024). As shown in Section 3, this parameter has no consequences for our results.

decreasing returns to scale of $\alpha + \gamma + v = 0.90$—well within the range estimated in the literature (e.g., Basu and Fernald, 1997)—and a standard division of income between capital (40%) and labor (60%), this implies output elasticities of $\alpha = 0.346$ for capital, $\gamma = 0.518$ for labor, and $v = 0.036$ for energy.[33] The annual depreciation rate of capital is set to 7%, implying a model-period depreciation rate of $\delta_K \approx 0.52$. Finally, we normalize mean log-productivity to one, $e^{\mu_a} = 1$ (equivalently, $\mu_a = 0$), and set the standard deviation of firm-level productivity σ_a to 0.6, consistent with our Compustat sample. To ensure compactness, we truncate the productivity distribution at 3 standard deviations.

The remaining parameters govern the climate block of the economy. The semi-elasticity of the damage function ϕ and mean carbon intensity μ_z jointly determine the aggregate cost of climate change by governing how energy use translates into productivity losses. We therefore normalize mean carbon intensity to zero, $\mu_z = 0$, and calibrate $\phi = 0.0118$ to generate a 2% loss in productivity for each 1°C increase in temperature. Consistent with the median estimate in IPCC (2021), we use a conversion factor of $4.6 \cdot 10^{-4}$°C per additional $GtCO_2e$. This damage estimate lies within the range of historical estimates used in the literature (e.g., Nordhaus and Moffat, 2017), but is substantially more conservative than recent estimates (Bilal and Känzig, 2024; Caggese et al., 2025). Section 5.4 examines the sensitivity of our results to this choice. We set the carbon depreciation rate δ_G to 0.013, so as to match the impulse response function of atmospheric carbon from the three-reservoir model in Golosov et al. (2014) to an exogenous $GtCO_2e$ increase in emissions. Figure C.1 documents a close fit over time. The stock of initial carbon in the atmosphere G_0 is set assuming emissions have grown by 2% per year over the past 170 years, i.e., since the pre-industrial period.[34] Finally, we set the standard deviation of carbon intensity σ_z to 1.6, equal to the residualized higher-quality ICE sample for the U.S. (Lemma 1),[35] and truncate the distribution at 3 standard deviations.[36] Firm-level productivity and carbon intensity are assumed to be uncorrelated ($\rho_{az} = 0$), consistent with the evidence in Section 2. The inverse elasticity of abatement with respect to carbon prices θ is set to 2, in line with the estimates from Känzig (2023) on the effects of carbon tax shocks on green technologies. Table C.1 in Appendix C.1 summarizes the parameters.

[33] These values convert income shares into gross-output elasticities under decreasing returns to scale: $\alpha = 0.40 \cdot 0.90 \cdot (1 - 0.04)$, $\gamma = 0.60 \cdot 0.90 \cdot (1 - 0.04)$, and $v = 0.04 \cdot 0.90$.

[34] The initial capital stock K_0 is set to its steady-state value conditional on G_0.

[35] Notice that, for the U.S., the original decentralized equilibrium is "incentive compatible"—firms have no direct mis-reporting incentives—as the U.S. does not operate a carbon tax.

[36] This implies a minimum carbon intensity of -4.79, which lies well within the support of the demeaned distribution of energy intensities shown in Panel (b) of Figure 2.

Table 2: Decomposition of consumption-equivalent changes

	Ω	$\mathcal{O}$	$\mathcal{A}$	$\mathcal{K}^{\alpha}$	$\mathcal{E}^{\nu}$	$\mathcal{C}$
Naïve planner	-0.00	$+0.00$	$+0.45$	$+0.17$	$+0.02$	$+0.64$
Mirrleesian planner	-0.02	$+0.42$	$+2.17$	$+0.59$	-0.47	$+2.68$
Full-information planner	-0.05	$+0.26$	$+2.35$	$+0.73$	-0.25	$+3.03$

Note: Present-value consumption-equivalent welfare relative to the decentralized equilibrium in %.

5.2 Quantification and decomposition

Welfare decomposition. We quantify the welfare consequences of firms' reporting incentives. To do so, we compare results based on the Mirrleesian optimal tax schedule with those from the first-best case, where the planner has full-information about firms' carbon intensities. We also compare our results to those from the naïve-taxation case, which does not internalize firms' reporting incentives. To assess the determinants of differences in allocations, we decompose consumption—the driver of household welfare—relative to the decentralized equilibrium into 5 components, using the break-down in Proposition 2:

$$\underbrace{\frac{C_t^{\mathcal{P}}}{C_t^{DE}}}_{\equiv \mathcal{C}_t} \equiv \Omega_t^{\mathcal{P}} \cdot \underbrace{\left(\frac{\Theta_t^{\mathcal{P}}}{\Theta_t^{DE}}\right)}_{\equiv \mathcal{O}_t} \cdot \underbrace{\left(\frac{A_t^{\mathcal{P}}}{A_t^{DE}}\right)}_{\equiv \mathcal{A}_t} \cdot \underbrace{\left(\frac{K_t^{\mathcal{P}}}{K_t^{DE}}\right)^{\alpha}}_{\equiv \mathcal{K}_t^{\alpha}} \cdot \underbrace{\left(\frac{E_t^{\mathcal{P}}}{E_t^{DE}}\right)^{\nu}}_{\equiv \mathcal{E}_t^{\nu}}, \tag{32}$$

where $\mathcal{P} = \{FB, MR, NV\}$. Recall that the 5 components are, respectively: differences in (i) factor misallocation Ω_t (equal to 1 in the decentralized equilibrium), (ii) the intermediate input share $\mathcal{O}_t$, (iii) productivity $\mathcal{A}_t$, (iv) capital use $\mathcal{K}_t$, and lastly (v) energy use $\mathcal{E}_t$. Recall further that because of log utility, the log of the consumption ratio in (32) can directly be interpreted as the consumption-equivalent welfare difference between the two allocations for period t. Table 2 presents the results of the decomposition in which we account for the full transition dynamics of the model starting from 2024.

Three points are worth highlighting. First, consistent with our theoretical results, the full-information planner delivers the largest consumption-equivalent welfare increase and the largest increase in productivity (+3.03% and +2.35%, respectively). This is followed by the Mirrleesian planner (+2.68% and +2.17%) with the naïve planner substantially further behind (+0.64% and +0.45%). Overall, the potential benefits from Mirrleesian or full-information taxation are thus large—close to 3% and close to the overall productivity loss often assumed from another 1.5°C increase in temperature.[37]

[37] This corresponds to the baseline case in, for example, IPCC (2021) when combined with the assumption

Second, notice that the Mirrleesian planner captures the bulk of the potential, first-best welfare benefits (around 88% of them), while the naïve planner captures substantially less (only around 20%). Combined, our quantitative results thus suggest that Mirrleesian taxation can recover much of the theoretical benefits of first-best carbon taxation, even with the additional constraints in Equation (23). Accounting for firms' reporting incentives does not substantially constrain the potential welfare benefits of carbon taxation. However, ignoring firms' reporting incentives altogether does render the standard Pigouvian approach largely ineffective at addressing the welfare consequences of climate change.

Third, and crucially, Table 2 shows that the majority of the increase in welfare achieved by the Mirrleesian and full-information planner is caused by productivity increases (c. 80%), which in turn are driven by declines in emissions. The use of additional capital, as it becomes more productive, contributes some to the overall welfare increase, yet much less than the increase in productivity. Furthermore, despite the increase in productivity and the accompanying decline in emissions, overall energy use does not change much. The Mirrleesian and full-information allocations combine large declines in emissions with small declines in energy, which is why these allocations create large welfare benefits in the first place. Appendix C.3 studies the firm-level drivers behind these economy-wide shifts. Finally, recall that the Mirrleesian and full-information planners introduce misallocation into the economy and alter the economy's intermediate input share. Table 2, nevertheless, shows that these forces are quantitatively small—in each case less than 1% of consumption. This is mainly because energy accounts for a small share of output.

In sum, our results demonstrate that most of the potential benefits of carbon taxation can be recovered in the Mirrleesian case, and that most of the benefits from carbon taxation arise due to emission reductions increasing aggregate productivity. In the next subsection, we unpack these findings: we first analyze the sources behind the reduction in emissions, and then examine why the Mirrleesian planner is able to almost attain most of the first-best benefits. Our aim is to both discern what causes the above benefits of carbon taxation—abatement or the reallocation of energy?—and to analyze the circumstances that drive Mirrleesian taxation to substantially outperform the naïve case.

Emission decomposition. Conceptually, emissions can decline because of declines in overall energy use, because of increased abatement, or because of a better allocation of energy and abatement among firms. Let $G_{\text{flow},t} \equiv \sum_{ij\ell} w_{ij\ell} \cdot g_{ij\ell,t}$ denote aggregate emissions. The following equation then breaks down changes in aggregate emissions relative

of a 2% loss in productivity for each 1°C increase in temperature.

Table 3: Decomposition of changes in aggregate emissions

	$\mathcal{Z}_x$	$\mathcal{Z}_e$	$\mathcal{X}$	$\mathcal{E}$	$\mathcal{G}$
Naïve planner	−17.04	+ 0.00	−0.20	+0.42	−16.81
Mirrleesian planner	−71.99	+ 0.00	−1.48	−7.46	−80.93
Full-information planner	−69.56	−13.39	−0.96	−3.63	−87.54

Note: The table shows the %-difference in the present value of emission flows $\mathcal{G}_t$ between the different planner allocations and the decentralized equilibrium. The table uses a Shapley decomposition to additively break down emissions into the three factors. The discount factor β is applied to emission flows.

to those in the decentralized equilibrium into these 3 components (Appendix B.3):

$$\underbrace{\frac{G^{\mathcal{P}}_{\text{flow},t}}{G^{DE}_{\text{flow},t}}}_{\equiv \mathcal{G}_t} = \underbrace{\underbrace{\left(\frac{\sum_{ij\ell} \varpi^{\mathcal{P}}_{ij\ell,t} \cdot \exp^{z_j - x^{\mathcal{P}}_{ij\ell,t} + X^{\mathcal{P}}_t}}{\sum_{ij\ell} \varpi^{\mathcal{P}}_{ij\ell,t} \cdot \exp^{z_j}}\right)}_{\equiv \mathcal{Z}_{x,t}} \cdot \underbrace{\left(\frac{\sum_{ij\ell} \varpi^{\mathcal{P}}_{ij\ell,t} \cdot \exp^{z_j}}{\sum_{ij\ell} \varpi^{DE}_{ij\ell,t} \cdot \exp^{z_j}}\right)}_{\equiv \mathcal{Z}_{e,t}}}_{\equiv \mathcal{Z}_t} \cdot \underbrace{\left(\exp^{-X^{\mathcal{P}}_t}\right)}_{\equiv \mathcal{X}_t} \cdot \underbrace{\left(\frac{E^{\mathcal{P}}_t}{E^{DE}_t}\right)}_{\equiv \mathcal{E}_t} \quad (33)$$

where $\varpi^{\mathcal{P}}_{ij\ell,t} \equiv \omega_{ij\ell} \cdot \frac{e^{\mathcal{P}}_{ij\ell,t}}{E^{\mathcal{P}}_t}$ and $\mathcal{P} = \{FB, MR, NV\}$. Changes in aggregate abatement and energy use are here denoted by $\mathcal{X}_t$ and $\mathcal{E}_t$, respectively. We have already seen that the latter component is not the main reason for the decline in emissions. The remaining term $\mathcal{Z}_t = \mathcal{Z}_{x,t} \cdot \mathcal{Z}_{e,t}$ captures changes in the cross-firm allocation of these inputs. In particular, $\mathcal{Z}_{e,t}$ measures the emission reduction from reallocating energy toward less carbon-intensive firms, holding aggregate energy use fixed, while $\mathcal{Z}_{x,t}$ measures the emission reduction from directing abatement toward firms with higher emissions, holding aggregate abatement fixed. Table 3 presents the results of the decomposition, where we for ease of exposition focus on the present-value of emissions flows from $t = 0$.

The table shows that the *abatement component* $\mathcal{Z}_{x,t}$ is by far the dominant force behind the estimated reduction in emissions. In the full-information and Mirrleesian cases, it accounts for around 80-90% of the overall decline. The *energy reallocation component* $\mathcal{Z}_{e,t}$ is, by contrast, 0 in both the Mirrleesian and naïve cases, as these allocations are bunched across carbon-intensity types, as is the case in the decentralized equilibrium. As a result, there is no between-firm energy reallocation effect here. In the full-information case, however, the social planner does allocate less energy towards high-z firms. Table 3 shows that the size of this effect is quantitatively meaningful (−13%), although not nearly the size of the abatement component, and accounts for most of the difference between the Mirrleesian and the full-information cases. Finally, we note that there is only a small rise in aggregate abatement; the extent of its reallocation is the main driver.

We conclude that—in both the Mirrleesian and the full-information case—most of the welfare and environmental benefits from carbon taxation derive from $\mathcal{Z}_{x,t}$; that is, from the planner incentivizing abatement by high-energy and/or high-carbon-intensity firms. By contrast, most of the differences in outcomes between the Mirrleesian and the full-information cases stem from $\mathcal{Z}_{e,t}$; that is, from the additional ability of the full-information planner to tailor a firm's energy use to its carbon-intensity type.

Allocation of abatement. To understand what drives the overall size of the *abatement component* $\mathcal{Z}_{x,t}$—as well as why it is quantitatively similar in the Mirrleesian and full-information cases—we exploit the following approximation (Appendix B.3):

$$\log \mathcal{Z}_{x,t} \approx -\mathbb{C}ov\left(\frac{g_{ij\ell,t}}{G_{\text{flow},t}}, \frac{\exp^{x_{ij\ell,t}}}{\exp^{X_t}}\right) < 0. \tag{34}$$

This result shows that the abatement component is numerically large when relative emissions $g_{ij\ell,t}/G_{\text{flow},t}$ co-move strongly with relative abatement efforts $\exp^{x_{ij\ell,t}} / \exp^{X_t}$, which is indeed the case in both the full-information and the Mirrleesian allocation. Consider the first-order condition for abatement (Equations 18 and 25) in the two cases:

$$\left(x^{FB}_{ij\ell,t}\right)^{\theta} = \Lambda_{G,t} \cdot g^{FB}_{ij\ell,t} \quad \text{and} \quad \left(x^{MR}_{ij\ell,t}\right)^{\theta} = \Lambda_{G,t} \cdot g^{MR}_{ij\ell,t} \cdot \frac{\mathbb{E}[\exp^{z_j} \mid a_i]}{\exp^{z_j}}. \tag{35}$$

These equations show that abatement is a *direct* function of emissions in both cases. Combined with the fact that energy allocations at the firm level are primarily driven by firm-level productivity differences rather than carbon intensity (see further below and Appendix C.3), and thus that emissions at the firm level are similar in the two cases, this explains the high (and similar) correlation between relative emissions and relative abatement. Both the full-information planner and its Mirrleesian counterpart create a strong correlation at the firm level between emissions and abatement, which in turn drives the large contribution of the abatement component to the overall decline in emissions.

Allocation of energy. Although the above shows the reason behind the large reduction in emissions, it does not fully explain why the Mirrleesian planner performs nearly as well as the full-information planner. As documented earlier, the difference between them is primarily driven by the *energy reallocation component* $\mathcal{Z}_{e,t}$. To address why this component is comparatively small, we next examine what determines its strength.

A few simple derivations show that (Appendix B.3):

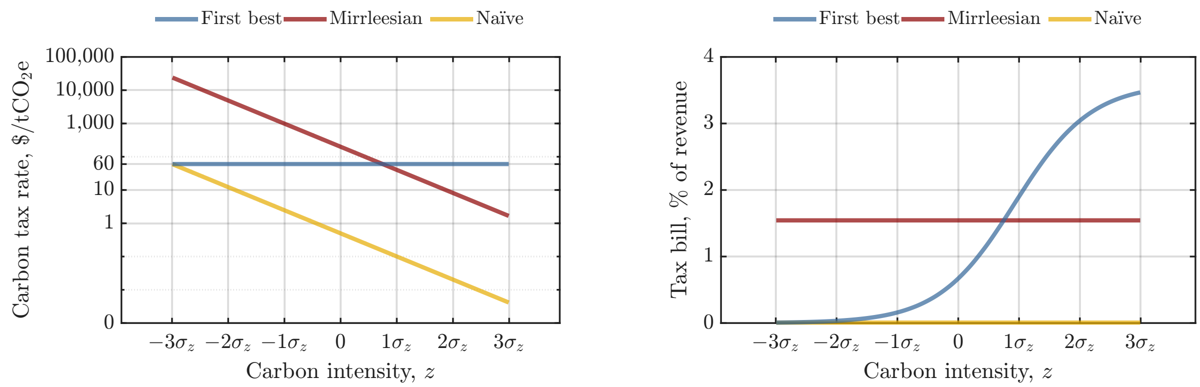
$$\mathcal{Z}_{e,t}^{FB} - \mathcal{Z}_{e,t}^{MR} = \mathbb{C}ov\left(\frac{e_{ij\ell,t}^{FB}}{E_t^{FB}},\ \frac{\exp^{z_j}}{\mathbb{E}\left[\exp^{z_j}\right]}\right) < 0. \tag{36}$$

This equation decomposes differences in the energy reallocation component into the covariance between firm-level energy shares in the full-information allocation and the relative carbon-efficiency type of the firm. Notice that this covariance is always negative, as firm-level energy shares co-move negatively with relative carbon intensities (Lemma 2). The gap between $\mathcal{Z}_{e,t}^{FB}$ and $\mathcal{Z}_{e,t}^{MR}$ is, as a consequence, larger in absolute value when firms' energy use in the full-information allocation is mainly driven by carbon-intensity differences rather than productivity differences. By contrast, when the variation in energy shares is mainly due to productivity differences, the covariance is small, and the Mirrleesian planner performs nearly as well as its full-information counterpart.

Thus, the reason the Mirrleesian planner can reap most of the first-best benefits from carbon taxation is that—in the first-best allocation—firm productivity is *quantitatively* the dominant driver of firms' optimal energy use. Although carbon intensities clearly matter under our baseline calibration, firm-level productivity is *the* quantitative driver of firms' optimal energy use, highlighting the importance of considering both sources of heterogeneity in our analysis. Appendix C.3 shows graphically the similarities in energy shares in the Mirrleesian and full-information cases.

In summary, in this section, we have shown that abstracting from firms' reporting incentives renders the standard (naïve) approach to carbon taxation largely ineffective at reducing emissions—with substantial consumption losses as a result. Clearly, the extent of firms' misreporting is, ultimately, close to unknowable; yet one advantage of our quantitative-theoretical framework is that we can estimate the potential losses from misreporting. Our analysis, all else equal, shows that the extent and the cost of misreporting can be large. Conversely, accounting for firms' reporting incentives does not appear to substantially constrain the potential welfare and environmental benefits of carbon taxation. The Mirrleesian optimal tax attains more than 80% of the potential, first-best benefits in our baseline analysis. The root cause of this comparative performance is that the first-best carbon tax attains its welfare improvement largely by re-allocating abatement efforts across firms—a task the Mirrleesian approach does almost equally well when firm productivity optimally determines a large share of the variation in firms' energy shares.

Figure 7: Calibrated carbon tax rates



(a) Marginal carbon tax rate

(b) Overall carbon tax bill

Note: Panel (a) shows the carbon tax that implements the different planner allocations. Panel (b) depicts the associated carbon tax bill as a share of firm revenue. We plot both of these as a function of a firm's carbon intensity z_j. We use 2024 world GDP and emission numbers to convert estimates into 2024 \$ per tCO_2e.

5.3 Calibrated carbon taxes

Before closing with several extensions of our baseline framework, we briefly turn to the implied carbon taxes that implement the different planner allocations. Common estimates of the social cost of carbon—equivalent to the $t = 0$-tax that implements the first-best full-information allocation—is between \$40-200 per tCO_2e.[38] Panel (a) in Figure 7 shows that our baseline calibration produces estimates within this range. The marginal tax that implements the full-information allocation is around \$60 per tCO_2e and constant across the carbon-intensity distribution, corroborating the soundness of our calibration strategy.

Turning to the optimal Mirrleesian tax, notice that—in line with Proposition 1—the optimal tax is log-linear in z and monotonically declining. Indeed, Panel (a) shows that for a firm that is 2 standard deviations *below* the mean carbon-intensity type, its value is around \$4,500 per tCO_2e; for a firm that is 2 standard deviations *above*, it is around \$10 per tCO_2e. At the ± 1 standard deviation level, the range is between \$1,000-40 per tCO_2e, with a median value of around \$200 per tCO_2e. Unlike the first-best full-information tax, there is substantial heterogeneity in Mirrleesian optimal carbon rates, and their median value substantially exceeds the social cost of carbon (Proposition 1).

A fear one might have is that by enacting such high rates on select firms, one may heavily penalize their profits, potentially to the detriment of future innovation. Panel (b) in Figure 7 reports the overall tax bill as a share of firm revenue. Although the marginal

[38]See, for example, IWG (2021), Ricke et al. (2018), Pindyck (2019), and Tol (2023), among others.

carbon tax can be high for an individual firm, the overall tax bill in the Mirrleesian case is always around 1.5% of revenue. Clearly, changes to the incentive structure are what drives firms' behavior rather than the size of the climate taxes themselves.

Panel (b) in Figure 7, furthermore, highlights a fundamental difference between the full-information and the Mirrleesian cases. The full-information planner, unconstrained by incentive-compatibility considerations, sets carbon taxes so that the tax bill as a share of revenue rises steeply with a firm's carbon intensity. The planner, as such, penalizes heavily more carbon-intensive firms, as this is the most cost-effective way to reduce emissions. Yet this, in turn, creates a strong incentive for carbon-intensive firms to misreport their emissions. By contrast, the Mirrleesian planner internalizes this incentive and, to eliminate it, charges all firms the same tax-bill share of revenue (conditional on their productivity type; see Proposition 1).

Clearly, there are other implementation mechanisms that could support the planner allocations than a carbon tax. In Section 4.4, we discussed the possibility of a linear tax on energy combined with a non-linear subsidy on abatement. Appendix C.5 shows that this alternative implementation yet inherits several of the properties of the carbon tax—chiefly, the equalization of profits across the carbon-intensity distribution.

Finally, Appendix C.4 shows the differences between the first-best full-information tax bill and the Mirrleesian tax bill along the productivity dimension. Although the marginal carbon tax is independent of productivity in both cases, the tax bill's share of revenue declines with it. In the presence of carbon taxes, more productive firms emit less per unit of revenue, as they have the strongest incentives to abate. As noted earlier, the marginal benefit from abatement scales with firm size (i.e., productivity). Consequently, both the first-best and the Mirrleesian planner find it most cost-effective to shift the burden from high- to low-productivity firms. This is, in part, also related to our earlier discussion on the reasons behind the relative performance of the Mirrleesian planner.

We conclude that the Mirrleesian tax schedule amends the full-information tax schedule by making the marginal carbon tax decline in a firm's carbon-intensity type. The marginal tax for most firms declines from around \$4,500 to \$10 per tCO_2e, and is on average equal to around 3.5 times the social cost of carbon. The average tax paid by firms is, however, small and estimated to be around 1.5% of revenue. The Mirrleesian carbon tax, as such, combines strong incentives with low average costs to firms.

Table 4: Overview of quantitative refinements

	Consumption/Welfare				*Emissions*			
	FB	MR	NV	MR/FB	FB	MR	NV	MR/FB
Baseline	+3.03	+2.68	+0.64	88%	−87.54	−80.93	−16.81	92%
Extensions/robustness								
Temperature and emissions model	+3.83	+3.34	+0.74	87%	−87.24	−80.15	−15.15	92%
Renewable and fossil energy	+3.22	+2.86	+0.65	89%	−92.63	−87.37	−17.03	94%
Productive abatement	+2.63	+1.94	+0.03	74%	−80.40	−68.48	− 0.80	85%
Higher (50%) planner information, η	+3.03	+2.94	+2.02	97%	−87.54	−82.97	−53.33	95%
Higher (50%) damage semi-elasticity, ϕ	+4.59	+4.13	+1.07	90%	−89.68	−84.13	−19.18	94%
Lower (50%) inverse abatement elasticity, θ	+3.07	+2.71	+0.20	88%	−89.35	−83.82	− 5.22	94%
Lower (50%) carbon intensity dispersion, σ_z	+1.04	+0.99	+0.64	95%	−75.37	−72.58	−40.32	96%

Notes: This table reports present-value %-deviations in consumption (i.e., consumption-equivalent welfare) and emissions relative to the decentralized equilibrium for various model extensions. The columns FB, MR, and NV denote the full-information, Mirrleesian, and naïve planner allocations, respectively. MR/FB reports the share of the first-best gains achieved under the Mirrleesian allocation.

5.4 Quantitative refinements and discussion

We briefly turn to several alternative versions of our benchmark framework. The overall aim of these is to demonstrate that our main result—the relative performance of Mirrleesian carbon taxation—is robust to alternative assumptions about the climate block of the economy, the productive role of abatement, the information structure, and the particular calibration choices of the climate parameters. We also here discuss "ballpark estimates" of the implied monitoring costs of verification that our results entail. We recalibrate (where appropriate) all alternative models to target the same moments as our benchmark model. Table 4 and Appendix C summarize the results.

Alternative climate specifications. A potential concern with the above findings is that they rely on a reductionist characterization of climate dynamics that abstract from the dynamics of temperature—a characterization that, nevertheless, matches well the dynamic evolution of carbon in the atmosphere (Appendix C.1). To address this concern, we extend our baseline model with a low-dimensional, temperature-augmented climate block similar to Miftakhova et al. (2020), disciplined by the climate dynamics in Folini et al. (2024). This approach has become increasingly prominent in recent macro-climate applications (e.g., Dietz et al., 2021; Fernández-Villaverde et al., 2024) because it combines DICE-style tractability with grounding in state-of-the-art climate models (e.g., Joos et al., 2013; Geoffroy et al., 2013; Taylor et al., 2012). Specifically, we assume aggregate productivity depends on temperature, $A = \exp^{-\phi \cdot T}$, rather than directly on emissions, and that temperature T evolves dynamically with carbon in the atmosphere according to:

$$T_{t+1} = (1 - \xi_T) \cdot T_t + \xi_T \cdot (\mathcal{T} + \Xi_G \cdot G_t), \tag{37}$$

where $1 - \xi_T$ is the persistence in global temperature, $\mathcal{T}$ is its historical mean, Ξ_G is the conversion factor mapping emissions into temperature, and G_t is the carbon stock that follows (4). The calibration of the alternative climate model is described in Appendix D.1.

Table 4 and Appendix D.1 summarize our findings. As in our baseline specification, the Mirrleesian planner allocations follow closely the full-information outcomes. Relative to the decentralized equilibrium, the full-information (Mirrleesian) planner, for example, achieves a reduction in the present value of emissions of -87.24% (-80.15%) and an increase in consumption-equivalent welfare of +3.83% (+3.34%) . The Mirrleesian planner once more captures the bulk of the full-information welfare and environmental benefits (around 90%). In contrast, the naïve planner yields only minor improvements.

On balance, our results from the extended climate model thus reinforce the insights from our baseline framework: the Mirrleesian planner consistently exhausts most of the potential gains from climate taxation. The reason is that the climate block affects a planner's problems primarily through the social cost of carbon, and therefore merely proportionally scales both the first-best and the Mirrleesian optimal tax schedules (Proposition 1). Our main conclusions are, as a result, robust to alternative climate dynamics.

Renewable and fossil energy. One reason why firms, even within narrowly-defined sectors, exhibit differences in emission intensities is the different availability of renewable versus fossil energy in their area of operation. Other technological differences—such as differences in filters used and burner designs, mentioned in Section 2—lead to differences in emission intensities, in addition to those which arise from differences in the use of different fuels. To explore the implications of the introduction of a renewable energy sector into our model, we enrich our baseline framework by assuming that energy $e_{ij\ell,t}$ is comprised of *renewable* $e_{\text{green},ij\ell,t}$ and *fossil energy*, $e_{\text{brown},ij\ell,t}$, in accordance with:

$$e_{ij\ell,t} = \left(e_{\text{green},ij\ell,t}^{\frac{\varepsilon-1}{\varepsilon}} + e_{\text{brown},ij\ell,t}^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}}, \tag{38}$$

where both fossil and renewable energy are produced as in the baseline model. We assume only the use of fossil energy creates emissions. Although conceptual issues with the CES structure adopted in (38) are known, it has proven to be a both popular and tractable means to model firms' choices of renewable versus fossil energy (e.g, Hassler et al., 2016).

Appendix D.2 provides details of the calibration.

Tables 4 and Appendix D.1 show that the main insights from the baseline model extend to the economy with a renewable energy sector. The Mirrleesian tax once again recovers around 90% of the potential environmental and welfare gains, substantially outperforming the naïve benchmark. Relative to the full-information allocation, however, the Mirrleesian planner induces a stronger shift toward renewable energy. The reason is that reallocating energy toward the renewable sector lowers emissions without requiring knowledge of firms' carbon intensities. Unable to tailor energy and abatement efforts, the planner substitutes towards a margin of adjustment that does not depend upon firm type.

Productive role of abatement. The baseline model treats abatement as expenditure aimed solely at reducing emissions—for example, through improved filtration or carbon-capture technologies.[39] However, some abatement investments may also enhance production by reallocating resources from old (unproductive), carbon-intensive activities toward newer (more productive) and cleaner ones (e.g., Capelle et al., 2023; Lanteri and Rampini, 2025; Bertolotti et al., 2026). In this subsection, we consider this possibility by augmenting the firm-level production function from Section 3 as follows:

$$y_{ij\ell,t} = A(G_t) \cdot \exp^{a_i} \cdot \Psi(x_{ij\ell,t}; \kappa) \cdot F(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}), \tag{39}$$

where $\Psi(x; \kappa) = \exp[\kappa \cdot (1 - \exp(-x))]$ with $\kappa \geq 0$ captures the productive role of abatement. The key difference relative to the baseline model is that the optimal level of abatement for each firm in the decentralized equilibrium is now strictly positive rather than zero. We discuss the calibration of the extended model in Appendix D.3.

Table 4 and Figure D.2 present the results. Figure D.2 shows that, across different values of κ ($\kappa = 0$ corresponds to the baseline model), the Mirrleesian planner consistently attains most of the first-best benefits, whereas the naïve planner captures only a small share. Indeed, focusing on $\kappa = 0.25$, which corresponds to a semi-elasticity of 0.25% per %-points of emissions abated (cf. Adhvaryu et al., 2020), the Mirrleesian planner captures once more close to 3/4 of the present-value welfare and emissions-reduction gains attainable, while the naïve planner captures almost none of them (Table 4).

Partial information. In our baseline analysis, the information friction between the planner and firms takes a particularly simple form: each firm knows its carbon intensity z_j,

[39] Notice that, unlike the earlier work by Kwerel (1977) and Roberts and Spence (1976), we treat abatement costs as known by the policymaker. The problem with unknown emissions and abatement costs is interesting but outside the scope of this paper.

while the planner knows only the distribution from which z_j is drawn. To account for the possibility of additional planner information, we enrich the baseline model with an additional planner signal. In particular, the signal is such that the social planner observes the share $\eta \in [0, 1]$ of a firm's carbon intensity z_j without error; the planner is only uncertain about the remaining $1 - \eta$ share. Varying η from 0 to 1 thus traces out the entire range of possible information frictions. Appendix D.4 shows the modified equilibrium conditions.

Table 4 and Figure D.3 report the corresponding quantitative results. Notice that, for all values of η, the Mirrleesian planner attains most of the potential welfare and environmental gains. Indeed, decreasing the information friction causes the Mirrleesian planner to attain a *larger* share of the possible welfare and environmental benefits. Clearly, increasing η also improves the share of the potential benefits that the naïve planner attains. However, even assuming a 50% reduction in the planner's uncertainty about firms' carbon intensity, the naïve planner still attains 30% fewer welfare and environmental benefits than the Mirrleesian planner (Table 4). The Mirrleesian planner, by contrast, nearly exhausts all of the potential benefits of carbon taxation. Accounting for firms' disclosure incentives remains important, even when policymakers possess substantial information.

Ballpark estimates of monitoring costs. The previous exercise increased the planner's information about firms' carbon intensities but treated that information as costless. However, in practice, reducing such uncertainty requires verification of firms' carbon disclosures, at a cost. To size this cost against the potential benefits of an explicitly Mirrleesian approach, we below consider the concrete verification technology often debated in the literature (European Commission, 2023b): mandating that firms install a *continuous emissions monitoring system* (CEMS) on every emitting stack. A CEMS measures pollutant concentrations and exhaust flow directly at the source; it thus replaces the calculation-based approach, under which emissions are inferred from reported fuel quantities, emission factors, and technological adjustments, with direct physical measurement.

Direct measurement, however, does not come cheap. Installation costs are estimated at \$120,000–350,000 per stack, with a useful life of around 15 years; annual operating costs are estimated between \$12,000–30,000 (Optical Scientific, 2026). The lower bound of both of these ranges appears conservative (U.S. EPA, 2024). Assuming one stack per firm and using our calibrated discount factor, the present-value cost of a CEMS system is thus estimated to be between \$1.38–3.70M. Furthermore, such technologies cannot measure all forms of carbon-equivalent emissions, such as *fugitive emissions*, which must instead be monitored through alternative methods, presumably at additional cost. This consideration, together with the high costs discussed above, may help explain why only 1.7% of installations in the EU ETS report using a CEMS (European Commission, 2023b).

We can compare this cost range to the *discounted break-even monitoring cost* implied by our model: i.e., the discounted consumption gain from moving from the Mirrleesian to the full-information allocation, divided by discounted first-best emission flows. Converting this per-ton threshold into a per-firm value using the average firm's discounted emissions under the first-best allocation yields an estimated lifetime break-even value of $0.87M per firm, below the comparable cost. This comparison is conservative, moreover, as our cost calculation assumes only one emitting stack per firm and no fugitive emissions. Obviously, there is sizable uncertainty surrounding both estimates; but—conditional on the above-estimated benefits from Mirrleesian carbon taxation—emission verification does not easily become cost-effective. This is, in a sense, not surprising: verification buys little when the modified tax schedule already does most of the work.

Climate parameter uncertainty. Finally, our baseline model relies on estimates of the environmental damage parameter ϕ, the inverse elasticity of abatement θ, and the standard deviation of the carbon-intensity distribution σ_z. As with the processes governing emissions, these parameters are subject to considerable uncertainty. Table 4 and Figure D.4–D.6 examine the robustness of our results to alternative parameter values.

Our baseline calibration conservatively sets ϕ to deliver a 2% productivity loss per additional 1°C increase in temperature. Although this value is consistent with traditional estimates of the cost of climate change (e.g., Nordhaus, 2007), some modern estimates, however, project substantially larger losses (e.g., Bilal and Känzig, 2024, Caggese et al., 2025). Table 4 and Table D.4 show that—although the benefits of both full-information and Mirrleesian taxation increase substantially as we push ϕ closer to the higher losses estimated in recent approaches—the overall take-away remains the same: Mirrleesian taxation recovers a large share of the potential first-best benefits.

Our conclusions likewise do not hinge on the precise value of θ used. Our baseline calibration of the inverse tax elasticity of abatement is informed by Känzig (2023) estimate of the response of green technologies to carbon tax shocks. While not an ideal measure of abatement behavior, it provides a useful benchmark given the scarcity of available estimates. Table 4 and Figures D.5 show that our results are relatively insensitive to the choice of θ. Two offsetting forces drive this finding. On one hand, a larger θ reduces the abatement response to carbon taxes; but, on the other hand, it also lowers the overall cost of abatement, $Q(x) = \frac{1}{1+\theta} \cdot x^{1+\theta}$. These effects largely cancel each other in equilibrium, so that the overall benefits of climate taxation do not depend much on the value of θ. We view this fact as a strength of our framework, given the limited evidence on θ.

Lastly, in our baseline calibration, we discipline the standard deviation of the carbon-intensity distribution σ_z using data from an economy without a carbon tax—the U.S.—where firms have little incentive to misreport emissions. Table 4 and Figure D.6 examine the robustness of our results to alternative values of σ_z. On balance, the results show that our conclusions remain robust to plausible departures from the baseline calibration. Although a lower dispersion in carbon intensities reduces firms' potential misreporting, even when σ_z is reduced by 50%, the Mirrleesian planner continues to recover a large share of the first-best gains, whereas the naïve planner captures only a small fraction.

In summary, calibrating our model framework to microdata, we estimate that the Mirrleesian approach attains around 75-90% of the potential welfare and environmental benefits from climate taxation. This stands in stark contrast to the naïve approach, which does not internalize firms' reporting incentives. We estimate that the latter often attains only around 20% of the potential benefits. Crucially, we have documented that this relative performance holds across a range of model extensions that vary the climate block of the economy and considers alternative parameter values, among others. We view this robustness as crucial given the uncertainty that exists about the best climate specification and information structure to use in macro-climate contexts.

6 Final remarks

Addressing the economic damages from climate change is to first-order a problem of curtailing firms' direct emissions. In response to this challenge, considerable efforts have recently gone into the design and implementation of (Pigouvian) carbon prices via taxes or markets that correct firms' climate externality. However, as acknowledged by practitioners and policymakers alike, a central challenge to their implementation is the largely self-reported nature of emission disclosures. Carbon disclosures are in most cases based on firm-level information that is either difficult to ascertain or costly to verify for policymakers (International Sustainability Standards Board, 2023; SEC Carbon Disclosures, 2024). Combined with a sizable carbon price, this creates a strong incentive for firms to misreport their carbon disclosures—consistent with recent legal evidence and court cases.

In this paper, we have formulated a neoclassical general-equilibrium model in which firms' fossil-energy use generates climate damages to production. The model crucially incorporates substantial cross-firm heterogeneity in carbon efficiency that we documented using a novel firm-level dataset spanning 151 countries. Firms choose energy and other inputs and self-report emissions that are otherwise privately observed. Our main result has been a simple formula for the optimal carbon tax that accounts for firms' incentives to distort their carbon disclosures. The optimal (Mirrleesian) tax varies markedly across firms as a function of their carbon intensity and the social cost of carbon. The information friction makes firm-level taxes *heterogeneous across firms* but *constant within a firm*.

We evaluated this formula quantitatively, using the U.S. distribution of carbon intensities, and found that the optimal carbon tax varies substantially. Indeed, we found that the optimal carbon tax for a firm that is 1 standard deviation below the mean carbon-intensity level is around \$1,000 per tCO_2e; a firm that is 1 standard deviation above the mean carbon-intensity level, by contrast, faces an optimal tax slightly above \$40 per tCO_2e. The median carbon tax paid is around \$200 per tCO_2e. Despite this heterogeneity, the overall tax bill faced by firms in equilibrium is the same conditional on productivity and never exceeds 2% of revenue. Instead, we have shown that it is changes in firm-level reporting incentives that drive the bulk of the welfare benefits of carbon taxation.

Our tax contrasts with other carbon-tax estimates that do not internalize firms' reporting incentives. Our model—presuming full-information about firms' carbon emissions—provides an optimal carbon tax, equal to the social cost of carbon, that is homogenous across firms and equal to around \$60 per tCO_2e. Although there are substantial uncertainties about the true social cost of carbon, this estimate is close to common estimates used in the literature. Our results, thus, do not rest on the presumption of a large social cost of carbon: accounting for firms' reporting incentives increases most carbon taxes paid by firms and by itself leads to substantial heterogeneity in optimal tax rates.[40]

Crucially, we have documented that the optimal Mirrleesian tax recovers most of the potential benefits from carbon taxation. In our baseline calibration, accounting for firms' reporting incentives allows us to recover more than 80% of the potential welfare and environmental benefits from carbon taxation. A naïve approach—which implements the full-information tax without accounting for firms' reporting incentives—in contrast recovers only around 20%. We have showed that this comparative performance rests on the ability of a well-constructed tax to incentivize abatement efforts by carbon-intensive firms. As a result, we have shown that the comparative performance of the optimal Mirrleesian tax extends to several variations of our model framework. In all cases, the optimal Mirrleesian tax recovers more than 3/4 of the potential welfare and environmental benefits; the naïve approach by contrast recovers substantially less—and sometimes close to zero.

An important feature of our proposed tax is its simplicity. The optimal carbon tax corresponds to a simple, proportional adjustment of the standard social cost of carbon

[40]We may also relate our estimates to the price of emission rights in the European Union Emission Trading System, covering large CO_2 emitters in the EU. The price is currently around \$90 per $tC0_2$ and equal across all firms, and thus well-below our estimate for the optimal carbon tax rates paid by most firms.

estimate, based on a firm's location in the carbon-intensity distribution. Given our formula, a firm obtains no benefit by misreporting its location in the distribution. As such, our proposed carbon tax retains many of the advantages in terms of simplicity and implementation of standard carbon tax proposals. Although it should be clear from our discussions throughout that several extensions of the present setting are desirable, one advantage of our framework is that such extensions should come at a low cost.

Finally, beyond the analysis in this paper, our approach suggests that firms' reporting incentives can meaningfully alter the optimal Pigouvian tax on firms. We see important scope for extending these ideas to other circumstances in which firms' actions have externalities on the broader economy and their efforts are hard to monitor—such as is the case with industrial policy. Overall, we view the research in this paper as a useful step towards a unified framework for the Mirrleesian taxation of firms, and the current model framework as a useful basis for extensions that incorporate dynamic investments in abatement, jurisdictional leakages, and other reporting and verification environments.

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Online Appendix to
"Mirrleesian Carbon Taxation"

Andrea Chiavari   Alexandre Kohlhas

August 2026

Contents

Table A.1: Descriptive statistics (ICE-Trucost)

	Obs.	Mean	Std.	Median
ICE				
Scope 1 emission (All)	402,523	409,770	5,801,209	2,307
Scope 1 emission (High-quality)	32,599	1,461,794	5,139,210	29,200
Scope 1 emission intensity (High-quality)	32,599	181	568	8
Trucost				
Scope 1 emission	49,011	541,617	2,042,981	26,061
Scope 1 emission intensity	49,012	131	357	22

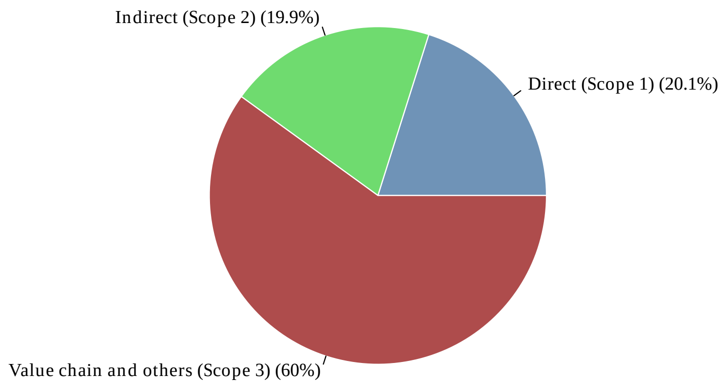
Notes: The table reports descriptive statistics for the sample of firms in the ICE and Trucost databases. The ICE *high-quality* sample is restricted to observations with third-party-validated disclosures, identified by ICE's reporting flag, that are self-reported. The Trucost sample does not contain an analogous observation-level validation indicator. The units of the first two rows and fourth one are tons of CO_2e emissions (tCO2e). The other rows are tCO2e per million $ in revenue. For both the ICE and Truecost samples, observations in the top and bottom 1% of the sample distributions have been removed.

A Motivating evidence

A.1 Data construction

ICE-ESG database. Our main analysis uses data from the ICE's ESG company database. The total data set covers the period 2009-2024 for 402,956 firm-years spanning 46,928 firms across 151 countries. The sample includes model-estimated emissions, reported emissions, and, among the reported emissions, those that have been validated by an independent third party. To construct the sample that we use in Figures 2, 3, A.3, A.4, and A.5, we discard ICE-model-generated emission estimates (derived from company data) and focus only on firms' self-reported emissions. We also discard observations that are not marked by ICE as having "high-quality reporting data" behind them. In our ICE data set, this restriction identifies firm-year observations with third-party-validated emissions disclosures. Table 1 uses all reported emissions, but not model-estimated ones, as its purpose is to compare reporting differences between firms whose emissions receive third-party validation and those whose emissions do not. Figure 1 uses the full sample of observations, as they aim to capture representative aggregates. Direct emission intensities are measured by reported scope 1 emissions (tCO_2e) per million $ of revenue (`code: Reported Emissions Intensity Scope 1 (tCO2e/$m Revenue)`). Firms' direct emissions are measured by the numerator in this expression (`code: Reported Emissions Scope 1 (tCO2e)`). We remove observations in the top and bottom 1% of the emission intensity distribution.

Figure A.1: Breakdown of ICE emissions



Note: The figure shows the breakdown of the average firm's emissions into scope 1, 2 and 3 emissions. We consider only ICE observations with third-party-validated non-model-generated disclosures. We limit the sample to firms from Europe or North America, where scope 2 and 3 emissions are standardized.

S&P Trucost database. We supplement the data from ICE with S&P's Trucost Database on publicly-listed firms self-reported emissions. The overall sample covers the period 2009-2019 for 50,013 firm-years spanning 9,227 firms across 39 different countries. Emission intensities are again measured by self-reported scope 1 emissions (tCO_2e) per million $ of revenue (`code: di319407`). Overall, emissions are measured by the numerator of this expression (`code: di319413`). Unlike ICE, Trucost does not contain an observation-level indicator for third-party validation. We once more remove observations in the top and bottom 1% of the emission intensity distribution.

Compustat database. Finally, we use the following variables from Compustat Fundamentals Annual: revenue (`code: sale`), capital (`code: ppent, ppegt`), investment (`code: capx`), employment (`code: emp`), and industry classification (`code: naics and sic`). We deflate nominal variables where appropriate with US CPI (`code: CPIAUCLS` from FRED) and compute (revenue-based) TFP as in Ottonello and Winberry (2020).

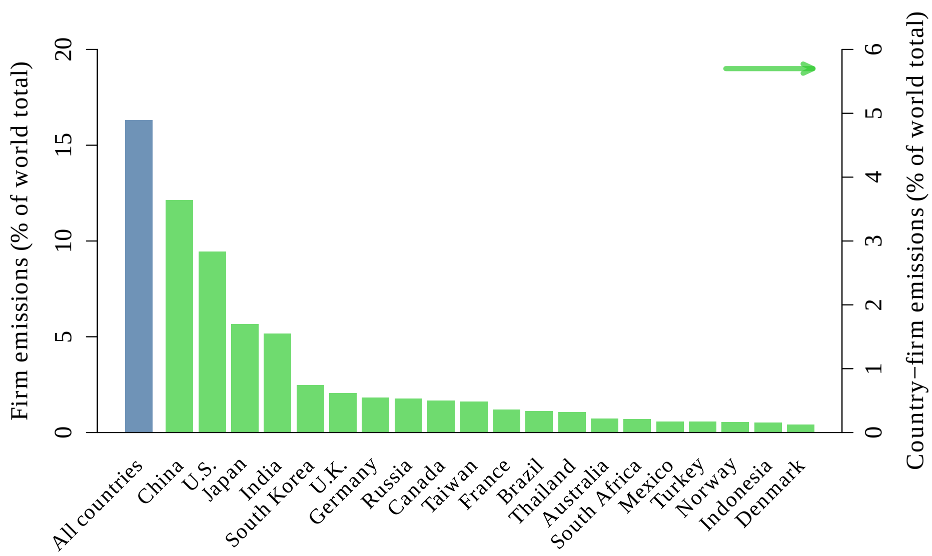
Descriptive statistics. Table A.1 reports descriptive statistics for the emissions data. The table highlights two features that are important for our empirical section. First, the ICE-ESG database has substantially broader firm coverage than Trucost, which is why the aggregate emission shares in Figure 1 are larger than their Trucost counterpart in Figure A.2. Second, emission intensities are highly dispersed even after trimming the tails of the distribution and, in ICE, restricting to third-party-validated disclosures. This motivates the focus in the main text on heterogeneity in firms' carbon use.[41]

Figure A.1 describes the scope decomposition of reported emissions in the ICE-ESG sample. For the average firm, direct production emissions are only one component of total emissions, with value-chain and other emissions accounting for the largest share. This decomposition is useful for interpreting our sample selection. The model and empirical analysis focus on Scope 1 emissions, as these are generated within the firm and are the natural object of interest for a production-based carbon tax. Scope 2 and Scope 3 emissions are also important, but using them to aggregate firm-level emissions would mechanically double-count emissions produced elsewhere in the economy and would shift the focus toward emissions over which firms have limited control.

A.2 Additional evidence

This subsection provides additional descriptive evidence supporting the two empirical claims in Section 2: firms account for a large share of aggregate emissions, and emission intensities are highly heterogeneous across and within economically meaningful groups.

Figure A.2: Estimated share of overall emissions—Trucost



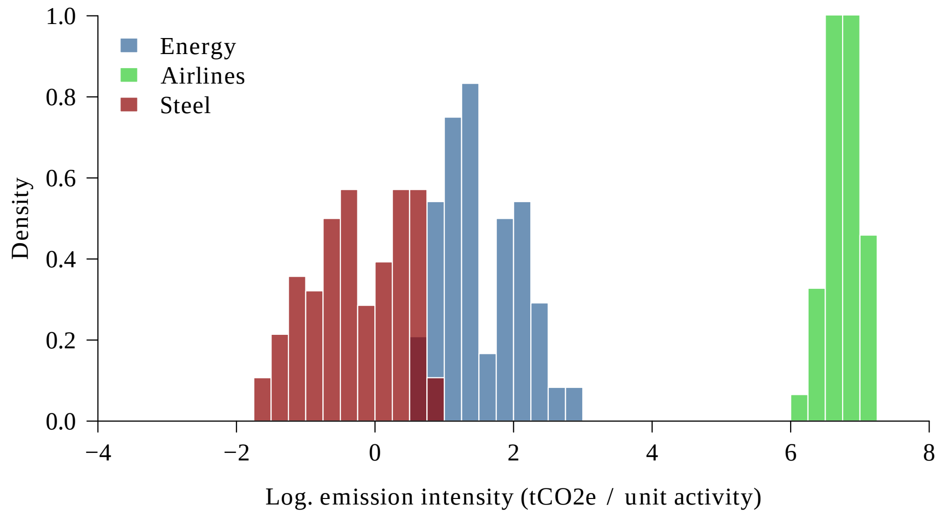
Note: Data from the Trucost sample. The figure shows overall firm emissions (measured by tons of CO_2e emissions) divided by total world emissions. Estimates for world emissions are taken from the *Global Carbon Budget* website (2024 estimates). Top-20 country-contributions are shown on the right-side axis.

Recall that Figure A.2 shows that publicly-listed firms in Trucost account for a smaller share of world emissions than firms in ICE-ESG. This difference is expected, as Trucost

[41] The large standard deviations for total scope 1 emissions reflect firm-size heterogeneity: the rows are in tCO_2e and therefore are not directly comparable to the normalized emission-intensity rows.

covers a narrower set of firms. That said, the magnitude of Trucost emissions remains substantial (c. 17%), and the ranking of large-emitting countries is similar to that in the ICE-ESG sample. The comparison thus supports the interpretation in the main text (Figure 1): firms remain a substantial driver of global emissions even in the smaller Trucost sample, and—given the representativeness of our data sets—these measures should, in all cases, be viewed as conservative lower bounds.

Figure A.3: Heterogeneity in emission intensities—output measures

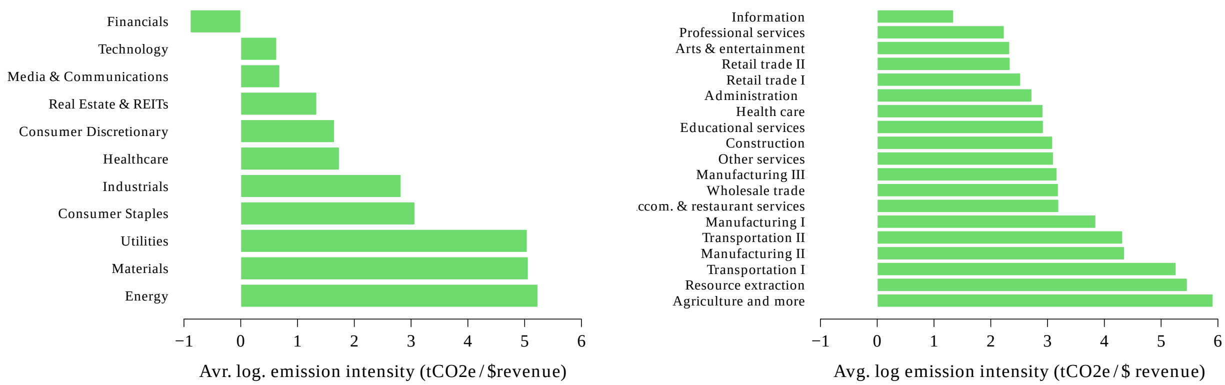


Note: Data from the ICE-ESG sample. The figure shows the distribution of log. emission intensities within the U.S. energy, airlines and steel sectors. Emission intensities are measured by tons of CO_2e emissions divided by output. We measure output by: mega jules of energy produced, thousands of return flights flown and tons of steel produced, respectively. We consider only ICE observations with third-party-validated disclosures (Appendix A.1). We trim emission intensities at the top/bottom 1% level.

Figure A.3 shows that large differences in emission intensity are also present when the denominator in the expression is physical units of output within narrowly-defined sectors instead of firm revenue. Emission intensities among U.S. energy producers, airlines, and steel firms differ sharply from each other; but, more importantly for our analysis, each narrow sector also displays substantial within-sector dispersion. This evidence reinforces the interpretation of Figure 2: the dispersion in emission efficiency is not primarily driven by differences among broadly defined sectors, nor by comparisons across firms producing different goods, or charging different prices.

Figure A.4 focuses in on the sectoral component of the dispersion in emission intensities. Emission intensities are highest in sectors such as energy, materials, utilities, agriculture, resource extraction, transportation, and manufacturing. These patterns are consistent with the conventional wisdom on sectoral emissions and also show why sector

Figure A.4: Sectoral heterogeneity in emission intensities



(a) ICE-ESG sample **(b)** Trucost sample

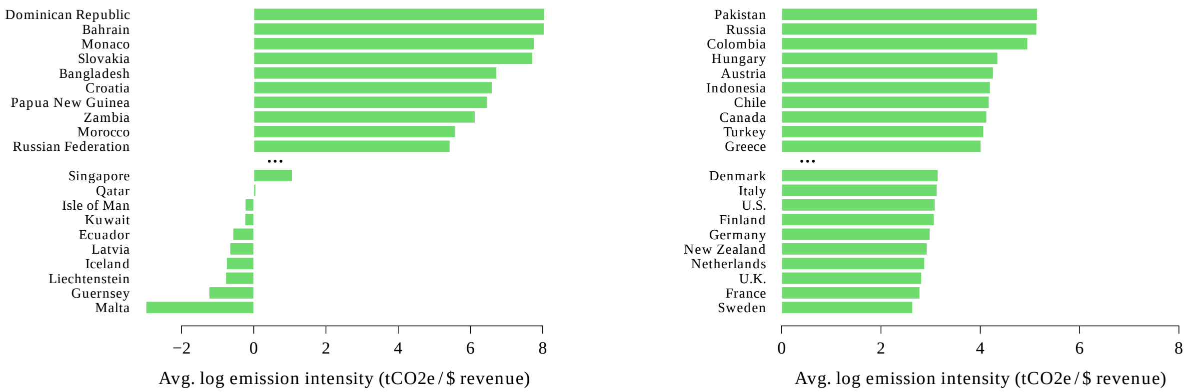
Note: Data from the ICE-ESG and TruCost samples. The figure shows the avr. log emission intensity within ICE and Trucost sectors (NAICS level 2). Emission intensities are measured by tons of CO_2e emissions divided by revenue (millions of $). For the ICE, we consider only observations with third-party-validated disclosures that are self-reported (Appendix A.1); Trucost does not provide an analogous validation flag. We trim emission intensities in both samples at the top/bottom 1% level.

fixed effects are important in Figure 2. The fact that substantial dispersion remains after absorbing these sectoral differences motivates the model's firm-level heterogeneity in carbon intensity.

Figure A.5 instead focuses on the country-level differences are also present in the sample. The figure reports the most and least emission-intensive countries in each dataset, omitting the middle of the distribution for readability. The cross-country ranking presents some differences across samples, reflecting differences in firm coverage, but both panels show that country composition is also an important source of raw variation. This motivates the country fixed effects used in the main empirical analysis.

Finally, Table A.2 relates emission intensity to firm productivity in the merged Truecost-Compustat sample. More productive firms are less emission intensive, both unconditionally and after absorbing year, sector, and country fixed effects, although the standard errors are large. Quantitatively, however, productivity differences accounts for little of the overall dispersion: the R^2 remains close to 5% Carbon intensity therefore captures a distinct dimension of firm heterogeneity—rather than simply reflecting standard productivity differences among firms.

Figure A.5: Country heterogeneity in emission intensities



(a) ICE-ESG sample

(b) Trucost sample

Note: Data from the ICE-ESG and Trucost samples. The figure shows the average log. emission intensity for the top and bottom 10 countries. Labels for a subset of countries are given. Emission intensities are measured by tons of CO_2e emissions divided by revenue (millions of $). For the ICE, we consider only observations with third-party-validated disclosures that are self-reported (Appendix A.1); Trucost does not provide an analogous validation. We trim emission intensities in both samples at the top/bottom 1% level.

A.3 Sources of heterogeneity, disclosure standards, and misreporting

The above-documented heterogeneity in emission intensities is consistent with a large body of evidence from the engineering and sustainability literature and is reflected in the emissions calculation standards adopted by all major ETS markets, as well as by the carbon tax framework introduced in Chile. We now explain, through the example of the European ETS, which firm-level elements—fuel composition and technology choices—drive the wide heterogeneity in emissions outcomes and how they are disclosed.

Sources of heterogeneity and disclosure standards. Emissions disclosure must (i) define an installation boundary, (ii) identify emission sources and source streams, (iii) specify the monitoring methodology, and lastly (iv) document procedures for collecting, verifying, and retaining data (see European Commission Directorate-General for Climate Action, 2025; UK Department for Energy Security and Net Zero, 2025). *Emissions are rarely measured* at the stack: in the EU ETS, only 145 installations, or 1.7 percent of installations across 22 countries, reported using continuous emissions monitoring systems (CEMS) in 2022 (European Commission, 2023b).[42] Instead, reported emissions are *calculated from*

[42]CEMS require continuous measurement of gas concentrations and exhaust flows, missing-data substitution protocols, certified equipment, along with regular quality assurance and surveillance tests by accredited laboratories. They are infeasible where emissions are diffuse or lack a well-defined release point. As

Table A.2: Carbon intensity and productivity

	Log. emission intensity	
	(1)	(2)
Log. TFP	-0.556***	-0.567***
	(0.150)	(0.153)
Fixed effects		
Time		✓
Sector		✓
Country		✓
R-squared	0.047	0.058
Observations	12,508	12,508

Notes: Least-squares estimates from the merged Trucost-Compustat sample. The top and bottom 1 percent of emission intensities (tons of CO_2e emissions divided by revenue in millions of $) and total factor productivity are removed. TFP is estimated as in Ottonello and Winberry (2020). NAICS-4 sectoral definitions are used throughout. Robust clustered (sectors) standard errors in parentheses. Sample: 2010–2024.

firm-specific disclosures—fuel quantities, emission factors, process parameters, and abatement records—and validated by a third party.

A useful way to summarize this accounting framework is through the equation:

$$g = \underbrace{\left(\sum_i \omega_i^e \cdot (1 - x_i)\right)}_{\text{Technological factors}} \cdot \underbrace{\left(\sum_i \omega_i^e \cdot z_i\right)}_{\text{Average emission factor}} \cdot \underbrace{\left(\sum_i e_i\right)}_{\text{Energy amount}}, \tag{A1}$$

where g denotes reported emissions in CO_2e units, e_i is energy use from source stream i, $\omega_i^e \equiv e_i / \sum_j e_j$ is the corresponding energy share, z_i is the emissions factor of that source stream, and x_i summarizes source-stream-specific technological and process adjustments that reduce reportable emissions. The equation is a parsimonious version of the calculation-based reporting approach used by the EU's ETS: reported emissions equal total energy use, multiplied by the average emissions factor of the energy bundle, and adjusted for technologies or processes that reduce the share of potential emissions.

• *Energy amount.* The last term, $\sum_i e_i$, in Equation (A1) captures reported energy use. The installation must identify all relevant source streams: physical quantities of fuels and materials and stock changes. The index i runs over a fine partition of inputs—

a result, they are costly, technically demanding, and require a well-defined emission point, making them typically applicable only to a few large plants with concentrated emission streams (see also Section 5.4).

distinguishing coal ranks and batches, natural gas, refinery gas, fuel oil, coke, waste gas, biomass, mixed fossil-biomass fuels, and other process energy inputs. An installation's reported emissions therefore depend on whether the operator has disclosed the quantity and composition of every fuel used in production, before any emissions factor or technological adjustment is applied.

• *Average emission factor.* The middle term, $\sum_i \omega_i^e \cdot z_i$, in Equation (A1) is the average warming content of the energy bundle, where z_i maps one unit of energy from source stream i into CO_2e. Depending on the source stream, z_i may be drawn from legislated default factors, supplier-guaranteed values, or installation-commissioned laboratory analysis. The margin is quantitatively meaningful even for relatively homogeneous fuels: the default factors for stationary combustion of coal range from 0.9-1.4tCO_2 per tonne for lignite to 2.5-2.7tCO_2 per tonne for anthracite, and more disaggregated evidence shows that factors vary further by origin, destination, and sample composition (Quick, 2010). Heterogeneous inputs exhibit even wider dispersion. Biomass and waste-derived fuels can span the full range from zero to fossil-like emission factors, because ETS accounting assigns zero to sustainable biomass while non-biogenic fractions—such as plastics in refuse-derived fuels—carry emission factors comparable to coal (European Commission, 2018a).

• *Technological factors.* The first term, $\sum_i \omega_i^e \cdot (1 - x_i)$, in Equation (A1) is the energy-weighted share of potential emissions remaining after source-stream-specific technological and process adjustments. The correction x_i is not a single technology but a composite of abatement choices that can differ substantially across installations using identical fuel mixes. Carbon capture at the stack targets CO_2 directly, with commercial post-combustion systems designed around capture rates near 90% (International Energy Agency, 2024). Incomplete combustion and oxidation corrections reduce reportable emissions when carbon exits as ash, slag, or unburnt residue rather than CO_2. Process modifications—such as chemical additives, altered conversion rates, or waste-gas recovery—reduce emissions by changing the stoichiometry of combustion itself. For non-CO_2 greenhouse gases, selective catalytic reduction and similar post-combustion controls can reduce nitrous oxide emissions by 80–90 % (U.S. Environmental Protection Agency, 2002), while methane oxidation and flaring govern the CO_2e contribution of fugitive gas streams. The implication is that two installations operating with identical fuel quantities and composition can report substantially different emissions depending on which subset of these technologies is installed—a feature which our model framework in Section 3 directly captures.

Reporting margins. Equation (A1) further clarifies the main margins of misreporting and how they can occur with carbon disclosures.

First, operators can omit energy sources entirely. In the Brikel case, discussed in the main text, the installation was documented burning waste streams—plastics and petroleum-derived industrial waste—alongside coal without the corresponding permit updates or public disclosure (ClientEarth, 2020; Bivol, 2020). Because such materials carry fossil-carbon content comparable to or exceeding that of coal, omitting them from reported source streams directly suppressed emissions.

Second, firms can misreport emission factor. Brikel, allegedly burning lignite coal with reference net-calorific factors between 96.1 and 104.2 kg CO_2/GJ, reported factors of 90.5, 79.5, and 76.2 in 2018, 2019, and 2020, respectively (Organized Crime and Corruption Reporting Project, 2021). The plant supported these figures with laboratory analyses that are difficult to challenge without independent physical access to the fuel sample.

Third, firms can evade the reporting boundary altogether. Since only installations above a threshold are often required to participate in the disclosures, misreporting activity data or fragmenting operations across sites can place an installation below the regulatory perimeter. Diageo operated three Scottish whisky-production sites carrying out regulated combustion activity without the required permits for multiple years. By never registering these sites, their emissions fell entirely outside the reporting population until the breach was identified. ExxonMobil's Mossmorran case shows the same logic applied within a registered installation: specific emission sources inside the plant were omitted from the 2008 EU ETS return, understating emissions by approximately 33,000t.

Finally, notice that the Brikel disclosures were certified by a third-party verifier, and the European Public Prosecutor's Office later opened a fraud investigation into a verification company alleged to have submitted false data and documentation. None of the these cases were uncovered through routine regulatory oversight: investigative journalists, NGOs, whistle-blowers, and later prosecutors played the central role in each. This pattern is consistent with the disclosure framework above. Regulators have access only to reported data—monitoring plans, audited records, and verified calculations—and do not conduct systematic on-site inspections of fuel composition or emission sources (European Commission, 2018b). Without independent physical access to the installation, detecting manipulation requires either an anomaly large enough to trigger scrutiny or an external informant. The documented cases are therefore likely the tip of the iceberg.

B Model framework and results

B.1 Proofs and derivations

Proof of Lemma 1. We start with firm (i, j, ℓ)'s problem:

$$\max_{\{n_{ij\ell,\tau}, k_{ij\ell,\tau}, e_{ij\ell,\tau}, x_{ij\ell,\tau}\}_{\tau=t}^{\infty}} v_{ij\ell,t} = \pi_{ij\ell,t} + \mathbb{1}_{\{\ell<L\}} \cdot \beta \cdot \frac{u'(C_{t+1})}{u'(C_t)} \cdot v_{ij,\ell+1,t+1}, \tag{A2}$$

where:

$$\pi_{ij\ell,t} = A\left(G_t\right) \cdot \exp^{a_i} \cdot F\left(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}\right) - W_t \cdot n_{ij\ell,t} - R_t \cdot k_{ij\ell,t} - P \cdot e_{ij\ell,t} - Q(x_{ij\ell,t}). \tag{A3}$$

Since the firm can costlessly reoptimize input choices at any point in time, it follows that the optimal choices maximize per-period profits and hence satisfy the expressions:

$$mrpn_{ij\ell,t} \equiv A(G_t) \cdot \exp^{a_i} \cdot F_n\left(\cdot\right) = W_t \quad \text{and} \quad mrpk_{ij\ell,t} \equiv A(G_t) \cdot \exp^{a_i} \cdot F_k\left(\cdot\right) = R_t, \tag{A4}$$

$$mrpe_{ij\ell,t} \equiv A(G_t) \cdot \exp^{a_i} \cdot F_e\left(\cdot\right) = P_t \quad \text{and} \quad Q'(x_{ij\ell,\tau}) = 0. \tag{A5}$$

Furthermore, notice that from the condition for energy:

$$\frac{\partial y_{ij\ell,t}}{\partial e_{ij\ell,t}} \cdot \frac{e_{ij\ell,t}}{y_{ij\ell,t}} = \frac{\partial \log y_{ij\ell,t}}{\partial \log e_{ij\ell,t}} = P \cdot \frac{e_{ij\ell,t}}{y_{ij\ell,t}}. \tag{A6}$$

Thus,

$$\log\left(\frac{g_{ij\ell,t}}{y_{ij\ell,t}}\right) = \log\left(\frac{\exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t}}{P \cdot \frac{\partial \log e_{ij\ell,t}}{\partial \log y_{ij\ell,t}} \cdot e_{ij\ell,t}}\right) = z_j + \log\left(\frac{\upsilon}{P}\right) + \wr(1), \tag{A7}$$

since $x_{ij\ell,t} = 0$ from Equation (A5) and $\frac{\partial \log y_{ij\ell,t}}{\partial \log e_{ij\ell,t}} = \upsilon + \wr(1)$. ■

Proof of Lemma 2. The full-information planner maximizes (16) subject to subject to Equations (3), (4), (5), (12), and (13). This yields the following Lagrangian:

$$\mathcal{L} = \sum_t \beta^t \cdot \Bigg[u(C_t) + \Lambda_{C,t} \cdot \Bigg(-C_t - K_{t+1} + (1-\delta_K) \cdot K_t \tag{A8}$$
$$+ \sum_{ij\ell} w_{ij\ell} \cdot \left(A(G_t) \cdot \exp^{a_i} \cdot F\left(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}\right) - P \cdot e_{ij\ell,t} - Q(x_{ij\ell,t}) \right) \Bigg)$$
$$- \Lambda_{C,t} \cdot \Lambda_{G,t} \cdot \Bigg((1-\delta_G) \cdot G_t + \sum_{ij\ell} w_{ij\ell} \cdot \exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t} - G_{t+1} \Bigg)$$
$$+ \Lambda_{C,t} \cdot \Lambda_{N,t} \cdot \Bigg(1 - \sum_{ij\ell} w_{ij\ell} \cdot n_{ij\ell,t} \Bigg) + \Lambda_{C,t} \cdot \Lambda_{K,t} \cdot \Bigg(K_t - \sum_{ij\ell} w_{ij\ell} \cdot k_{ij\ell,t} \Bigg) \Bigg].$$

• The first-order conditions for C_t, K_{t+1}, and G_{t+1} are:

$$\frac{\partial \mathcal{L}}{\partial C_t} = u'(C_t) - \Lambda_{c,t} = 0 \tag{A9}$$
$$\frac{\partial \mathcal{L}}{\partial K_{t+1}} = -\beta^t \cdot \Lambda_{C,t} + \beta^{t+1} \cdot \Lambda_{C,t+1} \cdot ((1-\delta_K) + \Lambda_{K,t+1}) = 0 \tag{A10}$$
$$\frac{\partial \mathcal{L}}{\partial G_{t+1}} = \beta^t \cdot \Lambda_{C,t} \cdot \Lambda_{G,t} + \beta^{t+1} \cdot \Lambda_{C,t+1} \cdot \frac{A'(G_{t+1})}{A(G_{t+1})} \cdot \mathcal{Y}_{t+1} \tag{A11}$$
$$- \beta^{t+1} \cdot \Lambda_{C,t+1} \cdot \Lambda_{G,t+1} \cdot (1-\delta_G) = 0,$$

which combined give the Euler equations for consumption and the social cost of carbon:

$$u'(C_t) = \beta \cdot u'(C_{t+1}) \cdot ((1-\delta_K) + \Lambda_{K,t+1}) \tag{A12}$$
$$\Lambda_{G,t} = \beta \cdot \frac{u'(C_{t+1})}{u'(C_t)} \cdot \left(-\frac{A'(G_{t+1})}{A(G_{t+1})} \cdot \mathcal{Y}_{t+1} + \Lambda_{G,t+1} \cdot (1-\delta_G) \right). \tag{A13}$$

• The first-order conditions for a firm's input $\left(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t},\right)$ are:

$$\frac{\partial \mathcal{L}}{\partial n_{ij\ell,t}} = \beta^t \cdot \Lambda_{C,t} \cdot w_{ij\ell} \cdot mrpn_{ij\ell,t} - \beta^t \cdot \Lambda_{C,t} \cdot \Lambda_{N,t} \cdot w_{ij\ell} = 0 \tag{A14}$$
$$\frac{\partial \mathcal{L}}{\partial k_{ij\ell,t}} = \beta^t \cdot \Lambda_{C,t} \cdot w_{ij\ell} \cdot mrpk_{ij\ell,t} - \beta^t \cdot \Lambda_{C,t} \cdot \Lambda_{K,t} \cdot w_{ij\ell} = 0 \tag{A15}$$
$$\frac{\partial \mathcal{L}}{\partial e_{ij\ell,t}} = \beta^t \cdot \Lambda_{C,t} \cdot (w_{ij\ell} \cdot mrpe_{ij\ell,t} - w_{ij\ell} \cdot P) - \beta^t \cdot \Lambda_{C,t} \cdot \Lambda_{G,t} \cdot w_{ij\ell} \cdot \exp^{z_j - x_{ij\ell,t}} = 0 \tag{A16}$$
$$\frac{\partial \mathcal{L}}{\partial x_{ij\ell,t}} = -\beta^t \cdot \Lambda_{C,t} \cdot w_{ij\ell} \cdot Q'(x_{ij\ell,t}) + \beta^t \cdot \Lambda_{C,t} \cdot \Lambda_{G,t} \cdot w_{ij\ell} \cdot \exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t} = 0. \tag{A17}$$

Thus,

$$mrpn_{ij\ell,t} = \Lambda_{N,t} \quad \text{and} \quad mrpk_{ij\ell,t} = \Lambda_{K,t}, \tag{A18}$$

and:

$$mrpe_{ij\ell,t} = P + \Lambda_{G,t} \cdot \exp^{z_j - x_{ij\ell,t}} \quad \text{and} \quad Q'\left(x_{ij\ell,t}\right) = \Lambda_{G,t} \cdot \exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t}. \tag{A19}$$

• The market-clearing conditions for labor and capital, the law-of-motion for the carbon stock, and the aggregate resource constraint, in turn, follow from the relevant first-order conditions for $\Lambda_{C,t}, \Lambda_{G,t}, \Lambda_{N,t}$, and $\Lambda_{K,t}$, and complete the description of the full-information planner allocation.

Finally, the last part of the statement follows from the $mrpe_{ij\ell,t}$ being monotonically declining in $e_{ij\ell,t}$, while abatement costs $Q(\cdot)$ increase in $x_{ij\ell,t}$. This is when combined with $\Lambda_{G,t} > 0$ in Equation (A19) and (A5), and that $R_t = \Lambda_{K,t}$ and $W_t = \Lambda_{N,t}$ in Equation (A18) and (A4). This completes the proof. ■

Proof of Corollary 1. Notice that both types' allocations are evaluated at the common multipliers $(\Lambda_{N,t}, \Lambda_{K,t})$ and solve each firm's problem. The result then follows directly from the second part of Lemma 2. Notice that, for any (i, ℓ, t), and any j', j with $z_{j'} > z_j$:

$$e^{FB}_{ij'\ell,t} < e^{FB}_{ij\ell,t} < e^{DE}_{ij\ell,t} \quad \text{and} \quad x^{FB}_{ij'\ell,t} > x^{FB}_{ij\ell,t} > x^{DE}_{ij\ell,t} = 0. \tag{A20}$$

As a result, for any t, $\pi^{FB}_{ij\ell,t} > \pi^{FB}_{ij'\ell,t}$ by revealed preference, and hence $v^{FB}_{ij\ell,t} > v^{FB}_{ij'\ell,t}$. ■

Proof of Lemma 3. The Mirrleesian planner solves the problem:

$$\max_{\{C_t, K_{t+1}, G_{t+1}, R_t, W_t, \{k_{ij\ell,t}, n_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t}, v_{ij,\ell+1,t+1}\}_{i,j,\ell}\}_{t=0}^{\infty}} \mathcal{U} = \sum_{t=0}^{\infty} \beta^t \cdot u\left(C_t\right) \tag{A21}$$

subject to Equations (3), (4), (5), (12), and (13). In addition, the Mirrleesian planner faces the incentive compatibility constraint, also detailed in the main text:

$$\begin{aligned} v_{ij\ell,t} = \sum_{\tau=0}^{L-\ell} m_{t,t+\tau} \cdot \pi(n_{ij\ell+\tau,t+\tau}, k_{ij\ell+\tau,t+\tau}, e_{ij\ell+\tau,t+\tau}, x_{ij\ell+\tau,t+\tau}) \geq \\ \sum_{\tau=0}^{L-\ell} m_{t,t+\tau} \cdot \pi(n_{ij'\ell+\tau,t+\tau}, k_{ij'\ell+\tau,t+\tau}, e_{ij'\ell,t+\tau}, x_{ij'\ell+\tau,t+\tau}) = v_{ij'\ell,t}, \quad \forall j' \neq j, \end{aligned} \tag{A22}$$

where $m_{t,t+\tau} \equiv \beta^{\tau} \cdot u'(C_{t+\tau})/u'(C_t)$ is the discount factor.

We abstract below from the individual rationality constraint, as this constraint is automatically slack due to the assumption of decreasing returns to scale and the Inada condition. Finally, the planner also faces the following implementability constraint:

$$\Lambda_{N,t} = W_t \quad \text{and} \quad \Lambda_{K,t} = R_t, \tag{A23}$$

where $\Lambda_{N,t}$ and $\Lambda_{K,t}$ are the shadow costs of labor and capital.

Since profits do not depend directly on z_j, firms can freely (and credibly) report any carbon intensity $z_j \in \mathcal{Z}$ that gives them the highest profit conditional on their productivity type a_i and their age ℓ. Thus, incentive compatibility requires:

$$\sum_{\tau=0}^{L-\ell} m_{t,t+\tau} \cdot \pi(n_{ij\ell+\tau,t+\tau}, k_{ij\ell+\tau,t+\tau}, e_{ij\ell+\tau,t+\tau}, x_{ij\ell+\tau,t+\tau}) = v_{i\ell,t}, \tag{A24}$$

where $v_{i\ell,t} \in \mathbb{R}_+$ is the common value for firms of type $\{i, \ell\}$ at time t.

Now, starting from the last year of a firm's life, $\ell = L$, it follows that:

$$\pi(n_{ijL,t}, k_{ijL,t}, e_{ijL,t}, x_{ijL,t}) = v_{i\ell,t} \equiv \tilde{\pi}_{iL,t}. \tag{A25}$$

Thus, at $\ell = L - 1$, we have:

$$\begin{aligned} \pi(n_{ijL-1,t}, k_{ijL-1,t}, e_{ijL-1,t}, x_{ijL-1,t}) + m_{t,t+1} \cdot \pi(n_{ijL,t}, k_{ijL,t}, e_{ijL,t}, x_{ijL,t}) &= v_{iL-1,t} \\ \pi(n_{ijL-1,t}, k_{ijL-1,t}, e_{ijL-1,t}, x_{ijL-1,t}) + m_{t,t+1} \cdot v_{iL,t} &= v_{iL-1,t} \\ \pi(n_{ijL-1,t}, k_{ijL-1,t}, e_{ijL-1,t}, x_{ijL-1,t}) = v_{iL-1,t} - m_{t,t+1} \cdot v_{iL,t} &\equiv \tilde{\pi}_{iL-1,t}, \end{aligned} \tag{A26}$$

and so on until $\ell = 1$. This implies that an incentive-compatible allocation satisfying (A24) also satisfies equivalently the following set of per-period constraints:

$$\pi_{ij\ell,t} = \tilde{\pi}_{i\ell,t}, \tag{A27}$$

which we use below.

The Lagrangian for the Mirrleesian planner's problem can therefore be stated as:

$$
\begin{aligned}
\mathcal{L} = \sum_t \beta^t \cdot \Bigg[u(C_t) + \Lambda_{C,t} \cdot \Bigg(& - C_t - K_{t+1} + (1-\delta_K) \cdot K_t \\
& + \sum_{ij\ell} w_{ij\ell} \cdot \Big(A(G_t) \cdot e^{a_i} \cdot F(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}) - P \cdot e_{ij\ell,t} - Q(x_{ij\ell,t}) \Big) \Bigg) \\
& - \Lambda_{C,t} \cdot \Lambda_{G,t} \cdot \Bigg((1-\delta_G) \cdot G_t + \sum_{ij\ell} w_{ij\ell} \cdot e^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t} - G_{t+1} \Bigg) \\
& + \Lambda_{C,t} \cdot \Lambda_{N,t} \cdot \Bigg(1 - \sum_{ij\ell} w_{ij\ell} \cdot n_{ij\ell,t} \Bigg) \\
& + \Lambda_{C,t} \cdot \Lambda_{K,t} \cdot \Bigg(K_t - \sum_{ij\ell} w_{ij\ell} \cdot k_{ij\ell,t} \Bigg) \\
& + \Lambda_{C,t} \cdot \sum_{ij\ell} w_{ij\ell} \cdot \Lambda_{\pi,ij\ell,t} \cdot \Big(\pi_{ij\ell,t} - \tilde{\pi}_{i\ell,t} \Big) \Bigg]. \qquad \text{(A28)}
\end{aligned}
$$

• The first-order conditions for a firm's inputs $(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t})$ are thus:

$$
\begin{aligned}
mrpn_{ij\ell,t} &= \Lambda_{N,t} && \text{(A29)} \\
mrpk_{ij\ell,t} &= \Lambda_{K,t} && \text{(A30)} \\
mrpe_{ij\ell,t} &= P + \frac{\Lambda_{G,t}}{1 + \Lambda_{\pi,ij\ell,t}} \cdot \exp^{z_j - x_{ij\ell,t}} && \text{(A31)} \\
Q'(x_{ij\ell,t}) &= \frac{\Lambda_{G,t}}{1 + \Lambda_{\pi,ij\ell,t}} \cdot \exp^{z_j - x_{ij\ell,t}} \cdot e_{ij\ell,t}, && \text{(A32)}
\end{aligned}
$$

where (A31) and (A32) can be combined to yield:

$$
mrpe_{ij\ell,t} - P = \frac{Q'(x_{ij\ell,t})}{e_{ij\ell,t}}. \qquad \text{(A33)}
$$

Now, notice that Equations (A27), (A29), (A30), and (A33) jointly characterize the firm's choice of $(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t})$ independently of z_j—i.e., $(n_{ij\ell,t}, k_{ij\ell,t}, e_{ij\ell,t}, x_{ij\ell,t}) = (n_{i\ell,t}, k_{i\ell,t}, e_{i\ell,t}, x_{i\ell,t})$. There is bunching across carbon-intensity types j.

• The first-order condition with respect to $\tilde{\pi}_{i\ell,t}$ takes the form:

$$
\sum_j w_{j|i\ell} \cdot \Lambda_{\pi,ij\ell,t} = 0, \qquad \text{(A34)}
$$

where $w_{j|i\ell} \equiv \frac{w_{ij\ell}}{\sum_j w_{ij\ell}}$. Combining (A31) and (A32) with (A34), and exploiting bunching across carbon-intensity types j, then provides us with:

$$mrpe_{i\ell,t} = P + \sum_j w_{j|i\ell} \cdot \Lambda_{G,t} \cdot \exp^{z_j - x_{i\ell,t}} \tag{A35}$$

$$Q'(x_{i\ell,t}) = \sum_j w_{j|i\ell} \cdot \Lambda_{G,t} \cdot \exp^{z_j - x_{i\ell,t}} \cdot e_{i\ell,t}. \tag{A36}$$

Since further

$$w_{j|i\ell} \equiv \frac{w_{ij\ell}}{\sum_j w_{ij\ell}} = \frac{\omega_{ij} \cdot L^{-1}}{\sum_j \omega_{ij} \cdot L^{-1}} = \frac{\omega_{ij}}{\sum_j \omega_{ij}}; \tag{A37}$$

is independent of ℓ, it follows that the entire system of Equations (A29), (A30), (A35), and (A36) does not directly depend on ℓ either—i.e., $(n_{i\ell,t}, k_{i\ell,t}, e_{i\ell,t}, x_{i\ell,t}) = (n_{i,t}, k_{i,t}, e_{i,t}, x_{i,t})$. There is bunching across firm age ℓ, in addition to carbon intensity j.

• The first-order conditions with respect C_t, K_{t+1}, and G_{t+1} are equal to those from the full-information case. Notice that the derivative of the incentive-constraint part of (A28) with respect G_{t+1} is zero, as these are the decentralized equilibrium profits of each atomistic firm. This completes the description of the Mirrleesian allocation.

Finally, the last part of the statement follows from the $mrpe_{ij\ell,t}$ being monotonically declining in $e_{ij\ell,t}$, while abatement costs $Q(\cdot)$ increase in $x_{ij\ell,t}$. This is when combined with $\Lambda_{G,t} > 0$ in Equation (A35), (A36), and (A5), and that $R_t = \Lambda_{K,t}$ and $W_t = \Lambda_{N,t}$ in Equation (A29) and (A30). This completes the proof. ■

Proof of Proposition 1. Follows directly from (27) and the proofs of Lemma 2 and 3. ■

Proof of Proposition 2. We express all first-order conditions as follows:

$$mrpk_{ij\ell,t} = R_t, \qquad mrpn_{ij\ell,t} = W_t, \tag{A38}$$

$$mrpe_{ij\ell,t} = (1 + \Delta^{\mathcal{P}}_{ij\ell,t}) \cdot P; \tag{A39}$$

where $\mathcal{P} \in \{DE, FB, MR, NV\}$, and $R_t = \Lambda_{K,t}$ and $W_t = \Lambda_{N,t}$ in the full-information and Mirrleesian cases, respectively. The wedge term $\Delta^{\mathcal{P}}_{ij\ell,t}$ equals:

$$\Delta^{DE}_{ij\ell,t} = 0, \qquad \Delta^{FB}_{ij\ell,t} = \frac{\Lambda_{G,t} \cdot \exp^{z_j - x_{ij\ell,t}}}{P}, \tag{A40}$$

$$\Delta^{MR}_{ij\ell,t} = \frac{\Lambda_{G,t} \cdot \mathbb{E}[\exp^{z_j - x_{ij\ell,t}} | a_i]}{P}, \qquad \Delta^{NV}_{ij\ell,t}, = \frac{\Lambda_{G,t} \cdot \exp^{z - x_{ij\ell,t}}}{P}. \tag{A41}$$

Then, we can express each input choice as follows:

$$k_{ij\ell,t} = (A \cdot \exp^{a_i})^{\frac{1}{\zeta}} \cdot \left(\frac{\alpha}{R_t}\right)^{\frac{1-\gamma-\nu}{\zeta}} \cdot \left(\frac{\gamma}{W_t}\right)^{\frac{\gamma}{\zeta}} \cdot \left(\frac{\nu}{(1+\Delta^{\mathcal{P}}_{ij\ell,t}) \cdot P}\right)^{\frac{\nu}{\zeta}} \tag{A42}$$

$$n_{ij\ell,t} = (A \cdot \exp^{a_i})^{\frac{1}{\zeta}} \cdot \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{\zeta}} \cdot \left(\frac{\gamma}{W_t}\right)^{\frac{1-\alpha-\nu}{\zeta}} \cdot \left(\frac{\nu}{(1+\Delta^{\mathcal{P}}_{ij\ell,t}) \cdot P}\right)^{\frac{\nu}{\zeta}} \tag{A43}$$

$$e_{ij\ell,t} = (A \cdot \exp^{a_i})^{\frac{1}{\zeta}} \cdot \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{\zeta}} \cdot \left(\frac{\gamma}{W_t}\right)^{\frac{\gamma}{\zeta}} \cdot \left(\frac{\nu}{(1+\Delta^{\mathcal{P}}_{ij\ell,t}) \cdot P}\right)^{\frac{1-\alpha-\gamma}{\zeta}} \tag{A44}$$

implying the following expression for the output of a firm:

$$y_{ij\ell,t} = (A \cdot \exp^{a_i})^{\frac{1}{\zeta}} \cdot \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{\zeta}} \cdot \left(\frac{\gamma}{W_t}\right)^{\frac{\gamma}{\zeta}} \cdot \left(\frac{\nu}{(1+\Delta^{\mathcal{P}}_{ij\ell,t}) \cdot P}\right)^{\frac{\nu}{\zeta}}. \tag{A45}$$

Aggregate gross output then equals:

$$\begin{aligned}
\mathcal{Y}_t &\equiv \sum_{ij\ell} w_{ij\ell} \cdot y_{ij\ell,t} \\
&= A_t^{\frac{1}{\zeta}} \cdot \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{\zeta}} \cdot \left(\frac{\gamma}{W_t}\right)^{\frac{\gamma}{\zeta}} \cdot \left(\frac{\nu}{P}\right)^{\frac{\nu}{\zeta}} \cdot \sum_{ij\ell} w_{ij\ell} \cdot (\exp^{a_i})^{\frac{1}{\zeta}} \cdot \left(\frac{1}{1+\Delta^{\mathcal{P}}_{ij\ell,t}}\right)^{\frac{\nu}{\zeta}} \\
&= \left(\frac{A_t^{\frac{1}{\zeta}} \cdot \left(\frac{\alpha}{R_t}\right)^{\frac{\alpha}{\zeta}} \cdot \left(\frac{\gamma}{W_t}\right)^{\frac{\gamma}{\zeta}} \cdot \left(\frac{\nu}{P}\right)^{\frac{\nu}{\zeta}} \cdot \sum_{ij\ell} w_{ij\ell} \cdot (\exp^{a_i})^{\frac{1}{\zeta}} \cdot \left(\frac{1}{1+\Delta^{\mathcal{P}}_{ij\ell,t}}\right)^{\frac{\nu}{\zeta}}}{\left(\sum_{ij\ell} w_{ij\ell} \cdot k_{ij\ell,t}\right)^{\alpha} \cdot \left(\sum_{ij\ell} w_{ij\ell} \cdot n_{ij\ell,t}\right)^{\gamma} \cdot \left(\sum_{ij\ell} w_{ij\ell} \cdot e_{ij\ell,t}\right)^{\nu}} \right) \cdot K_t^{\alpha} \cdot N_t^{\gamma} \cdot E_t^{\nu} \\
&= \underbrace{\frac{\left[\sum_{ij\ell} \left(\frac{w_{ij\ell} \cdot \exp^{a_i/\zeta}}{\sum_{i'j'\ell'} w_{i'j'\ell'} \cdot \exp^{a_{i'}/\zeta}}\right) \cdot \left(\frac{1}{1+\Delta^{\mathcal{P}}_{ij\ell,t}}\right)^{\frac{\nu}{\zeta}}\right]^{\nu+\zeta}}{\left[\sum_{ij\ell} \left(\frac{w_{ij\ell} \cdot \exp^{a_i/\zeta}}{\sum_{i'j'\ell'} w_{i'j'\ell'} \cdot \exp^{a_{i'}/\zeta}}\right) \cdot \left(\frac{1}{1+\Delta^{\mathcal{P}}_{ij\ell,t}}\right)^{\frac{\nu+\zeta}{\zeta}}\right]^{\nu}}}_{\equiv \Omega_t^{\mathcal{P}}} \cdot A_t \cdot \underbrace{\left(\sum_{ij\ell} w_{ij\ell} \cdot \exp^{a_i/\zeta}\right)^{\zeta}}_{\equiv \exp^{a}} \cdot K_t^{\alpha} \cdot N_t^{\gamma} \cdot E_t^{\nu}.
\end{aligned} \tag{A46}$$

This can be expressed in terms of consumption as follows:

$$
\begin{aligned}
C_t &= \mathcal{Y}_t - P \cdot E_t - \sum_{ij\ell} w_{ij\ell,t} \cdot Q(x_{ij\ell,t}) - I_t \\
&= \Omega_t^{\mathcal{P}} \cdot \underbrace{\left(1 - \frac{P \cdot E_t + \sum_{ij\ell} w_{ij\ell,t} \cdot Q(x_{ij\ell,t}) + I_t}{\mathcal{Y}_t}\right)}_{\equiv \Theta_t} \cdot A_t \cdot \exp^a \cdot K_t^\alpha \cdot N_t^\gamma \cdot E_t^\nu,
\end{aligned}
\tag{A47}
$$

which concludes the proof. ■

B.2 Alternative implementations

Suppose firm (i, j, ℓ) is charged an energy tax $\tau^e_{ij\ell,t}(P \cdot e_{ij\ell,t})$ and receives an abatement subsidy equal to $\tau^x_{ij\ell,t}(Q(x_{ij\ell,t}))$. Firm $(i, j, \ell)'$s after-tax profits then become:

$$
\pi_{ij\ell,t} = y_{ijt,t} - W_t \cdot n_{ij\ell,t} - R_t \cdot k_{ij\ell,t} - P \cdot e_{ij\ell,t} - Q(x_{ij\ell,t}) - \tau^e_{ij\ell,t}(P \cdot e_{ij\ell,t}) + \tau^x_{ij\ell,t}(Q(x_{ij\ell,t})).
$$

The first-order conditions for the firm's input choices are:

$$
mrpk_{ij\ell,t} = R_t, \qquad mrpn_{ij\ell,t} = W_t.
$$

$$
mrpe_{ij\ell,t} = P + P \cdot \tau^{e\prime}_{ij\ell,t}(P \cdot e_{ij\ell,t}), \qquad \tau^{x\prime}_{ij\ell,t}(Q(x_{ij\ell,t})) \cdot Q'(x_{ij\ell,t}) = Q'(x_{ij\ell,t}).
$$

Similar steps to before show that the resulting allocation is independent of firm age ℓ.

• *First-best planner.* Comparing the first-order conditions to those from the full-information planner in Section 4.2 now directly shows that:

$$
\tau^{e\prime}_{ij,t}(P \cdot e_{ij,t}) = \frac{\Lambda_{G,t} \cdot \exp^{z_j - x_{ij,t}}}{P}, \qquad \tau^{x\prime}_{ij,t}(Q(x_{ij,t})) = \frac{\Lambda_{G,t} \cdot \exp^{z_j - x_{ij,t}} \cdot e_{ij,t}}{Q'(x_{ij,t})}
$$

implements the first-best allocation. Recall that the aggregate conditions are the same in the decentralized equilibrium as in the full-information case.

• *Mirrleesian planner.* Comparing instead the first-order conditions to those from the Mirrleesian case in Section 4.3 further shows that:

$$
\tau^{e\prime}_{i,t}(P \cdot e_{i,t}) = \frac{\Lambda_{G,t} \cdot \mathbb{E}[\exp^{z_j} | a_i] \cdot \exp^{-x_{i,t}}}{P}, \quad \tau^{x\prime}_{i,t}(Q(x_{i,t})) = \frac{\Lambda_{G,t} \cdot \mathbb{E}[\exp^{z_j} | a_i] \cdot \exp^{-x_{i,t}} \cdot e_{i,t}}{Q'(x_{i,t})}.
$$

implements the Mirrleesian allocation. Recall that the decentralized aggregate conditions are also the same as the aggregate conditions in the Mirrleesian case.

• *Naïve planner.* Finally, comparing the first-order conditions to those from the naïve planner case in Section 4.2 shows that:

$$\tau^{e\prime}_{ij,t}(P \cdot e_{ij,t}) = \frac{\Lambda_{G,t} \cdot \exp^{\underline{z} - x_{ij,t}}}{P}, \qquad \tau^{x\prime}_{ij,t}(Q(x_{ij,t})) = \frac{\Lambda_{G,t} \cdot \exp^{\underline{z} - x_{ij,t}} \cdot e_{ij,t}}{Q'(x_{ij,t})}$$

implements the naïve allocation.

• *Discussion of tax schedules.* Notice that all of the above energy taxes are *linear*, while the abatement subsides are *non-linear*. Indeed, the above marginal abatement subsidies can be integrated to, in the three cases, yield the tax schedules:

$$\tau^{x,FB}_{ij,t}\left(Q(x)\right) = \Lambda_{G,t} \cdot \exp^{z_j} \cdot e_{ij,t} \cdot (1 - \exp^{-x}) \tag{A48}$$

$$\tau^{x,MR}_{i,t}\left(Q(x)\right) = \Lambda_{G,t} \cdot \mathbb{E}[\exp^{z_j} \mid a_i] \cdot e_{i,t} \cdot (1 - \exp^{-x}) \tag{A49}$$

$$\tau^{x,NV}_{ij,t}\left(Q(x)\right) = \Lambda_{G,t} \cdot \exp^{\underline{z}} \cdot e_{ij,t} \cdot (1 - \exp^{-x}), \tag{A50}$$

which we quantitatively evaluate in Section 5.3 and Appendix C.5. Notice that the Mirrleesian tax and subsidy rates are independent of a firm's carbon intensity z_j, and that the subsidy rate on abatement is above 100% for levels below the optimal Mirrleesian level.

B.3 Aggregate emission decomposition

Here, we present the derivation of the aggregate emissions decomposition, as well as the derivations of each of its firm-level components.

Aggregate decomposition. Start from the definition of aggregate emissions:

$$\begin{aligned} G_{\text{flow},t} &= \sum_{ij\ell} w_{ij\ell} \cdot g_{ij\ell,t}, \\ &= \left(\frac{\sum_{ij\ell} w_{ij\ell} \cdot \frac{e_{ij\ell,t}}{E_t} \cdot \exp^{z_j - x_{ij\ell,t} + X_t}}{\sum_{ij\ell} w_{ij\ell} \cdot \frac{e_{ij\ell,t}}{E_t} \cdot \exp^{z_j}} \right) \cdot \left(\sum_{ij\ell} w_{ij\ell} \cdot \frac{e_{ij\ell,t}}{E_t} \cdot \exp^{z_j} \right) \cdot \exp^{-X_t} \cdot E_t. \end{aligned} \tag{A51}$$

In the decentralized equilibrium, this becomes:

$$G^{DE}_{\text{flow},t} = \left(\sum_{ij\ell} w_{ij\ell} \cdot \frac{e^{DE}_{ij\ell,t}}{E^{DE}_t} \cdot \exp^{z_j} \right) \cdot E^{DE}_t. \tag{A52}$$

Taking the ratios of $G_{\text{flow},t}$ and $G^{DE}_{\text{flow},t}$ then provides us with Equation (33).

Abatement allocation component. Starting from the definition:

$$\mathcal{Z}_{x,t} \equiv \frac{\sum_{ij\ell} w_{ij\ell} \cdot e_{ij\ell,t} \cdot \exp^{z_j - x_{ij\ell,t} + X_t}}{\sum_{ij\ell} w_{ij\ell} \cdot e_{ij\ell,t} \cdot \exp^{z_j}} = \frac{1}{\mathbb{E}\left[\frac{g_{ij\ell,t}}{G_{\text{flow},t}} \cdot \exp^{x_{ij\ell,t} - X_t}\right]}. \tag{A53}$$

Now, the denominator can be approximated as follows:

$$\begin{aligned}
\mathbb{E}\left[\frac{g_{ij\ell,t}}{G_{\text{flow},t}} \cdot \exp^{x_{ij\ell,t} - X_t}\right] &= \mathbb{E}\left[\frac{g_{ij\ell,t}}{G_{\text{flow},t}}\right] \cdot \mathbb{E}\left[\exp^{x_{ij\ell,t} - X_t}\right] + \mathbb{C}\left(\frac{g_{ij\ell,t}}{G_{\text{flow},t}}, \exp^{x_{ij\ell,t} - X_t}\right) \\
&= \mathbb{E}\left[\exp^{x_{ij\ell,t} - \mathbb{E}[x_{ij\ell,t}]}\right] + \mathbb{C}\left(\frac{g_{ij\ell,t}}{G_{\text{flow},t}}, \exp^{x_{ij\ell,t} - X_t}\right) \\
&\approx 1 + \mathbb{C}\left(\frac{g_{ij\ell,t}}{G_{\text{flow},t}}, \exp^{x_{ij\ell,t} - X_t}\right).
\end{aligned} \tag{A54}$$

It follows that:

$$\log \mathcal{Z}_{x,t} = -\log\left(1 + \mathbb{C}\left(\frac{g_{ij\ell,t}}{G_{\text{flow},t}}, \frac{\exp^{x_{ij\ell,t}}}{\exp^{X_t}}\right)\right) \approx -\mathbb{C}\left(\frac{g_{ij\ell,t}}{G_{\text{flow},t}}, \frac{\exp^{x_{ij\ell,t}}}{\exp^{X_t}}\right). \tag{A55}$$

Energy reallocation component. Using the definition of the energy reallocation component, and some rearranging yields:

$$\begin{aligned}
\mathcal{Z}_{e,t}^{FB} - \mathcal{Z}_{e,t}^{MR} &= \frac{\sum_{ij\ell} w_{ij\ell} \cdot \frac{e_{ij\ell,t}^{FB}}{E_t^{FB}} \cdot \exp^{z_j}}{\sum_{ij\ell} w_{ij\ell} \cdot \frac{e_{ij\ell,t}^{DE}}{E_t^{DE}} \cdot \exp^{z_j}} - \frac{\sum_{ij\ell} w_{ij\ell} \cdot \frac{e_{ij\ell,t}^{MR}}{E_t^{MR}} \cdot \exp^{z_j}}{\sum_{ij\ell} w_{ij\ell} \cdot \frac{e_{ij\ell,t}^{DE}}{E_t^{DE}} \cdot \exp^{z_j}} \\
&= \frac{\mathbb{E}\left[\frac{e_{ij\ell,t}^{FB}}{E_t^{FB}}\right] \cdot \mathbb{E}\left[\exp^{z_j}\right] + \mathbb{C}\left(\frac{e_{ij\ell,t}^{FB}}{E_t^{FB}}, \exp^{z_j}\right) - \mathbb{E}\left[\frac{e_{ij\ell,t}^{MR}}{E_t^{MR}}\right] \cdot \mathbb{E}\left[\exp^{z_j}\right] - \mathbb{C}\left(\frac{e_{ij\ell,t}^{MR}}{E_t^{MR}}, \exp^{z_j}\right)}{\mathbb{E}\left[\frac{e_{ij\ell,t}^{DE}}{E_t^{DE}}\right] \cdot \mathbb{E}\left[\exp^{z_j}\right] + \mathbb{C}\left(\frac{e_{ij\ell,t}^{DE}}{E_t^{DE}}, \exp^{z_j}\right)}.
\end{aligned}$$

Thus,

$$\mathcal{Z}_e^{FB} - \mathcal{Z}_e^{MR} = \frac{\mathbb{C}\left(\frac{e_{ij\ell,t}^{FB}}{E_t^{FB}}, \frac{\exp^{z_j}}{\mathbb{E}\left[\exp^{z_j}\right]}\right) - \mathbb{C}\left(\frac{e_{ij\ell,t}^{MR}}{E_t^{MR}}, \frac{\exp^{z_j}}{\mathbb{E}\left[\exp^{z_j}\right]}\right)}{1 + \mathbb{C}\left(\frac{e_{ij\ell,t}^{DE}}{E_t^{DE}}, \frac{\exp^{z_j}}{\mathbb{E}\left[\exp^{z_j}\right]}\right)} = \mathbb{C}\left(\frac{e_{ij\ell,t}^{FB}}{E_t^{FB}}, \frac{\exp^{z_j}}{\mathbb{E}\left[\exp^{z_j}\right]}\right) \tag{A56}$$

since $\mathbb{C}\left(\frac{e_{ij\ell,t}^{DE}}{E_t^{DE}}, \frac{\exp^{z_j}}{\mathbb{E}\left[\exp^{z_j}\right]}\right) = \mathbb{C}\left(\frac{e_{ij\ell,t}^{MR}}{E_t^{DE}}, \frac{\exp^{z_j}}{\mathbb{E}\left[\exp^{z_j}\right]}\right) = 0$, as energy in these two equilibria is independent of carbon intensity, and a_i and z_j are independent in the baseline calibration.

C Quantitative results

C.1 Parametrization

Table C.1 reports the model parameters, their calibrated values, and the corresponding calibration targets and sources, as discussed in Section 5.1. Figure C.1 compares the impulse response of the atmospheric carbon stock to an emissions shock in our model with that implied by the three-reservoir model in Golosov et al. (2014).

Table C.1: Parameters and calibration targets or sources

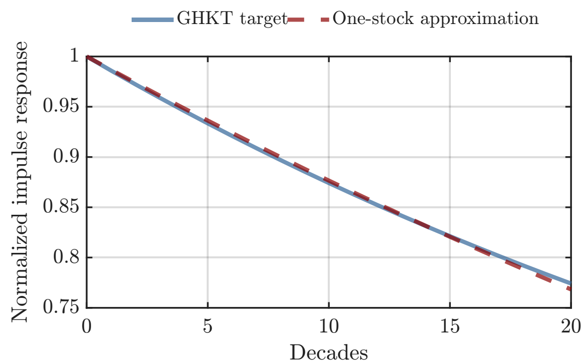
Description	Parameter	Value	Target or source
Economic block			
Discount rate	β	0.985^{10}	
Capital elasticity	α	0.346	
Labor elasticity	γ	0.518	Standard
Energy elasticity	ν	0.036	
Capital depreciation rate	δ_K	$1-(1-0.07)^{10}$	
Energy price	P	1	Normalization
Mean productivity	μ_a	0	Normalization
Std productivity	σ_a	0.6	Compustat data estimates
Firm lifespan	L	1	World Bank (2024)
Climate block			
Carbon depreciation rate	δ_G	0.0131	Golosov et al. (2014)
Damage semi-elasticity	ϕ	0.0118	Productivity loss from 1 $GtCO_2e$
Corr between a and z distributions	ρ_{az}	0	ICE ESG data estimates
Mean carbon intensity	μ_z	0	Normalization
Std carbon intensity	σ_z	1.6	ICE ESG data estimates
Abatement cost inverse elasticity	θ	2	Känzig (2023)
Initial steady state			
Initial stock of carbon in atmosphere	G_0	1.4394	2% annual emissions growth over 170 years
Initial stock of capital	K_0	1.7655	Steady state consistent with G_0

C.2 Sketch of the solution algorithm

At each date, the transition is characterized by four aggregate equilibrium conditions: the law-of-motion for the carbon-stock, the consumption Euler equation, the goods market clearing condition, and the recursion for the social cost of carbon. These correspond, respectively, to Equations (4), (7), (13), and Definition 1. Importantly, these dynamic conditions are *planner-independent*. Given the initial capital and carbon stock, and imposing convergence of consumption and the cost of carbon to their regime-specific steady-state values at the terminal date, they define a nonlinear two-point boundary-value problem for the sequences of capital, the carbon stock, consumption, and the social cost of carbon.

The planner-specific allocation is determined within each period by a *static block* conditional on the aggregate states K_t and G_t, and the social cost of carbon $\Lambda_{G,t}$. Factor-market clearing implies $W_t(\text{or } \Lambda_{N,t}) = \gamma \cdot \mathcal{Y}_t$ and $R_t(\text{or } \Lambda_{K,t}) = \alpha \cdot \mathcal{Y}_t / K_t$. Given a candidate

Figure C.1: IRF match to emission shock in Golosov et al. (2014)



Note: The figure compares the normalized impulse response of atmospheric carbon to a one-time emissions pulse under the carbon-cycle specification of Golosov et al. (2014) and our one-stock baseline model. We calibrate δ_G by minimizing the squared deviations between the two impulse responses over the first 20 decadal observations. The resulting carbon depreciation rate is $\delta_G = 0.0131$ per decade.

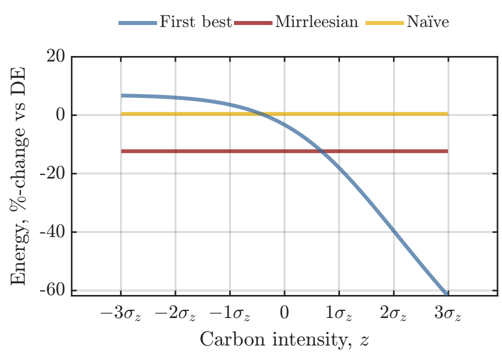
value of aggregate output $\mathcal{Y}_t$, shadow prices are therefore determined, and each firm's capital and labor choices solve the relevant version of (10) in all four models. Each firm's energy and abatement choices then solve the first-order conditions: (11) in the decentralized equilibrium, (17) for the full-information planner, (25) for the Mirrleesian planner, and (20) for the naïve planner. These conditions thus determine the firm's effective marginal cost of energy, given by the decentralized energy price plus the planner-specific marginal social cost of energy. Finally, gross-output aggregation, $\mathcal{Y}_t = \sum_{ij\ell} w_{ij\ell} \cdot y_{ij\ell,t}$, closes the one-dimensional fixed point. The static block therefore returns aggregate output, emissions, energy use, and abatement costs in one sweep.

The transition is then obtained by solving the planner-independent dynamic system jointly with the relevant planner-specific static block over the entire horizon. We solve the resulting nonlinear system globally with a damped Newton method, using a sparse, block-banded Jacobian assembled analytically from the derivatives of the static block. The initial conditions fix $t = 0$ capital stock and carbon stock, while the terminal conditions anchor consumption and the social cost of carbon at their long-run steady-state values. The method thus determines the entire path of endogenous variables jointly. We set the horizon to ($T = 800$), which ensures that the slowly decaying carbon stock is sufficiently close to its steady-state level at the terminal date.

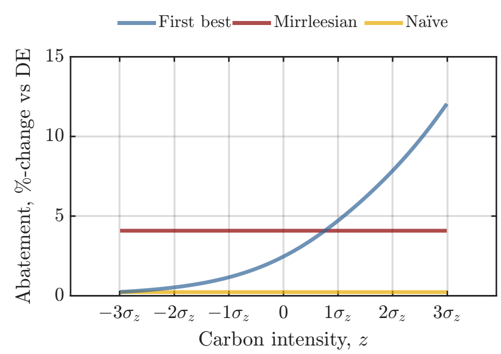
C.3 Firm-level outcomes

Figures C.2 and C.3 plot firm-level energy use, abatement, and emissions under the different planner allocations as %-deviations from the decentralized equilibrium along the carbon intensity and productivity dimensions. Figure C.2 starts by illustrating how the planner allocations vary with firms' carbon intensity. Under full information, the planner separates firms by carbon intensity, imposing progressively higher carbon bill on more carbon-intensive firms. As a result, these firms reduce energy use more, undertake greater abatement, and experience larger declines in emissions. By contrast, the Mirrleesian planner bunches firms along the carbon-intensity dimension to eliminate incentives to misreport. The naïve planner also induces bunching, as all firms optimally misreport the lowest carbon-intensity type. Because the resulting allocation fails to internalize the true variation in carbon intensities, it delivers only negligible reductions in emissions.

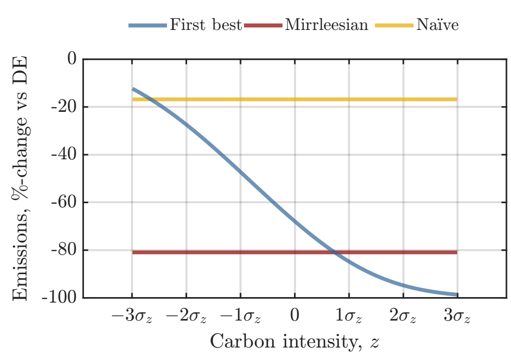
Figure C.2: Firm-level outcomes by carbon intensity



(a) Energy



(b) Abatement

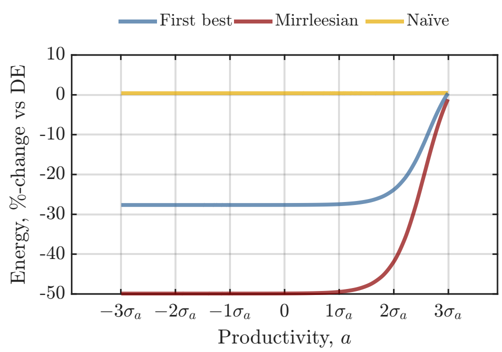


(c) Emissions

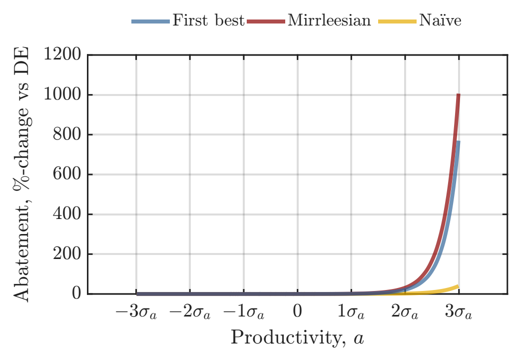
Note: The figure reports firm-level energy use, abatement, and emissions under the full-information, Mirrleesian, and naïve planner allocations as %-deviations from the decentralized equilibrium by carbon-intensity type. Higher values of z correspond to more carbon-intensive firms.

Figure C.3 shows how the planner allocations instead vary with productivity. Under full information, more productive firms use more energy because of their higher marginal revenue product of energy. Consequently, they also undertake more abatement and achieve larger reductions in emissions. The Mirrleesian allocation closely tracks these patterns across the productivity distribution, although it delivers somewhat smaller emission reductions, on average; incentive compatibility prevents the planner from fully conditioning allocations on carbon intensities. By contrast, the naïve planner fails to generate meaningful emission reductions because all firms optimally misreport the lowest carbon-intensity type, weakening the effective carbon price faced by high-emission firms.

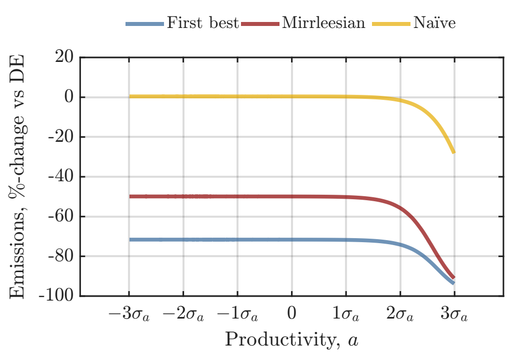
Figure C.3: Firm-level outcomes by productivity



(a) Energy



(b) Abatement



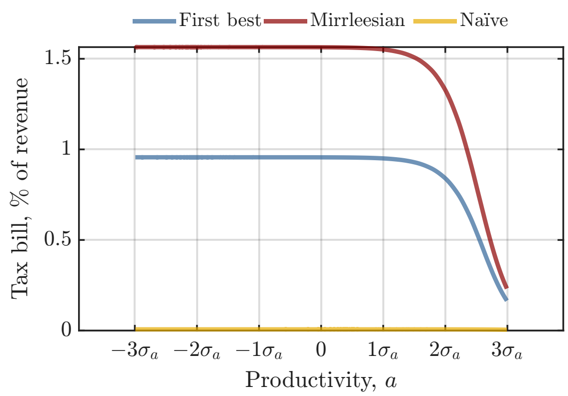
(c) Emissions

Note: The figure reports firm-level energy use, abatement, and emissions under the full-information, Mirrleesian, and naïve planner allocations as %-deviations from the decentralized equilibrium by productivity type. Higher values of z correspond to more productive firms.

C.4 Carbon tax

Figure C.4 reports the overall carbon tax bill as a share of firm revenue across the productivity distribution. Under both the full-information and Mirrleesian allocations, the tax burden declines with productivity, reflecting that more productive firms generate proportionally more revenue relative to their emissions since they undertake more abatement. The Mirrleesian allocation closely tracks the first-best allocation, although tax bills are slightly higher as the planner cannot fully condition allocations on firms' carbon intensity and therefore achieves somewhat less abatement, on average. By contrast, the naïve planner yields a substantially smaller tax burden because firms optimally misreport their carbon intensity, reducing the effective taxation of emissions.

Figure C.4: Carbon tax bill as a share of revenues by productivity

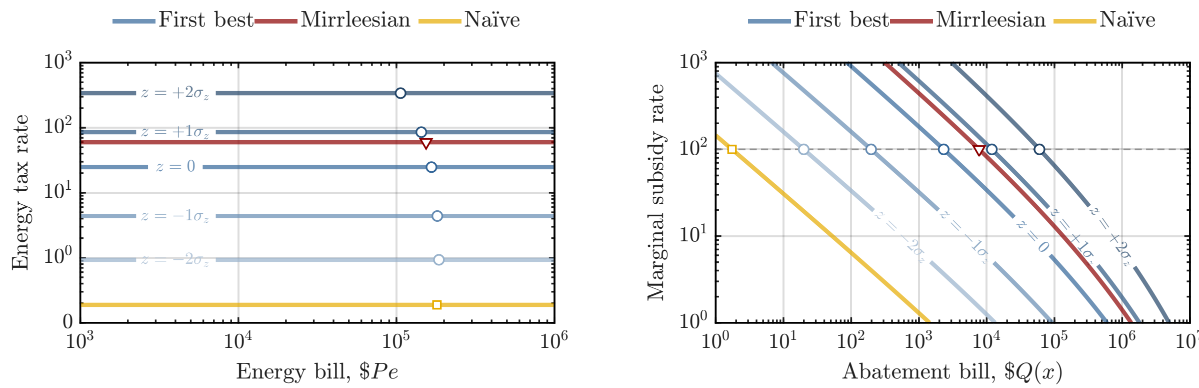


Note: The figure reports the overall carbon tax bill as a share of firm revenues by productivity type under the full-information, Mirrleesian, and naïve planner allocations.

C.5 Energy tax and abatement subsidy

Figure C.5 illustrates quantitatively the alternative implementation based on an energy tax and an abatement subsidy. The energy tax is linear, whereas the abatement subsidy is nonlinear and declining in abatement expenditure, as predicted by the theory (Appendix B.2). Because abatement does not generate any private benefits, the planner subsidizes its marginal cost; in particular, the marginal subsidy exceeds 100 percent for abatement levels below the firm's optimal choice and equals 100 percent at the optimum. The full-information planner varies both the energy-tax and abatement-subsidy schedules with carbon intensity, whereas the Mirrleesian planner uses schedules that are uniform across carbon-intensity types. The Mirrleesian energy-tax schedule is substantially higher than the median full-information schedule. Under full information, the median firm faces an energy-tax rate of approximately 20–30 percent, compared with roughly 1 percent for firms two standard deviations below the mean and slightly above 300 percent for firms two standard deviations above the mean. By contrast, the Mirrleesian planner sets a uniform energy-tax rate of approximately 60 percent.

Figure C.5: Calibrated energy tax and abatement subsidy



(a) Energy tax **(b)** Abatement subsidy

Note: Panel (a) shows the marginal energy-tax rate as a % of the firm's energy bill. Panel (b) shows the marginal abatement-subsidy rate as a % of the firm's abatement spending. Each schedule is drawn against the corresponding annual dollar bill, per firm. The first-best (full-information) schedule is carbon-intensity specific, so we plot it for $z_j \in \{0, \pm 1\sigma_z, \pm 2\sigma_z\}$ (blue); the Mirrleesian (red) and naïve (yellow) planners instead use a single, uniform schedule. Markers denote optimal choices.

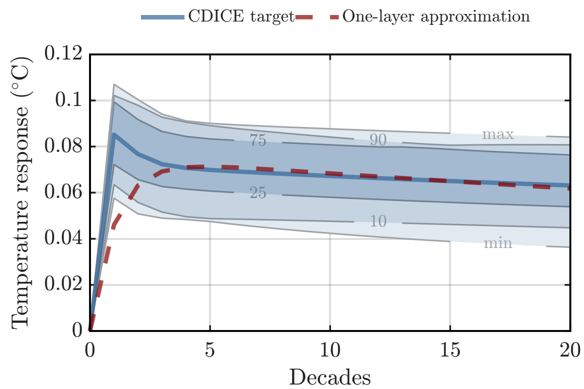
D Quantitative refinements

D.1 Alternative climate model

The parameters of the temperature-augmented model are: the damage semi-elasticity ϕ, the mean-reversion speed of temperature ξ_T, the long-run carbon-stock-to-temperature sensitivity Ξ_G, the carbon-stock depreciation rate δ_G, and the historical mean temperature $\mathcal{T}$. Because damages now operate directly through temperature, our target of a 2% productivity loss per 1°C of warming directly implies $\phi = 0.02$.

We calibrate the remaining climate block parameters $\{\delta_G, \xi_T, \Xi_G\}$ jointly by fitting the model's decadal temperature impulse response to that of the CDICE calibration of Folini et al. (2024), while also matching the IPCC AR6 best estimate of the transient climate response to cumulative CO_2e emissions, $\xi_T \cdot \Xi_G \approx 4.5 \cdot 10^{-4}$ °C/GtCO_2e. The calibration yields a carbon-stock depreciation rate of $\delta_G \approx 0.010$ per decade, a stock-to-temperature sensitivity of $\Xi_G \approx 7.4 \cdot 10^{-4}$ °C/GtCO_2e, and a temperature mean-reversion speed of $\xi_T \approx 0.62$ (i.e., $\xi_T \cdot \Xi_G \approx 4.61 \cdot 10^{-4}$). This corresponds to a per-decade temperature persistence of $1 - \xi_T \approx 0.38$, which lies between the persistence parameters estimated for the land and deep-ocean layers, 0.06 and 0.96, respectively. Figure D.1 shows that the resulting one-layer decadal system closely reproduces the CDICE temperature response, with the temperature impulse response to a 100-GtCO_2e emissions pulse at $t = 0$ peaking in the first decades, consistent with the evidence in Dietz et al. (2021).

Figure D.1: IRF match to emission shock in Folini et al. (2024)



Note: The figure compares the temperature impulse response to a 100-GtCO$_2$e emissions pulse at $t = 0$ in our one-layer temperature and carbon-stock model with the corresponding response in CDICE's multilayer temperature and carbon-stock model (Folini et al., 2024). The dashed line represents our model, while the solid line represents the CDICE benchmark. The shaded bands show the spread of CDICE-type responses across the 16 CMIP5 AOGCMs in Geoffroy et al. (2013), obtained by varying each model's two-layer temperature parameters while holding the carbon cycle fixed at the CDICE multimodel mean. From darkest to lightest, they represent the 25–75th, 10–90th, and min–max ranges across models, with each boundary labeled by its percentile.

Finally, the specification requires three initial conditions, $\{K_0, G_0, T_0\}$. We set the historical mean temperature to $\mathcal{T} = 14$°C and, rather than imposing the current temperature, derive it endogenously. As in the baseline calibration, we set the initial atmospheric carbon stock $G_0 = 1.33258$ by assuming that emissions have grown at 2% per year over the 170 years since the pre-industrial period. Given the calibrated climate block, this carbon stock implies approximately 1.5°C of warming relative to the historical mean temperature, i.e., $T_0 \approx 15.5$°C, consistent with the upper-end of current estimates of observed warming (e.g., IPCC, 2021). Finally, we set the initial capital stock $K_0 = 1.09952$ to equal the steady-state capital stock consistent with G_0 and T_0.

D.2 Green and brown energy inputs

The model requires us to calibrate the following parameters: $\{\epsilon, P_{\text{brown}}, \tau_{\text{green}}, \phi, G_0, K_0\}$. We calibrate the elasticity of substitution between brown and green energy ϵ to 5, consistent with the recent estimates of Petersen (2025) and near the upper end of the empirical range reported by Papageorgiou et al. (2017) and Jo (2025). We normalize the price of brown energy to $P_{\text{brown}} = 1$ and choose the renewable-energy price wedge τ_{green} so that green energy accounts for one quarter of total energy expenditure in the decentralized equilibrium in 2024. This target is consistent with the EIA's 2024 estimate for the United States and implies a relative price of green energy of $P_{\text{green}} = (1 + \tau_{\text{green}}) \cdot P_{\text{brown}} = 1.3161$.

As in the baseline calibration, we set the damage semi-elasticity to $\phi = 0.0157$, generating a 2% productivity loss for each additional 1°C of warming, using the TCRE estimate of 4.609×10^{-4} °C per additional GtCO_2e. We set the initial atmospheric carbon stock by assuming that emissions grew at 2% per year over the 170 years since the pre-industrial period. The initial capital stock is then set equal to the steady-state capital stock consistent with this carbon stock. This procedure yields $G_0 = 1.0841$ and $K_0 = 1.7729$, respectively.

Table D.1: Green versus brown energy

	$\mathcal{C}$	$\mathcal{A}$	$\mathcal{E}_{\text{brown}}$	$\mathcal{E}_{\text{green}}$	$\mathcal{K}$	$\mathcal{X}$
Naïve planner	+0.65	+0.46	+0.23	+1.26	+0.50	+0.00
Mirrleesian planner	+2.86	+2.34	−49.80	+63.94	+1.96	+0.54
Full-information planner	+3.22	+2.49	−27.48	+48.16	+2.33	+0.28

Note: The table shows the present-value of consumption, aggregate productivity, brown-energy use, green-energy use, capital, and abatement under the naïve, Mirrleesian, and full-information planner allocations, respectively. We report results in terms of the %-deviation from the decentralized-equilibrium path.

Table D.1 shows the present-value of consumption, aggregate productivity, brown-energy use, green-energy use, capital, and abatement across the different planner allocations. We report results in terms of %-deviation from the decentralized-equilibrium path. Allowing firms to substitute between brown and green energy substantially changes the energy mix under the optimal allocations, without materially affecting the relative performance of the different planners. As in the baseline model, the Mirrleesian planner captures most of potential welfare gains (c. 89%). By contrast, the naïve planner captures only a much smaller fraction of the available welfare gains (c. 20%).

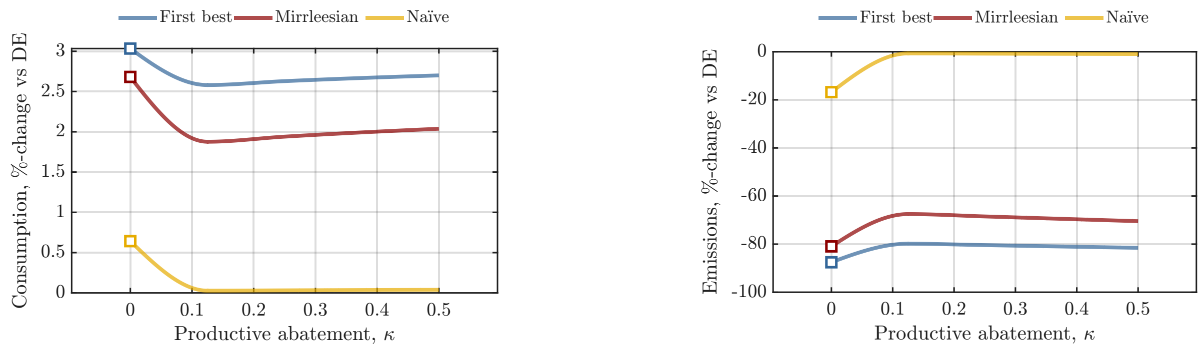
D.3 Productive abatement

To guarantee a well-behaved interior solution for the optimal abatement choice, we use the following functional form for the productive role of abatement:

$$\Psi(x; \kappa) = \exp[\kappa \cdot (1 - \exp(-x))]. \tag{A57}$$

All other parameters remain unchanged from the baseline calibration, except for ϕ, which is recalibrated for each value of κ to ensure that the semi-elasticity of damages with respect to aggregate emissions remains the same as in the baseline model.

Figure D.2: Different strength of the productive role of abatement κ



(a) Consumption **(b)** Emissions

Note: The figure plots the present-value of consumption and emissions under the full-information, Mirrleesian, and naïve planner allocations, expressed as %-deviations from the decentralized equilibrium, for different values of κ. Square symbols denote the baseline steady states.

Figure D.2 plots the present-value of consumption and emissions under the full-information, Mirrleesian, and naïve planner allocations, expressed as %-deviations from the decentralized equilibrium, for different values of the strength of the productive role of abatement κ. Here, $\kappa = 0$ corresponds to the baseline model, while $\kappa = 0.5$ represents an unlikely upper bound under which a firm that fully decarbonizes would experience a 65% increase in productivity. Overall, regardless of the strength of abatement's productive role, the Mirrleesian planner continues to attain most of the first-best benefits, whereas the naïve planner captures substantially less of them.

D.4 Alternative degrees of imperfect information

In this case, the wedge in the Mirrleesian planner's first-order conditions equals:

$$\lambda^{MR}_{G,ij,t} = \Lambda_{G,t} \cdot \exp^{\eta \cdot z_j} \cdot \mathbb{E}[\exp^{(1-\eta)\cdot z_j} \,|a_i] \cdot \exp^{-x_{ij,t}} \tag{A58}$$

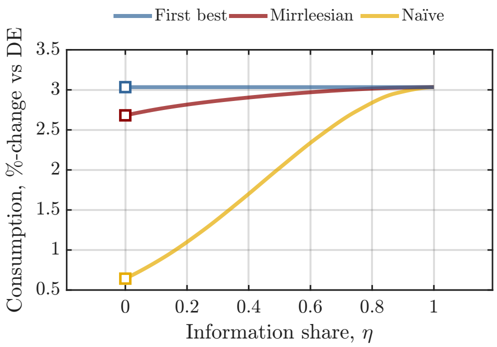
while the corresponding wedge for the naïve planner is

$$\lambda^{NV}_{G,ij,t} = \Lambda_{G,t} \cdot \exp^{\eta \cdot z_j} \cdot \exp^{(1-\eta)\cdot \underline{z}} \cdot \exp^{-x_{ij,t}} \,. \tag{A59}$$

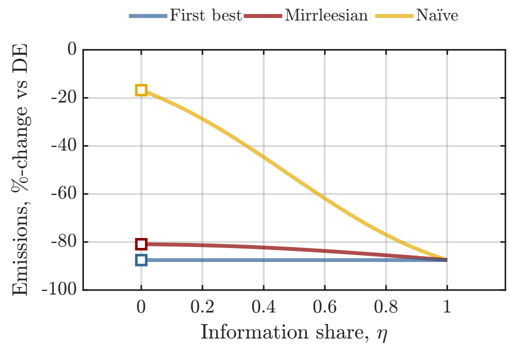
Clearly, as $\eta \to 1$, both expressions converge to the full-information case. The baseline specification corresponds to $\eta = 0$ under which the planner knows only the distribution of carbon intensities but has no firm-specific information.

Figure D.3 plots the present-value of consumption and emissions under the full- information, Mirrleesian, and naïve planner allocations, expressed as %-deviations from the

Figure D.3: Alternative degrees of imperfect information



(a) Consumption



(b) Emissions

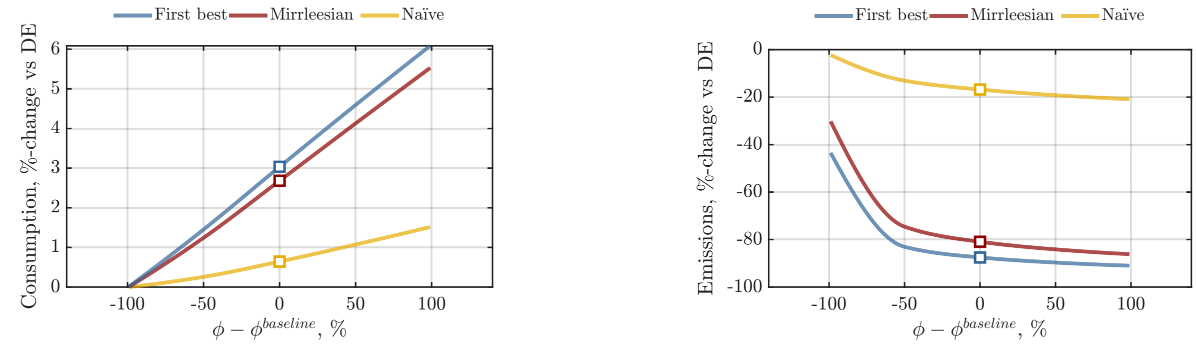
Note: The figure plots the present-value of consumption and emissions under the full-information, Mirrleesian, and naïve planner allocations, expressed as %-deviations from the decentralized equilibrium, for different values of the information parameter η. Square symbols denote the baseline steady states.

decentralized equilibrium, for different values of the information parameter η. Notice that accounting for firms' disclosure incentives remains important—almost irrespective of the planner's information set. Across all values of η, the Mirrleesian planner recovers most of the first-best welfare gains, whereas the naïve planner performs substantially worse. For example, even when the planner observes 50% of firms' carbon-intensity information perfectly, ignoring misreporting incentives still recovers 30% less of the associated first-best welfare gains, while the Mirrleesian planner continues to recover most of them.

D.5 Climate parameters uncertainty

In this section, we perform comparative statics with respect to three climate-related parameters: the environmental damage parameter ϕ, the inverse elasticity of abatement θ, and the standard deviation of the carbon-intensity distribution σ_z. Figures D.4-D.6 report the results for parameter values 100% above and below the baseline calibration. Overall, the main message remains unchanged: the Mirrleesian planner consistently recovers a large share of the first-best gains and substantially more than the naïve planner.

Figure D.4: Comparative static I: environmental externality

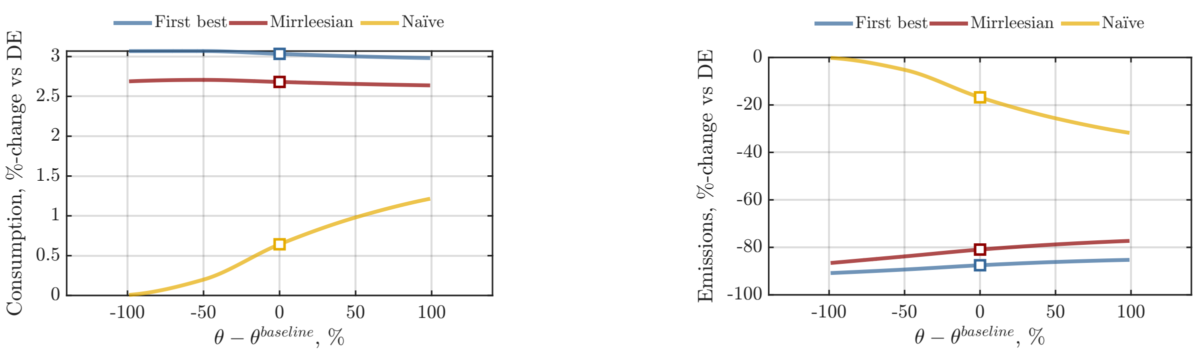


(a) Consumption

(b) Emissions

Note: The figure plots the present-value of consumption and emissions under the full-information, Mirrleesian, and naïve planner allocations, expressed as %-deviations from the decentralized equilibrium, for different values of the environmental externality ϕ. Square symbols denote the baseline steady states.

Figure D.5: Comparative static II: abatement elasticity

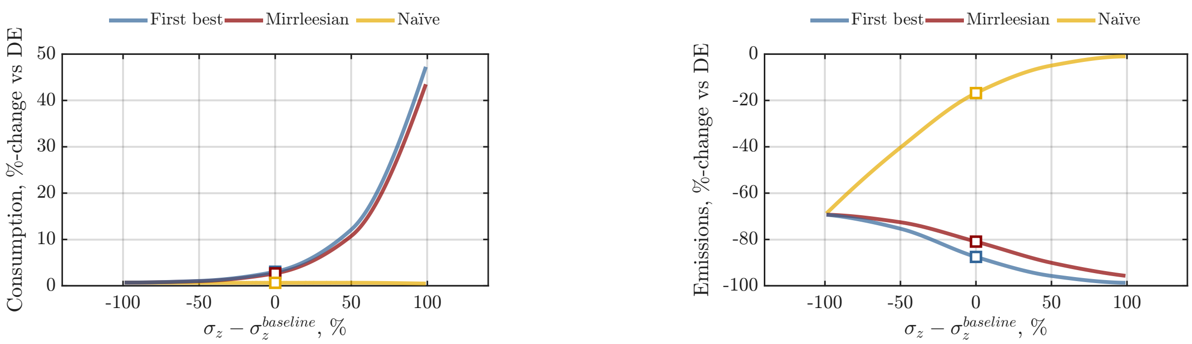


(a) Consumption

(b) Emissions

Note: The figure plots the present-value of consumption and emissions under the full-information, Mirrleesian, and naïve planner allocations, expressed as %-deviations from the decentralized equilibrium, for different values of the abatement elasticity $\theta > 0$. Square symbols denote the baseline steady states.

Figure D.6: Comparative static III: carbon-intensity dispersion



(a) Consumption

(b) Emissions

Note: The figure plots the present-value of consumption and emissions under the full-information, Mirrleesian, and naïve planner allocations, expressed as %-deviations from the decentralized equilibrium, for different values of the carbon intensity dispersion σ_z. Square symbols denote the baseline steady states.